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Impact of disasters, including pandemics, on cardiometabolic outcomes across the life-course: A systematic review

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Impact of disasters, including pandemics, on cardiometabolic outcomes across the life-course: A systematic review

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ABSTRACT

Objectives: Disasters, such as the current COVID-19 pandemic, disrupt daily life, increase uncertainty and stress, and may increase long-term risk of adverse cardiometabolic outcomes, including heart disease, obesity and diabetes. The objective was to conduct a systematic review to determine the impact of disasters, including pandemics, on cardiometabolic outcomes across the life-course.

Design: A systematic search was conducted in May 2020 using two electronic databases, EMBASE and Medline. All studies were screened in duplicate at title and abstract, and full-text level. Studies were eligible for inclusion if they assessed an association with population-level or community disaster and cardiometabolic outcomes. There were no restrictions on year of publication, country or population. Non-English and earthquake-related studies were excluded. Data were extracted on study characteristics, exposure (e.g., type of disaster, name of specific event, region, year), cardiometabolic outcomes, and measures of effect. Study quality was evaluated using the Joanna Briggs Institute critical appraisal tools.

Results: A total of 58 studies were included, with 24 studies reporting the effects of exposure to disaster during pregnancy/childhood and 34 studies reporting the effects of exposure during adulthood. Studies included exposure to natural (60%) and human-made (40%) disasters, with only 3 (5%) of these studies evaluating previous pandemics. Most studies were conducted in North America (62%). Most studies reported increased cardiometabolic risk, including increased cardiovascular disease incidence or mortality, diabetes, and obesity. Few studies investigated potential mechanisms or identified high risk subgroups.

Conclusions: Understanding the long-term consequences of disasters on cardiometabolic outcomes across the life-course may inform public health strategies for the current COVID-19 pandemic. This review found strong evidence of an increased association between disaster exposure and cardiometabolic outcomes across the life-course, although more research is needed to better understand the mechanisms and preventative efforts.

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Strengths and limitations of this study:

- This systematic review is one of the first to review the literature on disasters, including pandemics, and subsequent cardiometabolic outcomes throughout the life-course.
- A comprehensive search strategy was developed in consultation with Health Science Librarians at McMaster University, which resulted in 58 studies that were eligible for inclusion into the review.
- Due to the heterogeneity of the included studies, a meta-analysis was not conducted.
- This review contributes a synthesis of the literature on the impact of disasters and cardiometabolic outcomes, that can help to inform public health strategies for the current COVID-19 pandemic.

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BACKGROUND

Disasters as defined by the World Health Organization (WHO), are events that disrupt the daily functioning of a community or society causing material, economic or environmental losses, overwhelming local capacity (1). Disasters can be categorized into natural disasters, human-made disasters, and hybrid disasters (2). Natural disasters include natural phenomenon above and beneath the earths surface (e.g., tsunamis or landslides), meteorological phenomenon (e.g., tornadoes or floods) or biological phenomenon (e.g., epidemics and pandemics) (2). Human-made disasters include adverse transportation incidents, technological events (e.g., fire or toxic leaks), terrorism, warfare or conflict (2). A hybrid disaster results from both human error and natural forces, such as the clearing of a jungle that results in a landslide (2). All types of disasters can result in public health emergencies as they typically impact a significant proportion of people (3). Epidemics, defined as a greater than expected increase in cases of a disease, and pandemics, which cross countries and continents, are types of natural disasters with far-reaching global disruption (4). The COVID-19 pandemic is a present-day example of a global disaster that is unlike any disaster in recent history. Understanding the potential long-term health implications of the current COVID-19 pandemic and resulting public health mitigation strategies is urgently needed.

Previous systematic reviews have focused specifically on the psychological impact of quarantine during pandemics (5), the impact on health outcomes after disasters in older adults (6), medically unexplained physical symptoms following disasters (7), and chronic medical interventions following a natural disaster (8). However, there is currently no review specifically focusing on the impact of either disasters, or epidemics and pandemics on cardiometabolic outcomes across the life-course. Noncommunicable diseases (NCDs), including cardiovascular disease (CVD), obesity, and diabetes, are the leading cause of morbidity and mortality worldwide (9,10). NCDs are attributed to 71% of all global deaths annually, with approximately 14 million CVD-related deaths and 1.6 million diabetes-related deaths (10). Findings from the Global Burden of Diseases Study indicate that CVD and diabetes account for over 20% of the global burden of disability, with diabetes recently emerging as the fourth leading cause of disability globally (9). Exposure to disasters may cause cardiometabolic outcomes to emerge or worsen through several different mechanistic pathways including stress exposure (11), lack of access to health services (12), food security and behavioural changes such as alterations in physical activity, sleep, and diet (13). It is important to understand the impact of disasters on the incidence of new cardiometabolic diseases and changes in disease status in all populations and age groups. Particular subgroups of a population may be more or less susceptible to cardiometabolic outcomes and

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understanding this can inform targeted interventions. The primary objective of this review was to determine the impact of disasters, including pandemics such as COVID-19, on risk of cardiometabolic outcomes across the life-course. The secondary objectives were to determine how to reduce the impact of chronic disease outcomes following a disaster and to identify populations at highest risk of cardiometabolic outcomes following a disaster.

METHODS

A systematic review was conducted following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) (14). This review was registered on PROSPERO (CRD – 42020186074).

Search strategy

A systematic search was conducted in May 2020 using the electronic databases EMBASE and MEDLINE. The health research librarians at McMaster University assisted in developing the search strategy. The search broadly captured two concepts: disasters and cardiometabolic outcomes (e.g., diabetes, obesity, hypertension). The complete search strategy for EMBASE can be found in Table 1. The search strategy for MEDLINE can be found in the Appendix (Table A1). Reference lists of eligible studies and relevant systematic reviews were hand searched to identify additional articles.

Eligibility criteria

Studies were eligible for inclusion if they assessed the relationship between a population-level or community disaster and the risk of future cardiometabolic outcomes including CVD, diabetes or obesity or cardiometabolic risk scores (15). CVD included myocardial infarction, stroke, hypertension and angina. There were no restrictions on year of publication, country of disaster, or population. Only studies evaluating the impact of real-world disasters in humans were included. Due to the research team's capacity, only studies published in English were included. Observational and quasi-experimental study designs, including case-control, cohort and other longitudinal study designs or natural experiments were included. Outcomes that were not cardiometabolic-related or acute cardiometabolic events, such as an immediate complication (defined as <1 month after disaster), were excluded. Studies that assessed the exposure to a chemical as a result of the disaster, were excluded, as we were not interested in outcomes resulting from chemical exposure. Earthquake studies were also excluded since a

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systematic review was published in 2018 that assessed the impact of earthquakes on cardiometabolic outcomes (16).

Study selection

After running the search, all identified studies were imported into Covidence and duplicates were removed (17). Studies were screened at title and abstract-level, and then at full-text by two independent reviewers (VD, JL, MSA, YYM, ATA, ES, SI, JDM, RR LNA). Conflicts were resolved by a third reviewer who made the final decision regarding eligibility for inclusion.

Data extraction

A data extraction template was created, and pilot tested prior to data extraction. Data were then extracted from all studies by two independent abstractors (VD, JL, SMA, YYM, ATA, RR) and conflicts were resolved by a third independent abstractor. Study characteristics including year of publication, study design, country of disaster, sample size, and length of study were extracted where reported. Specific information on the exposure and outcome in each study were extracted including the type and name of the disaster, country and year of the disaster, the outcome of interest, and how the exposure and outcome were measured. Finally, any information on subgroups including age, population, sex and disaster type were also extracted, if applicable.

Critical appraisal

Critical appraisal was conducted using the Joanna Briggs Institute (JBI) Critical Appraisal Tools (18). This tool was chosen due to the availability of checklists for a wide range of study designs, including cohort, cross-sectional and quasi-experimental designs (18). The quasi-experimental study design checklist was used for natural experiments including time-series studies and pre/post designs, as it was decided this was the most appropriate tool. All studies were critically appraised independently by two individuals (VD, JL, MA, YYM, ATA, ES, SI) and a third individual was consulted for any discrepancies.

Data analysis

Data from the included studies were narratively synthesized. Results are presented by exposure period (perinatal/childhood versus adulthood) and by cardiometabolic outcome (obesity, CVD, and diabetes). Characteristics of studies are presented as frequencies and percentages. Due to the heterogeneity of studies, a meta-analysis was not conducted.

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RESULTS

A total of 4830 studies were identified through the electronic database search. An additional 12 studies were identified through manual searching of the reference lists of relevant studies. After removing duplicates (n=439), 4403 studies were screened at title and abstract level. After applying inclusion and exclusion criteria, 4068 studies were excluded, leaving 335 studies screened for full-text eligibility. A total of 58 studies were eligible for inclusion into the review. The complete screening process is described in Figure 1.

Description of studies

Table 2 provides a summary of included studies. Of 58 included studies, 24 studies (13,19–41) investigated exposure to disasters during pregnancy or childhood while the remaining 34 studies (12,42–74) investigated exposure to disaster during adulthood. Almost all studies (n=49) assessed cardiometabolic outcomes during adulthood, only two studies assessed outcomes during pregnancy (25,26) and seven studies assessed outcomes during childhood and adolescence (13,19–24). The length of studies, including prospective follow-up and retrospective assessment, ranged from 1 month to 95 years. Most studies (n=36) focused on disasters that occurred in North America (12,19–26,36,37,40,43–47,49,50,52,53,55–58,61–63,65–70,73,74), followed by Europe (n=13) (27–33,39,41,51,60,71,72). The remaining disasters occurred in Asia (n=7) (13,35,38,42,48,54,59) and Africa (n=2) (34,64). The characteristics of included studies and key findings are provided in Table 3 for disaster exposure during the perinatal period and childhood, and Table 4 for exposure during adulthood.

Studies on perinatal and childhood exposure to disaster and subsequent cardiometabolic outcomes

Of the 24 studies that evaluated perinatal and childhood exposure to disaster, 12 studies evaluated human-made disasters (27–35,38,39,45) and the remaining 12 evaluated natural disasters (13,19–26,36,37,40) of which two were pandemics (36,37). Most studies (n=15) assessed the disaster as the main exposure of interest (22,26–31,33–37,37–40). The remaining studies evaluated stress (e.g., maternal stress, disaster-related PTSD, subjective stress, objective hardship) (19–23,25,45), maternal weight and maternal nutrition status (13,79), cognitive appraisal (24), and coping strategies (25) that

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were the result of the disaster as the exposure variable. The age when cardiometabolic outcomes were assessed varied across studies, with 7 studies evaluating outcomes during childhood and adolescence (≤18 years of age) (13,19–24), 4 studies during young to mid adulthood (>18-40 years of age) (25,34,80,83), and 10 during later adulthood (≥50 years of age) (28,30–33,36–40). Ekamper et al. (2015), evaluated outcomes throughout the life-course, with age at outcome ranging from 18-63 years (29). Two studies (26,45) did not specifically identify the age at which outcomes were assessed. Detailed characteristics and findings of all studies that assessed perinatal and childhood exposures to disasters can be found in Table 3.

Studies that assessed human-made disasters found an increased association or increased change in obesity, metabolic syndrome, CVD related-outcomes and diabetes, however not all were statistically significant (OR range 1.15-3.11; hazard ratio (HR) range: 1.26-1.91) (27,29–35,38,45). Some studies also reported no association or a decreased association following exposure to human-made disasters and cardiometabolic outcomes (OR range: 0.94-1.02) (29,34). Among the studies that evaluated natural disasters, maternal stress, including objective hardship and objective prenatal maternal stress, were found to increase the odds of obesity using different adiposity measures during childhood and adolescence (19–21). However, weak correlations were found between subjective distress/stress and body mass index (BMI) (r=0.39; p=0.03), zBMI (r=0.40; p=0.02), and body fat percentage (r=0.33; p=0.09) in offspring at age 13 (22). The prevalence of heart disease, ischemic heart disease, emergency department visits for gestational hypertension, myocardial infarction, diabetes diagnoses and diabetes-related emergency department visits increased after exposure to Hurricanes Sandy, San Felipe and Ciprian, and the 1918 Influenza Pandemic (26,36,40). In addition, exposure to hurricanes and stress-related to hurricanes was noted to increase pregnancy induced hypertension and gestational diabetes (odds ratio (OR) range: 1.13-1.22) (25).

Studies on adult exposure to disaster and subsequent cardiometabolic outcomes

Thirty-four studies investigated the effects of exposure to disasters during adulthood on cardiometabolic outcomes. The length of follow-up ranged from 1 month to 13 years. There were 23 studies that examined natural disasters (12,42–44,46,49,50,52–54,56–59,62,63,65,71–74), and 11 studies that examined human-made disasters (41,47,48,51,55,60,61,66–70). Of these studies, only one evaluated the impact of an infectious disease epidemic (64). Most studies (12,41–44,46,48,50,52–54,56–

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64,66,69–74) considered the disaster as the main exposure of interest (n=27). The remaining 7 studies assessed disaster-related stress (51,55,65,67), including post-traumatic stress disorder (PTSD) and psychological strain, unemployment rates as a result of the disaster (49) and exposure to damaged or collapsed buildings during the World Trade Center disaster (68). Detailed characteristics and findings of all studies that assessed adult exposures to disasters are included in Table 4.

The studies that assessed exposure to human-made disasters during adulthood reported mixed results in terms of associations with outcomes and statistical significance. Most studies reported an increased association with outcomes, including heart disease-related mortality, CVD hospitalization, complaints of CVD, vascular problems, stroke, and incidence of hypertension (OR range: 1.21-3.30; HR range: 1.63-1.82) (41,43,47,51,55,60,61,67,68,70). No association or a decreased association with CVD hospitalizations was noted following exposure to the World Trade Centre attacks (47,66). In addition, time of evaluation from damaged buildings resulting from the World Trade Centre also had a decreased association with cardiometabolic outcomes (OR range: 0.7-0.9) (68). Among studies that evaluated the impact of exposure to natural disasters on subsequent cardiometabolic outcomes, nearly all studies reported an increased association (OR range: 1.08-5.65; relative risk (RR) range: 2.01-2.62; HR range: 1.72-3.29) (43,44,49,56,62,70). However, five studies reported no association or a decreased association after exposure to hurricanes among specific groups or at different timepoints following a disaster (47,56,58,62,70). Several studies evaluated the mean difference in outcomes including cardiovascular, diabetes and obesity-related outcomes, however, these results were very inconsistent, with some studies noting an increase, some a decrease and other reporting no change in outcomes following a disaster (12,46,52–54,57,59,63,65,73,74).

Mediation and modification of cardiometabolic outcomes

Across all studies, few evaluated effect modification or subgroups of a population that may be at a greater risk of negative health outcomes following disasters. Some studies stratified by sex (28,32,34,39,43,48,55,62), gestational timing of exposure (20,27,29–31,33,36), year of birth or age at outcome (40,62,67), urban or rural area (35), race (43,46) and socioeconomic status (65), however, results varied greatly due to the differences in exposure period, disaster type, cardiometabolic outcome and age at outcome. One study explored the possible mediators between cognitive appraisal following the Quebec Ice Storm and obesity. It was noted that negative cognitive appraisal was found to predict

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obesity via DNA methylation of diabetes-related genes (24). No studies evaluated or discussed possible interventions to mitigate risk of cardiometabolic disease following a disaster.

Critical appraisal

The critical appraisal assessment for all study designs are found in Tables A2-A4. Among the cohort studies, most studies met all criteria included in the checklist indicating high study quality. For instance, almost all cohort studies had comparable populations that were recruited in a similar way and exposures that were assessed in the same way across populations. However, across almost all cohort studies, information on follow-up or strategies to address incomplete follow-up were unclear or not addressed. The critical appraisal results for cross-sectional studies were inconsistent with a small number of studies meeting only some checklist requirements. For quasi-experimental studies, the checklist requirement for within person comparisons were not applicable for all studies, however, all studies clearly defined the cause and effect within the study.

DISCUSSION

Exposure to both natural and human-made disasters, including famine, war, terrorism, natural disasters, and infectious disease epidemics, during both the perinatal/childhood and adult periods were associated with increased cardiometabolic outcomes including obesity, hypertension, myocardial infarction, diabetes, and cardiac mortality in most studies. These findings emphasize that the burden of disasters extends beyond the known direct harm they cause, and attention is needed on the detrimental indirect long-term effects on cardiometabolic health and chronic disease. Meta-analysis was not possible due to the very heterogeneous study designs, measures of effect, outcomes, and timing of exposures. The included studies also reported varied follow-up periods and length of retrospective assessment. From an epidemiological standpoint, the difference in the follow-up periods, timing of exposure, year of disaster and geographic location may help to explain the variation in the reported results. It is important for future studies to explore these differences and how they may influence cardiometabolic outcomes.

Studies also reported insufficient data on subgroups that were at increased risk of worse cardiometabolic health outcomes and interventions that were implemented to mitigate risk of cardiometabolic outcomes. There were also few results that investigated sex and gender differences, or

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that applied an equity lens. It has been noted that those of different levels of socioeconomic status experience differential cardiometabolic outcomes (84,85). This signifies the importance of exploring associations between exposure to disasters and cardiometabolic outcomes stratified by these factors. Understanding how these associations differ will also help to identify groups of people who will experience worse outcomes following a disaster.

The potential mechanisms discussed whereby exposure to disaster could increase cardiometabolic outcomes include maternal objective hardship, subjective stress and various coping mechanisms, psychological strains including PTSD, anxiety and fear, and reduced health care service access. One study that explored mediators in the association between stress and obesity measures identified the role of DNA methylation in this association (24). The findings from this study are important to better understand a possible pathway of obesity development following exposure to a disaster. More studies are needed that evaluate other mechanisms, such as disruptions to healthcare or medication access, and changes in dietary intake or physical activity. Understanding the mechanistic pathway following exposure to disaster will allow for targeted public health strategies, contributing to a mitigation of risk.

This review comprehensively evaluated of the impact of a wide range of disaster exposures on various cardiometabolic outcomes across the life-course. The search strategy was developed in consultation with health science librarians at McMaster University to ensure the most comprehensive search was developed and relevant literature was identified. Very few studies have evaluated the long-term impacts of pandemics and epidemics on cardiometabolic outcomes, identifying a current gap in the literature. The timely findings of this synthesis are a strength of this review, given the current COVID-19 pandemic, which is affecting millions of people worldwide. While only a small proportion of the identified studies focused on pandemics and epidemics, the findings may serve to guide our understanding of expected outcomes and to develop future research to study the effects of COVID-19 on cardiometabolic outcomes. The results from this study can also be used to better understand the trade-off between the implementation of public health measures, such as physical distancing to reduce transmission of a virus, and the implications with access to healthcare, as the review found access to healthcare was limited following a disaster (12). The findings from this review can help to inform future decisions by policy makers regarding the implementation of various public health measures.

The overall findings of this review indicate exposure to any type of disaster may increase the risk of cardiometabolic outcomes including obesity, diabetes and CVD. Further, our results suggest that increased risk is observed for disaster exposure at any period over the life-course from the perinatal

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child and adult periods. However, more studies are needed to understand the mechanisms which may explain these associations and to identify subgroups of the population at highest risk of long-term cardiometabolic outcomes post-disaster. Findings from this review may inform research that should be conducted in the context of the current COVID-19 pandemic and may inform public health mitigation measures that should be considered post-pandemic to prevent long term cardiometabolic outcomes in the population.

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Table 1. Search strategy for EMBASE

1	social isolation.mp. or social isolation/	24963
2	quarantine.mp. or quarantine/	4752
3	*epidemic/	32686
4	*pandemic/	4387
5	disease outbreak.mp.	2321
6	disaster/	13321
7	*natural disaster/	968
8	humanitarian crisis.mp.	257
9	mass casuality.mp. or mass disaster/	3654
10	coronavirus.mp. or coronaviridae/	23106
11	cardiovascular disease.mp. or *cardiovascular disease/	357319
12	*diabetes mellitus/	210248
13	*cerebrovascular accident/	78444
14	*heart infarction/	99072
15	*angina pectoris/	22631
16	*obesity/	178134
17	public health emergency.mp.	1752
18	*body mass/	31459
19	*hypertension/	198593
20	1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 or 9 or 10 or 17	109105
21	11 or 12 or 13 or 14 or 15 or 16 or 18 or 19	1087681
22	20 and 21	2047
23	limit 22 to human	1832

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Table 2. Characteristics of included studies (n=58)

Characteristics	N (%)
Continent	
North America	36 (62%)
Europe	13 (22%)
Asia	7 (12%)
Africa	2 (3%)
Year of Publication	
2010-2020	44 (76%)
2000-2009	12 (21%)
1996-1999	2 (3%)
Study design	
Cohort/longitudinal	41 (71%)
Quasi-experimental[1]	10 (17%)
Cross-sectional	7 (12%)
Sample size	
≥10,000	19 (33%)
1,000- <10,000	10 (17%)
≤1,000	24 (41%)
Not specified	5 (9%)
Exposure life stage	
Pregnancy/Childhood	24 (41%)
Adulthood	34 (57%)
Outcome life stage	
Pregnancy	2 (3%)
Childhood [2]	7 (12%)
Adult	49 (84%)
Disaster	
Human-made	23 (40%)
Natural	35 (60%)
Cardiometabolic outcome [3]	
Cardiovascular disease [4]	41 (71%)
Diabetes [5]	11 (19%)
Obesity or BMI	12 (21%)
Mortality from cardiovascular disease	9 (16%)
Cardiometabolic risk during pregnancy [6]	2 (3%)

1. Includes pre/post study design, time series and natural experiments
2. Children defined as ≤18 years of age
3. Does not equal to 100% as studies report multiple cardiometabolic outcomes
4. Includes hypertension, coronary artery disease/heart disease, angina, heart attack/myocardial infarction, metabolic syndrome, cardiac disease related blood markers, stroke
5. Diabetes, blood glucose, metabolic syndrome
6. Gestational diabetes, gestational hypertension, pre-eclampsia

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Table 3. Characteristics of included studies investigating the association between exposure to a disaster during the perinatal and childhood periods and cardiometabolic outcomes across the life-course, by disaster type (n=24)

Study	Study design	Country	Name of disaster	Year	Sample size	Primary exposure and comparator	Average follow-up	Outcomes	Primary results
Human-made disaster									
Bercovich, 2014 (38)	Cross-sectional	Israel	Holocaust	1941-1945	300	European Jews born in 1940-1945 with exposure to the holocaust compared to European Jews during the same time period born	N/A	Adult outcomes: 1) Hypertension 2) Diabetes 3) Dyslipidemia 4) Any cardiovascular disease	1) Adjusted OR: 2.2, 95%CI: 1.2-3.8 2) Adjusted OR: 2.2, 95%CI: 1.2-4.2 3) Adjusted OR: 3.1, 95%CI: 1.7-5.7 4) Adjusted OR: 2.6, 95%CI: 1.4-4.7
de Rooij, 2007 (28)	Cohort	Netherlands	Dutch Famine	1944-1945	783	Prenatal exposure to Dutch famine defined as people born between January 7, 1945 and December 8 1945 compared to people born before Jan 7 1945 or conceived after Dec 8 1945	58 years	Metabolic syndrome at age 58	Metabolic syndrome OR: 1.2; 95% CI: 0.9, 1.7
Ekamper, 2015 (29)	Cohort	Netherlands	Dutch Famine	1944-1945	41096	Male military conscripts born between Jan 1944 and 1946 and compared to military conscripts born before 1944 or after 1946	63 years	Adult outcomes: 1) Heart disease mortality 2) Cerebrovascular disease mortality 3) Diabetes mellitus mortality	Hazard ratio (HR): 1) HR: 0.94; 95% CI: 0.77, 1.15 2) HR: 1.55, 95% CI: 0.95, 2.51 3) HR: 1.61, 95% CI: 0.91, 2.86
Huang, 2010 (35)	Cohort	China	1959-1961 Chinese Famine	1959-1961	35,025	County level famine intensity for women born during 1957-1962 compared to women born post-famine in 1963	32 years	Adult outcomes at age 32: 1) BMI among rural sample 2) BMI among urban sample 3) Hypertension among rural sample 4) Hypertension among urban sample	1) Average effect = 0.92, 95% CI: 0.32, 1.51 2) Average effect = 0.03, 95% CI: -2.82, 2.87 3) Log odds = 1.23, 95% CI: -0.38, 2.84 4) Log odds = 0.37 95% CI: -2.07, 2.80
Hult, 2010 (34)	Cohort	Nigeria	Biafran Famine	1967-1970	1,339	Individuals exposed to famine during early childhood (born 1965-1967) or exposed to famine in fetal life and infancy (born 1968-Jan 1970) compared to people born between 1971 and 1973	~40 years	Adult outcomes at age ~40 years: 1) Hypertension 2) Diabetes 3) Overweight (BMI>25 kg/m^2) 4) Obesity (BMI>30kg/m^2)	Adjusted OR (95% CI) 1) Childhood exposure OR: 1.42, 95% CI: 0.63, 3.13; Fetal-infant exposure: OR: 2.50, 95% CI: 1.19, 5.26 2) Childhood exposure OR: 1.81, 95% CI: 0.64, 5.15; Fetal-infant exposure OR: 2.56, 95% CI: 0.92, 7.17 3) Childhood exposure OR: 1.02, 95% CI: 0.77, 1.34; Fetal-infant exposure: OR: 1.41, 95% CI: 1.03, 1.93 4) Childhood exposure OR: 1.20, 95% CI: 0.87, 1.67; Fetal-infant exposure: OR: 1.30, 95% CI: 0.92, 1.85
Lumey , 2012[1] (30)	Cohort	Netherlands	Dutch Famine	1944-1945	1075	Infants whose mothers were exposed to famine during or immediately preceding pregnancy (born Feb 1, 1945, March 31, 1946) compared to individuals born in the same hospital before or after famine	~56-62 years	Adult outcomes at 56-62 years Coronary artery disease	Early gestation HR: 1.26, 95%CI: 0.59, 2.70 Late gestation HR: 1.31, 95%CI: 0.67, 2.57

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Painter, 2006 (31)	Cohort	Netherlands	The Dutch Famine	1944-1945	975	Infants who were born between January 1945 and December 1945 who were exposed to famine in utero compared to infants born before the famine (November 1943 and January 1945) and after the famine (December 1945 and February 1947)	~50-58 years	Adult outcomes at 50-58 years: Coronary artery disease	HR: 1.9, 95% CI: 1.0, 3.8
Ravelli, 1999[1] (32)	Cohort	Netherlands	The Dutch Famine	1944-1945	741	Infants exposed to famine during different periods of gestation (late, mid and early) whose maternal daily ration was <1000 kcal (born between January 1945 and December 1945)	50 years	Obesity adult outcomes at 50 years stratified by sex: 1) Weight 2) BMI 3) Waist circumference 4) Waist to hip ratio	Mean (95% CI) for late and early gestation: Men: 1) late: 0.8 (-3.1, 4.7); early: 1.5 (-3.5, 6.6) 2) late: 0.4 (-3.5, 4.5); early: 0.5 (-4.6, 6.0) 3) late: 1.8 (-1.4, 4.9); early: (1.8 (-2.4, 6.0) 4) late: 1.3 (-0.5, 3.1); early: 1.5 (-0.9, 3.9) Women: 1) Late: -1.8 (-6.1, 2.5); early: 7.9 (2.5, 13.2) 2) late: -2.1 (-7.0, 3.1); early 7.4 (0.7, 14.5) 3) late: -0.7 (-4.4, 3.0); early: 5.7 (1.1, 10.3) 4) late: 0.8 (-1.2, 2,8); early: 2.2 (-0.3, 4.7)
Roseboom, 2001[1] (27)	Cohort	Netherlands	The Dutch Famine	1944-1945	2414	Infants who were exposed to famine in utero whose mother had a daily ration <1000 calories during any 13-week period of gestion compared to infants who were born before or conceived after the famine period (before November 1943 or after February 1947)	~28 years	Adult outcomes at 28 years; 1) Plasma glucose (mmol/l) 2) Plasma insulin (pmol/l) 3) Total cholesterol (mmol/l) 4) HDL (mmmol/l) 5) LDL (mmol/l) 6) LDL/HDL cholesterol 7) BMI (kg/m) 8) CHD 9) Systolic BP (mmHg) 10) Diastolic BP (mmHg)	Mean values of outcomes for late gestation and early gestation: 1) Late: 6.3; early: 6.1 2) Late: 200;; early: 207 3) Late: 5.83; early: 6.13 4) Late: 5.83; early: 6.13 5) Late: 1.32; early: 1.26* 6) Late: 3.87; early: 3.26* 7) Late: 26.7; early: 28.1 8) Late: 2.5; early: 8.8* 9) Late: 127.4; early: 123.4 10) Late: 86.4; early: 84.8 * p<0.05
Schrier, 2011 (39)	Cohort	Finland	Winter War and Continuation War	1939-1940 (Winter War), 1941-1944 (Continuation War)	13,039	Individuals in utero who were exposed to bombings that occurred for 48 days between 1934-1944 compared to those who were not exposed in utero	~60-70	Adult outcomes: 1) Coronary heart disease 2) Cerebrovascular disease	Results are shown graphically Higher CHD survival rates among women 64+ and among men aged 50-54 exposed while in utero
Stein, 2006 (33)	Cohort	Netherlands	The Dutch Famine	1944-1945	971	Prenatal exposure to famine defined as the weeks post-last menstrual period that mother was exposed to an official ration of <900 kcal/week (gestation weeks: 1-10, 11-20, 21-30, or 31-delivery)	59	Adult outcomes: 1) systolic BP (mmHg) 2) Diastolic BP (mmHg) 3) Hypertension	Adjusted regression coefficients 1) 1-10 weeks: 1.20 (95% CI: -3.28, 5.69); 11-20 weeks: -1.19 (95% CI: -4.92, 2,55); 21-30 weeks: 1.33 (95% CI: -2.24, 4.90); 31-delivery: 2.02 (95% CI: -1.53, 5.57) 2) 1-10 weeks: 1.10 (95% CI: -1.36,

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									3.57); 11-20 weeks: -1.26 (95% CI: -3.32, 0.80); 21-30 weeks: 1.19 (95% CI: -0.78, 3.15); 31-delivery: 0.71 (95% CI: -1.24, 2.66) 3) 1-10 weeks: 1.14 (95% CI: 0.62, 2.11); 11-20 weeks: 0.98 (95% CI: 0.59, 1.65); 21-30 weeks: 1.23 (95% CI: 0.74, 1.05); 31-delivery: 1.42 (95% CI: 0.86, 2.35)
Trasande, 2018 (45)	Prospective cohort study	USA	World Trade Center attacks (9/11)	2001	402	New York children/teenagers enrolled in the World Trade Center Health Registry (birthdates: Sept 11, 1993 to September 10, 2001) compared to individuals born during the same time period who were ineligible for enrollment in the WTCHR	2 years	Youth and young adult outcomes: 1) BMI 2) zBMI 3) Triglycerides 4) Cholesterol 5) LDL 6) HDL	Regression coefficient and 95% CI: 1) BMI (kg/m2) (95% CI): -1.12 (-2.11, -0.12) 2) zBMI: -0.24 (95% CI: -0.49. -0.002) 3) logTrig: 0.02 (95% CI: -0.07, 0.12) 4) logChol (0.02 (95% CI: -0.02, 0.06) 5) log LDL: 0.06 (95% CI: -0.001, 0.12) 6) logHDL: -0.04 (95% CI: -0.10, 0.03)
Natural disaster									
Cao-Lei, 2016 (24)	Cohort	Canada	Quebec Ice Storm	1998	31	Negative cognitive appraisal of the impact of the ice storm among pregnant women compared to neutral or positive appraisal	13 years	Outcomes among children at age 13: 1) Central adiposity (waist to height ratio) 2) BMI	Mean (standard deviation) 1) Exposed: 20.86 (3.73); unexposed: 22.84 (5.19) 2) Exposed: 0.43 (0.04); unexposed: 0.45 (0.06)
Dancause, 2012 (21)	Cohort	Canada	Quebec Ice Storm	1998	111	Higher objective PNMS scores compared to lower scores among women who were pregnant or conceived within 3 months of the storm	5.5 years	Childhood obesity at 5.5 years of age	OR: 1.37, 95% CI: 1.06-1.77
Dancause, 2013 (22)	Cohort	Canada	Quebec Ice Storm	1998	32	Higher objective hardship compared to lower hardship scores reported among pregnant women exposed to the storm	13.4 years	Childhood insulin secretion at 13 years of age	Insulin secretion: Adjusted linear regression standardized coefficient = 0.52, p<0.01
Dancause, 2015 (20)	Cohort	USA	Iowa Flood	2008	106	Higher reported measures of objective hardship and subjective distress compared to lower scores among pregnant women during the floods	2.5-4 years	Childhood outcomes: 1) Child BMI z-scores at age 2.5 2) child BMI z-scores at age 4 3) Difference in BMI from age 2.5 to 4 4) Child adiposity (skinfolds) at age 2.5 5) Child adiposity (skinfolds) at age 4 6) Difference in adiposity from age 2.5 to 4	1) -0.07 (p=0.56) 2) -0.22 (p=0.07) 3) 0.11 (p=0.41) 4) 0.00 (p=0.97) 5) -0.06 (p=0.72) 6) 0.03 (p=0.82)
Goudet, 2011 (13)	Cohort	Bangladesh	1998 Bangladesh Flood	1998	220	Maternal malnutrition among mothers of infants and young children following flood exposure defined as underweight (BMI<18.5 kg/m2) compared to normal (BMI>=18.5)	12 months	Child outcomes at 12-36 months of age: 1) Underweight (weight for age z-score <-2) 2) Stunted (height for age z-score < -2) 3) Wasted (weight for height z-score < -2)	1) Adjusted OR = 3.509, 95%CI: 1.022,12.048) 2) Adjusted OR: 4.447, 95%CI: 1.044,18.943 3) Adjusted OR: 2.097, 95%CI: 0.507, 8.671
Kroska, 2018 (19)	Longitudinal study	USA	Iowa Flood	2008	103	Levels of maternal stress among those exposed to Iowa floods	2.5 years	Children outcomes at 2.5 years: BMI	Standardized coefficient: 0.2071 (p=0.0322)

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Liu, 2016 (23)	Longitudinal study	Canada	Quebec Ice Storm	1998	52-111 at different time points	Levels of maternal stress (objective hardship and subjective stress) among those exposed to Iowa floods	5.5-15.5 years	Children outcomes at 5.5-15.5 years: 1) BMI 2) Waist to height ratio	Estimate (95% CI) 1) Objective hardship: -0.059 (95% CI: -0.175, 0.058) 2) Objective hardship: 0.010 (95% CI: -0.002, 0.021) Subjective stress: 0.044 (95% CI: -0.010, 0.097)
Mazumder, 2010[2] (36)	Cohort	USA	1918 Influenza Pandemic	1918-1919	101,068	Infants who were born during the Influenza pandemic (third and fourth quarter of 1918, and the first, second and third quarter of 1919) compared to those born in the last quarter of 1919	~ 60-82 years	Adult outcomes at 60-82 years: 1) Diabetes 2) Heart disease	Excess cases of diabetes/heartburn: 1) 1918 Q4: 7.7% excess (95% CI: -10.6, 25.9); 1919 Q1: -5.2 (95% CI: -22.9, 12.5); 1919 Q2: 36.7% excess (95% CI: 18.9, 54.4); 2) 1918 Q4: 4.6% excess (95% CI: -4.3, 13.5); 1919 Q1: 10.9% excess (95% CI: 2.3, 19.6); 1919 Q2: 6.4 excess (95% CI: -2.2, 15.1);
Mryskyla, 2013[2] (37)	Cohort	USA	1918 Influenza Pandemic	1918-1919	81,571	Infants who were born during the Influenza pandemic (born during different quarters of 1917, 1918 and 1919) compared to those born in 1920-1924	~ 63-95 years	Adult outcomes at 63-95 years: Cardiovascular mortality	1918 Q1: HR: 1.05 (95% Ci: 0.94, 1.17); 1918 Q2: HR: 1.02 (95% Ci: 0.91, 1.14); 1918 Q3 HR: 0.99 (95% CI: 0.89, 1.10); 1918 Q4 HR: 0.97 (95% Ci: 0.87, 1.09); 1919 Q1 HR 1.07 (95% CI: 0.96, 1.19); 1919 Q2: HR:1.06 (95% CI: 0.95, 1.19);
Oni, 2015 (25)	Cross-sectional	USA	Hurricane Katrina	2005	146	Women who were pregnant during Hurricane Katrina or became pregnant immediately after hurricane compared to those who were not exposed to the hurricane; Women who experienced prenatal stress caused by Hurricane Katrina compared to those who did not experience stress	9 months	Childhood outcomes at 9 months: 1) Pregnancy induced hypertension 2) Gestational diabetes	1) Hurricane exposure: adjusted OR: 1.22 (95% CI: 0.81, 1.84); perceived stress: adjusted OR: 1.16 (95% CI: 1.05, 1.30); 2) Hurricane exposure: adjusted OR: 1.04 (95% CI: 0.69,1.57); perceived stress: adjusted OR: 1.13 (95% CI: 1.02, 1.25)
Sotomayer, 2013 (40)	Cohort Natural experiment	Puerto Rico	Hurricane San Felipe and San Cipiran	1928 and 1932	11990	Those born during 1929 and 1933 were defined as exposed to the hurricanes compared to individuals born outside of these years between 1920 and 1940	Not reported (average age = 70)	Outcomes at ~70 years of age: 1) Diabetes 2) Hypertension 3) High cholesterol 4) CVD 5) AMI 6) Coronary/angina 7) Stroke	Parametric estimates (SE) 1) San Felipe: 6.77 (0.98); San Ciprian: 4.22 (0.98) 2) San Felipe: 6.08 (1.07); San Ciprian: 5.63 (1.07) 3) San Felipe: 10.06 (1.21); San Ciprian: 4.67 (1.21) 4) San Felipe: -1.84 (2.02); San Ciprian: 2.23 (2.02) 5) San Felipe: 1.08 (1.25); San Ciprian: 2.95 (1.25) 6) San Felipe: 0.13 (2.45); San Ciprian: 0.60 (2.45) 7) San Felipe: 0 (0.83); San Ciprian: 0.96 (0.83)
Xiao, 2019 (26)	Time series/Quasi-experimental	USA	Hurricane Sandy	2012	Not reported	Exposure to Hurricane Sandy lasting impacts defined as the following 12 months after Sandy (November 2012-October 2013) compared to the November to October in other years during November 2005 to October 2014 among women who were pregnant	12 months	Outcomes in adults: 1) Emergency department visits for gestational hypertension 2) Emergency department visits for diabetes or abnormal glucose	1) Increased at 7 months: 7.3% (95% CI: 1.0%, 13.9%) 2) Increased at 8 months:26.3% (95% CI: 3.9%, 53.6%) *results for 12 month reported graphically

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1. Only presenting results for early and late gestation; results for mid gestation are not included in summary table but can be found in studies
2. Not all results presented for different exposure groups

Abbreviations: AMI: Acute myocardial infarction; BMI: Body mass index; BP: Blood pressure; CI: Confidence Interval; CHD: Coronary heart disease; Chol: Cholesterol; CVD: Cardiovascular disease; HDL: High-density lipoproteins; HR: Hazard ratio; LDL: Low-density lipoproteins OR: Odds Ratio; PMNS: Prenatal Maternal Stress; Q1-Q4: quarter; SE: Standard error; Trig; Triglycerides.

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Table 4. Description of studies investigating the association between exposure to a disaster during adulthood and cardiometabolic outcomes across the life-course, by disaster type (n=34)

Study	Study design	Country	Name of disaster	Year	Sample size	Primary exposure and comparator	Average follow-up	Outcomes	Primary results
Human-made disaster									
Brackbill, 2006 (68)	Cohort	USA	World Trade Center attacks (9/11)	2001	8,418	Adult survivors of 9/11 present at time of first airplane impact in a structure that was damaged compared to those that collapsed; time of evacuation before compared to after damage	1 year	1) Hypertension 2) Coronary heart disease 3) Angina 4) Heart attack 5) Diabetes 6) Stroke	1) Building type: adjusted OR: 1.2 (p<0.05); time of evacuation: adjusted OR: 0.9 (0.6, 1.3) 2) Building type: adjusted OR: 0.8 (0.4, 1.6); time of evacuation: adjusted OR: 0.5 (0.1, 2.2) 3) Building type: adjusted OR: 0.8 (0.4, 1.6); time of evacuation: adjusted OR: 0.7 (0.2, 3.1) 4) Building type: adjusted OR: 2.1 (0.9, 4.9); 5) Time of evacuation: adjusted OR0.7 (0.3, 1.7) 6) Building type: adjusted OR: 1.5 (0.6, 4.0)
Dirkzwager, 2007 (51)	Cohort	Netherlands	Fireworks depot explosion	2000	896	PTSD among those exposed to the fireworks disaster 19 months following the disaster compared to those with no PTSD exposed to the fireworks explosion	18 months	1) Cardiovascular 2) Vascular problems	1) Physical health problems OR: 1.23; 95% CI: 0.78, 1.94; New health problems (not present pre disaster): 1.11; 0.65, 1.89 2) Physical health problems OR: 2.12 95% CI: 1.23, 3.68; 1.92;New health problems (not present pre disaster) OR: 1.92, 95% CI: 1.04, 3.55.
Dorn, 2007 (41)	Cohort	Netherlands	Volendam Pub Fire	2001	2255	Parents of children with burns from fire parents of children without burns, bereaved parents compared to community controls who were not trapped in fire	4 years	Incidence of hypertension	Bereaved parents: OR: 2.42, 95%CI: 0.90, 6.55); parents of victims with burns: OR: 1.43, 95%CI: 0.97, 2.11; parents of victims without burns: OR:1.44, 95%CI: 0.92, 2.26)
Gerin, 2005 (69)	Pre/post design/quasi experimental	USA	World Trade Center attacks (9/11)	2001	528	Adults 2 months before 9/11 compared to 2 months after 9/11 across 4 cities (Chicago, Washington DC, New York and Mississippi)	4 months	Systolic blood pressure	Difference (SE) New York: 1.58 (0.82) p<0.05 Chicago: 2.15 (0.32) p<0.001 Mississippi: 2.92 (0.67) p<0.001 Washington DC: 8.67 (1.16) p<0.001
Huizink, 2006 (60)	Cohort	Netherlands	Amsterdam Air Disaster	1992	1996	Police officers and firefighters who performed at least one disaster-related task compared to professional colleagues who did not perform any disaster-related tasks	8.5 years	Cardiovascular complaints	Adulthood outcomes Police officers: OR: 1.76 (95% CI: 1.35, 2.29) Firefighters: OR: 3.3 (95% CI: 1.70, 6.41)
Jordan, 2011A (55)	Prospective cohort	USA	World Trade Center attacks (9/11)	2001	39,324	9/11-related PTSD compared to no PTSD	2.9 years	Heart disease	Women adjusted OR: 1.68 (95% CI: 1.33, 2.12) Men adjusted OR1.62 (95% CI: 1.34, 1.96)
Jordan, 2011B (61)	Prospective cohort study	USA	World Trade Center attacks (9/11)	2001	39324	Low, intermediate and high exposure to 9/11	2.9 years	Heart disease mortality	Intermediate exposure: HR: 1.21 (95% CI: 80, 1.83) High exposure: HR: 2.06 (95% CI: 1.10, 3.86).
Jordan, 2013[1] (47)	Cohort	USA	World Trade Center attacks (9/11)	2001	46,346	Low, intermediate and high exposure to 9/11	7 years	CVD hospitalizations	Rescue/recovery workers: women: high: adjusted HR: 3.29 (95% CI: 0.85, 12.69); men: high: 1.82 (95% CI: 1.06, 3.13) Non-rescue/recovery workers: women: high: adjusted HR: 0.88 (95% CI: 0.54, 1.43); men: high: adjusted HR: 0.94 (95% CI: 0.60, 1.47)

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Kong, 2019 (48)	Pre/post design/quasi experimental	South Korea	Sewol Ferry Disaster	2014	73,632	Exposure to the Sewol Ferry Disaster in one-week periods from May 21 through June 17, 2014 compared to the reference period (March 2015-April 2015)	8 weeks	Adulthood outcomes 1) Acute MI 2) Angina	1) 8 weeks after Sewol: IRR: 0.91 (95% CI: 0.81, 1.02) 2) 8 weeks after Seowl: IRR: 0.93 (95% CI: 0.85, 1.01)
Lin, 2010 (66)	Pre/post design/quasi experimental	USA	World Trade Center attacks (9/11)	2001	Not reported	Areas affected by 9/11 compared to areas not affected by 9/11	10 years	Adulthood outcomes for Cardiovascular disease hospitalizations	Prevalence ratio (95% CI): 08/14–09/10: 0.51 (95% CI: 0.26, 1.00) 09/11–09/17: 0.56 (95% CI: 0.28-1.11); 09/18-09/24: 0.77 (95% CI: 0.44, 1.32); 09/25-10/01: 0.49 (95% CI: 0.24, 1.00); 10/02-10/08: 0.98 (95% CI: 0.53, 1.87); 10/09-10/15: 1.09 (95% CI: 0.60, 1.98); 10/16–10/22: 0.50 (95% CI: 0.26-0.95); 10/23–10/29: : 0.45 (95% CI: 0.20-0.98); 10/30–11/05: 0.48 (95% CI: 0.23-0.97)
Yu, 2018 (67)	Cohort Study	USA	World Trade Center attacks (9/11)	2001	42,527	9/11-related PTSD compared to no PTSD	13 years	Stroke	Adjusted HR: 1.69 (95% CI: 1.42, 2.02)
Natural disaster									
An, 2015 (65)	Cross-sectional	USA	Hurricane Ike	2008	19	Psychological strains among Hurricane Ike survivors	3 months	1) Blood glucose 2) Obesity	Mean (high vs low) and standard deviation: PTSD symptom:22.44 (4.93) vs.12.86 (10.48); p=0.014; perceived stress:23.00 (5.03) vs. 28.11 (5.07) p=0.048 2) 28.43 kg/m^2 (3.92) vs. 20.83kg/m^2 (3.92) p=0.018
Baum, 2019 (12)	Cohort study	USA	Hurricane Sandy	2012	81 544	Veterans who used Manhattan VA Medical Center before Hurricane Sandy and experienced decreased access to health care services compared to veterans who used the VA Bronx, Brooklyn or West Haven medical centers	2 years	1) Uncontrolled hypertension 2) Systolic BP (mmHg) 3) Diastolic BP (mmHg) 4) Uncontrolled diabetes 5) Uncontrolled cholesterol 6) Weight	% differential change (95% CI): 1) 6 months: 19.3 (4.5, 8.7); 12 months: 4.5 (3.1, 5.9); 18 months: 5.0 (3.5, 6.5); 24 months: 2.1 (0.5, 3.6) 2) 6 months: 3.8 (3.1, 4.5); 12 months: 2.3 (1.7, 2.9); 18 months: 3.1 (2.5, 3.7); 1.5 (0.9, 2.1) 3) 6 months: 2.7 (2.3, 3.1); 12 months: 2.2 (1.9, 2.6); 18 months 2.9 (2.5, 3.3), 24 months: 2.0 (1.7, 2.4) 4) 6 months: 1.9 (-0.1, 4.0); 12 months: 1.7 (-0.3, 3.6); 18 months: 0.8 (-1.2, 2.8); 24 months: -0.2 (-2.2, 1.8) 5) 6 months: 1.3 (-0.1, 2.6); 12 months: 0.6 (-0.6, 1.8); 18 months: -0.7 (-2.0, 0.6); 24 months: -0.2 (-1.4, 1.0) 6) 6 months: -0.1 (-0.5, 0.2); 12 months: 0.2 (-0.2, 0.5); 18 months: -0.2 (-0.5, 0.2); 24 months: 0.5 (0.1, 0.9)
Becquart, 2018 (46)	Time series/quasi experimental	USA	Hurricane Katrina	2005	383 552	Exposure to hurricane before, during and after among older adults in Louisiana in the affected counties	1 year	Adult outcomes Hospitalizations due to CVD	Mean (SD) Orleans: T1: 7.25 (2.44); T2: 3.91 (1.45)*; T3; 18.47 (17.3)*; T4:13.76 (6.51)*; T5: 9.54 (2.78); T6: 4.69 (2.08) Jefferson: T1: 5.90 (1.90); T2: 5.01(1.52); T3: 8.118 (3.70)*; T4: 7.25 (2.15)*; T5: 5.26 (1.53); T6: 4.65 (1.57)* East BR: T1: 8.69 (2.74); T2: 9.11 (2.69); T3: 6.52 (2.58); T4: 6.55 (1.70)*; T5: 6.69 (2.42)*; T6: 7.39

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									(2.37)* * p<0.05
Bich, 2011 (59)	Cross-sectional	Vietnam	Historic flood in 2008	2008	781	Individuals who resided in households affected by flood in Hanoi in 2005 compared to non-affected households	1 month	Worsening hypertension after rain/flood	Rural: non flooded 33.3%; flooded: 51.2%; Urban: non flooded 20.3% flooded: 42.9%* * p<0.05
Fonseca, 2009 (57)	Cohort	USA	Hurricane Katrina	2005	1795	Adults with diabetes who were in the databases from 3 health care systems 6 months before the hurricane (Feb 28, 2005-Aug 27, 2005) compared to 6-16 months after the hurricane (March 1, 2006-December 31, 2006)	22 months	1) Glycemic control/A1C 2) Systolic BP (mmHg?) 3) Diastolic BP (mmHg) 4) HDL 5) LDL 6) Triglycerides	Difference in mean (SD) 1) 0.1 (1.6) (p<0.01) 2) 10.5 (20.4) (p<0.01) 3) 3.9 (13.1) (p<0.01) 4) 6.0 (35.5) (p<0.01) 5) -2.4 (9.2) (p<0.01) 6.) -2.1 (137.5) (p=0.60)
Gautam, 2009 (53)	retrospective cohort	USA	Hurricane Katrina	2005	396	Exposure to Hurricane Katrina compared to period before hurricane	4 years	Incidence of AMI admission	Pre-Katrina group: 150 admissions for AMI (0.71%) Post-Katrina group: 246 admission for AMI (2.18%) p<0.0001
Hendrickson, 2996 (62)	Pre/post design/quasi experimental	USA	Hurricane Iniki	1992	Not reported	Mortality data for residents of Kauai for 5-year period 1987-1991 prior disaster compared to the year immediately following the hurricane (Oct 1 1992-Sept 30, 1993)	6 years total	Mortality by: 1) Heart disease 2) Stroke 3) Diabetes mellitus	1) RR: 0.96 (95% CI: 0.79-1.17) 2) RR: 1.20 (95% CI: 0.81-1.78) 3) RR: 2.61 (95% CI: 1.44-4.74)
Husarewycz, 2014 (70)	Cross-sectional	USA	Natural disaster/terrorism	Lifetime disaster experience	34,653	Number of times directly experienced natural disaster/terrorism compared to no experiences	1 year	1) Cardiovascular disease 2) Hypertension/arteriosclerosis 3) Diabetes 4) Obesity	1) OR: 1.28 (95% CI: 1.10, 1.49) 2) OR: 1.08 (95% CI: 0.95, 1.24) 3) OR: 1.10 (95% CI: 0.94, 1,29) 4) OR: 1.01 (95% CI: 0.90, 1.14)
Jiao, 2012 (52)	retrospective cohort observational study	USA	Hurricane Katrina	2005	Not reported	2 years prior to the hurricane (August 29, 2003 - August 28, 2005) compared to the 3-year period post-Hurricane Katrina (February 14, 2006 - February 13, 2009)	5 years	Incidence of AMI	Pre-Katrina: 0.7% compared to post-Katrina: 2% (p<0.001)
Joseph, 2014 (49)	Cohort/longitudinal	USA	Hurricane Katrina	2005	215	African Americans who experienced acute unemployment due to Hurricane compared to those who remained employed	4 years	Cardiometabolic event	Adjusted OR = 5.65, p < .05
Karatzias , 2015 (42)	Cross-sectional	Hong Kong	Natural disaster	Not specified	1147	Experience of natural disaster across life-course compared to less or no experiences	Survey done from August to December 2012	1) Hypertension 2) Heart disease 3) Diabetes	Chi square (p-value) 1) X2: 3.3 p=0.047 2) X2: 3.6 p=0.056 3) X2: 2.5 p=0.088
Kim, 2017 (56)	Pre/post design/quasi experimental	USA	Hurricane Sandy	2012	Not reported	The month of Hurricane Sandy (October 28, 2012-November 27, 2012) compared to the same month in 2009-2011; Sandy quarter (October 28, 2012-January 27, 2013) compared to the same period in 2009-2011 among elderly people	Sandy month: Oct 28, 2012-Nov 27, 2012 Sandy quarter: Oct 28, 2012-Jan 27, 2013	Cardiovascular disease-related death	Sandy quarter: adjusted RR: 1.06; 95% CI: 1.02, 1.10 Sandy month: adjusted RR: 1.10; 95% CI: 1.02, 1.18
Koroma, 2019 (64)	Cross-sectional	Sierra-Leone	Ebola	2014-2015	10011	District facilities for six-month periods before Ebola (June-December 2012), during Ebola (June-December 2014) and post-Ebola (June-December 2015)	June-December 2012,2013,2014	1) Cardiovascular disease 2) Hypertension 3) Diabetes	1) Pre-Ebola: 355, Ebola: 300, Post-Ebola: 196 2) Pre-Ebola: 282, Ebola: 230, Post-Ebola: 457 3) Pre-Ebola: 3716, Ebola: 1851, Post-Ebola: 2463

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Lawrence, 2019 (43)	Prospective Cohort Study	USA	Superstorm Sandy	2012	651858	Residing in counties affected by Superstorm Sandy compared to non-affected counties; Superstorm period compared to reference periods (short-term and long-term (4 and 12 months))	1 year	Emergency department visits, outpatient visits, and hospital admissions for cardiovascular disease	4 months: Superstorm sandy period: RR: 2.10 (95% CI: 2.10, 2,10); Affected counties RR: 2.62 (95% CI: 2.62, 2.63) 12 months: Superstorm sandy period: RR: 2.01 (95% CI: 2.00, 2.01); Affected counties RR: 2.64 (95% CI: 2.64, 2.65)
McKinney, 2011 (50)	Time-series/quasi experimental	USA	Hurricane Charley, Frances, Ivan and Jeanne and Tropical Storm Bonnie	2004	Not reported	Counties in 2004 directly impacted by the hurricanes, ordered evacuated regardless of the level of damage that occurred and adjacent to the impact zone where direct deaths were reported compared to compared to the same areas in 2001-2006	5 years	Heart-related mortality	Results shown graphically Significantly elevated heart-related deaths
Moscona, 2019 (73)	Retrospective cohort study	USA	Hurricane Katrina	2005	2-year pre-Katrina - 21,079 10-year post Katrina - 84,751	Individuals who lived in New Orleans who went to the Tulane University Health Sciences Center compared to the two months prior to the Hurricane	12 years	1) Hospital admission for Incidence of AMI 2) Changes in CAD 3) Changes in diabetes mellitus 4) Changes in hypertension 5) Changes in hyperlipidemia	Pre-Katrina versus Post-Katrina 1) 0.7% vs 2.8% (p<0.001) 2) 36.4% vs. 47.9%, (p= 0.01) 3) 31.3% vs. 39.9% (p= 0.04) 4) 71.1% vs 80.6%, (p=0.12) 5) 45.4% vs. 59.3% (p = 0.005)
Nagayoshi, 2015 (54)	Pre/post design/quasi experimental	Japan	July 12, 2012 heavy rain and mudslides "mountain tsunamis"	2012	583	Individuals who were admitted at Aso Central Hospital from July 12 to August 31, 2012 compared to the 3-year period before flooding	3 years	1) Hospital admission for cardiovascular outcomes 2) Cardiovascular events	1) 4.5/month before compared to 16.8/month after; p < 0.01 2) 5.1/month before compared to 16.8/month after; p < 0.01).
Ng, 2011 (71)	Cohort	United Kingdom	Flood	June, 2007	1,743	Diabetics affected by floods compared to diabetics not affected by floods	2 years	glycemic control/HbA1c levels	Mean HbA1c before 7.6% (7.5–7.7) vs. After 7.9% (7.7–8.0); p = 0.002
Peters, 2013 (63)	Retrospective Cohort	USA	Hurricane Katrina	2005	698	Admission to Tulane University Health Sciences Centre in the 3-year period post-Katrina compared to the 6-year period pre-Katrina	9 years	Chronobiology of AMI onset	Pre-Katrina: 45% vs post-Katrina: 30.9%, p=0.002
Rey, 2007 (72)	Longitudinal	France	6 Heat Waves	1971-2003	Not reported	Time of heat wave compared to the expected mortality during the 3 years prior to the heat wave	N/A	Excess cardiovascular disease death	41% in 1975 to 23% in 2003
Sliva-Palacios, 2015 (58)	Pre/post design/quasi experimental	USA	Oklahoma Tornado	2013	22,607	Victims of the Oklahoma Tornado Outbreaks compared to the same people pre-tornado and same period one year prior	6 Months	Hospital admissions for CVE	One year prior: PR = 1.05 95% CI: 0.91 to 1.21, p = 0.50; 3-month pre-tornado: PR = 0.96, 95% CI: 0.83 to 1.21, p = 0.63
Thethi, 2010 (74)	Cohort	USA	Hurricane Katrina	2005	1523	Individuals exposed to Hurricane Katrina compared to 6-16 months pre-Hurricane Katrina (February 28, 2005-August 27, 2005)	6 months before Katrina and 6-16 months after Katrina and follow up 1 year after the first post-Katrina visit	1) LDL 2) HDL 3) Triglycerides 4) Cholesterol 5) Diastolic blood pressure 6) Systolic blood pressure	Pre-Katrina vs Post-Katrina mean: 1) 101.34 vs 107.44 2) 43.53 vs 41.08 3) 160.8 vs 158.65 4) 181.9vs 181.39 5) 70.99 vs 74.88 6) 130.73 vs 141.27

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Vanasse, 2016 (44)	Population-based retrospective cohort study with a time series design	Canada	Flood of Saint-Jean-sur-Richelieu	2011	111,317	Exposure to flood in spring 2011 and exposure to flooded area (Area 1) compared to same period in Spring 2010 and 2012 and non-flooded areas in the same town (Areas 2, 3 and 4)	4 months	Acute cardiovascular event	Spring 2010: aOR 1.25 (95%CI: 0.81, 1.92); spring 2012 aOR: 1.27 (95% CI: 0.82, 1.92); Non-flooded areas 2: aOR: 1.11 (95%CI: 0.79, 1.59), Non-flooded area 3: aOR: 0.94 (95% CI: 0.68, 1.32); Non-flooded area 4: aOR 1.08 (95%CI: 0.78, 1.47)

1. Only results for extreme outcomes are reported in table, remaining results can be found in the study.

Abbreviations: aOR: Adjusted odds ratio; AMI: Acute myocardial infarction BMI: Body mass index; BP: Blood pressure; CI: Confidence Interval; CAD: Coronary artery disease; CHD: Coronary heart disease; CVD: Cardiovascular disease; CVE: Cardiovascular events; HbA1c ; Hemoglobin A1C; HDL: High-density lipoproteins; HR: Hazard ratio; LDL: Low-density lipoproteins OR: Odds Ratio; PR: Prevalence Ratio; PTSD: Post-traumatic stress disorder; RR: Relative risk; SD: Standard deviation; SE: Standard Error; Trig; Triglycerides.; T1-T6; Time.

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References

1. World Health Organization. WHO | Definitions: emergencies [Internet]. WHO. World Health Organization; 2020 [cited 2020 May 12]. Available from: https://www.who.int/hac/about/definitions/en/

2. Mohamed SI. An overview on disasters. Disaster Prev Manag Int J. 2007 Jan 1;16(5):687–703.

3. Giorgadze T, Maisuradze I, Japaridze A, Utiashvili Z, Abesadze G. Disasters and their consequences for public health. Georgian Med News. 2011 May;(194):59–63.

4. Ordway D-M, February 1 JR, 2016. Global trends in human infectious disease: Rising number of outbreaks, fewer per-capita cases [Internet]. Journalist's Resource. 2016 [cited 2020 May 8]. Available from: https://journalistsresource.org/studies/society/public-health/global-rise-human-infectious-disease-outbreaks/

5. Brooks SK, Webster RK, Smith LE, Woodland L, Wessely S, Greenberg N, et al. The psychological impact of quarantine and how to reduce it: rapid review of the evidence. The Lancet. 2020 Mar 14;395(10227):912–20.

6. Bell SA, Horowitz J, Iwashyna TJ. Health Outcomes After Disaster for Older Adults With Chronic Disease: A Systematic Review. The Gerontologist [Internet]. [cited 2020 May 8]; Available from: https://academic.oup.com/gerontologist/advance-article/doi/10.1093/geront/gnz123/5565119

7. Jackson JL, George S, Hinchey S. Medically Unexplained Physical Symptoms. J Gen Intern Med. 2009 Apr;24(4):540–2.

8. Chan EYY, Sondorp E. Medical interventions following natural disasters: missing out on chronic medical needs. Asia Pac J Public Health. 2007;19 Spec No:45–51.

9. James SL, Abate D, Abate KH, Abay SM, Abbafati C, Abbasi N, et al. Global, regional, and national incidence, prevalence, and years lived with disability for 354 diseases and injuries for 195 countries and territories, 1990–2017: a systematic analysis for the Global Burden of Disease Study 2017. The Lancet. 2018 Nov;392(10159):1789–858.

10. Forouzanfar MH, Afshin A, Alexander LT, Anderson HR, Bhutta ZA, Biryukov S, et al. Global, regional, and national comparative risk assessment of 79 behavioural, environmental and occupational, and metabolic risks or clusters of risks, 1990–2015: a systematic analysis for the Global Burden of Disease Study 2015. The Lancet. 2016 Oct;388(10053):1659–724.

11. Tan SY, Yip A. Hans Selye (1907–1982): Founder of the stress theory. Singapore Med J. 2018 Apr;59(4):170–1.

12. Baum A., Barnett M.L., Wisnivesky J., Schwartz M.D. Association between a Temporary Reduction in Access to Health Care and Long-term Changes in Hypertension Control among Veterans after a Natural Disaster. JAMA Netw Open. 2019;2(11):e1915111.

13. Goudet S, Griffths P, Bogin BA. Mother's body mass index as a predictor of infant's nutritional status in the post-emergency phase of a flood. Disasters. 2011 Oct;35(4):701–19.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

14. Moher D, Liberati A, Tetzlaff J, Altman DG, PRISMA Group. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. Ann Intern Med. 2009 Aug 18;151(4):264–9, W64.

15. Kamel M, Smith BT, Wahi G, Carsley S, Birken CS, Anderson LN. Continuous cardiometabolic risk score definitions in early childhood: a scoping review. Obes Rev. 2018;19(12):1688–99.

14. Bazoukis, G., Tse, G., Naka, K.K., Kalfakakou, V., Vlachos, K., Saplaouras, A., Letsas, K. P., Korantzopoulos, P., Thomopoulos, C., Michelongona, P., Bazoukis, X., González-Salvado, V., Liu, T., Michalis L.K., Baranchuk, A., Itoh, T., Efremidis, M., Tsioufis, C., Stavrakis, C. Impact of major earthquakes on the incidence of acute coronary syndromes - A systematic review of the literature [Internet]. Vol. 59, Hellenic journal of cardiology : HJC = Hellenike kardiologike epitheorese. Hellenic J Cardiol; 2018 [cited 2020 Sep 10]. Available from: https://pubmed.ncbi.nlm.nih.gov/29807192/

17. Covidence - Better systematic review management [Internet]. Covidence. [cited 2020 Sep 10]. Available from: https://www.covidence.org/

18. Aromataris E, Munn Z, editors. JBI Manual for Evidence Synthesis [Internet]. JBI; 2020 [cited 2020 Sep 10]. Available from: https://wiki.jbi.global/display/MANUAL

19. Kroska EB, O'Hara MW, Elgbeili G, Hart KJ, Laplante DP, Dancause KN, et al. The impact of maternal flood-related stress and social support on offspring weight in early childhood. Arch Womens Ment Health. 2018 Apr 1;21(2):225–33.

20. Dancause KN, Laplante DP, Hart KJ, O'Hara MW, Elgbeili G, Brunet A, et al. Prenatal stress due to a natural disaster predicts adiposity in childhood: the Iowa Flood Study. J Obes. 2015;2015(101526295):570541.

21. Dancause KN, Laplante DP, Fraser S, Brunet A, Ciampi A, Schmitz N, et al. Prenatal exposure to a natural disaster increases risk for obesity in 51/2-year-old children. Pediatr Res. 2012;71(1):126–31.

22. Dancause KN, Veru F, Andersen RE, Laplante DP, King S. Prenatal stress due to a natural disaster predicts insulin secretion in adolescence. Early Hum Dev. 2013 Sep;89(9):773–6.

23. Liu GT, Dancause KN, Elgbeili G, Laplante DP, King S. Disaster-related prenatal maternal stress explains increasing amounts of variance in body composition through childhood and adolescence: Project Ice Storm. Environ Res. 2016;150:1–7.

24. Cao-Lei L, Dancause KN, Elgbeili G, Laplante DP, Szyf M, King S. Pregnant women's cognitive appraisal of a natural disaster affects their children's BMI and central adiposity via DNA methylation: Project Ice Storm. Early Hum Dev. 2016 Dec 1;103:189–92.

25. Oni O, Harville E, Xiong X, Buekens P. Relationships among stress coping styles and pregnancy complications among women exposed to Hurricane Katrina. J Obstet Gynecol Neonatal Nurs JOGNN. 2015;44(2):256–67.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

26. Xiao J, Huang M, Zhang W, Rosenblum A, Ma W, Meng X, et al. The immediate and lasting impact of Hurricane Sandy on pregnancy complications in eight affected counties of New York State. Sci Total Environ. 2019;678(uj0, 0330500):755–60.

27. Roseboom TJ, van der Meulen JH, Ravelli AC, Osmond C, Barker DJ, Bleker OP. Effects of prenatal exposure to the Dutch famine on adult disease in later life: an overview. Mol Cell Endocrinol. 2001 Dec 20;185(1–2):93–8.

28. de Rooij SR, Painter RC, Holleman F, Bossuyt PM, Roseboom TJ. The metabolic syndrome in adults prenatally exposed to the Dutch famine. Am J Clin Nutr. 2007 Oct 1;86(4):1219–24.

29. Ekamper P, van Poppel F, Stein AD, Bijwaard GE, Lumey LH. Prenatal Famine Exposure and Adult Mortality From Cancer, Cardiovascular Disease, and Other Causes Through Age 63 Years. Am J Epidemiol. 2015 Feb 15;181(4):271–9.

30. Lumey LH, Martini LH, Myerson M, Stein AD, Prineas RJ. No relation between coronary artery disease or electrocardiographic markers of disease in middle age and prenatal exposure to the Dutch famine of 1944-5. Heart Br Card Soc. 2012 Nov;98(22):1653–9.

31. Painter RC, de Rooij SR, Bossuyt PM, Simmers TA, Osmond C, Barker DJ, et al. Early onset of coronary artery disease after prenatal exposure to the Dutch famine. Am J Clin Nutr. 2006 Aug;84(2):322–7; quiz 466–7.

32. Ravelli AC, van Der Meulen JH, Osmond C, Barker DJ, Bleker OP. Obesity at the age of 50 y in men and women exposed to famine prenatally. Am J Clin Nutr. 1999 Nov;70(5):811–6.

33. Stein AD, Zybert PA, van der Pal-de Bruin K, Lumey LH. Exposure to famine during gestation, size at birth, and blood pressure at age 59 y: evidence from the Dutch Famine. Eur J Epidemiol. 2006;21(10):759–65.

34. Hult M, Tornhammar P, Ueda P, Chima C, Bonamy A-KE, Ozumba B, et al. Hypertension, diabetes and overweight: looming legacies of the Biafran famine. PloS One. 2010 Oct 22;5(10):e13582.

35. Huang C, Li Z, Wang M, Martorell R. Early life exposure to the 1959-1961 Chinese famine has long-term health consequences. J Nutr. 2010 Oct;140(10):1874–8.

36. Mazumder B, Almond D, Park K, Crimmins EM, Finch CE. Lingering prenatal effects of the 1918 influenza pandemic on cardiovascular disease. J Dev Orig Health Dis. 2010 Feb;1(1):26–34.

37. Myrskyla M, Mehta NK, Chang VW. Early life exposure to the 1918 influenza pandemic and old-age mortality by cause of death. Am J Public Health. 2013;103(7):e83-90.

38. Bercovich E, Keinan-Boker L, Shasha SM. Long-Term Health Effects in Adults Born during the Holocaust. 2014;16:5.

39. Schreier NK, Moltchanova EV, Blomstedt PA, Kajantie E, Eriksson JG. Prenatal exposure to wartime stress – long-term effect on coronary heart disease in later life. Ann Med. 2011 Nov 1;43(7):555–61.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

40. Sotomayor O. Fetal and infant origins of diabetes and ill health: evidence from Puerto Rico's 1928 and 1932 hurricanes. Econ Hum Biol. 2013 Jul;11(3):281–93.

41. Dorn T, Yzermans CJ, Guijt H, van der Zee J. Disaster-related stress as a prospective risk factor for hypertension in parents of adolescent fire victims. Am J Epidemiol. 2007;165(4):410–7.

42. Karatzias T, Yan E, Jowett S. Adverse life events and health: A population study in Hong Kong. J Psychosom Res. 2015 Feb;78(2):173–7.

43. Lawrence WR, Lin Z, Lipton EA, Birkhead G, Primeau M, Dong G-H, et al. After the Storm: Short-term and Long-term Health Effects Following Superstorm Sandy among the Elderly. Disaster Med Public Health Prep. 2019 Feb;13(1):28–32.

44. Vanasse A, Cohen A, Courteau J, Bergeron P, Dault R, Gosselin P, et al. Association between Floods and Acute Cardiovascular Diseases: A Population-Based Cohort Study Using a Geographic Information System Approach. Int J Environ Res Public Health. 2016;13(2):168.

45. Trasande L, Koshy TT, Gilbert J, Burdine LK, Marmor M, Han X, et al. Cardiometabolic profiles of adolescents and young adults exposed to the World Trade Center Disaster. Environ Res. 2018;160(ei2, 0147621):107–14.

46. Becquart NA, Naumova EN, Singh G, Chui KKH. Cardiovascular Disease Hospitalizations in Louisiana Parishes' Elderly before, during and after Hurricane Katrina. Int J Environ Res Public Health. 2018;16(1).

47. Jordan HT, Stellman SD, Morabia A, Miller-Archie SA, Alper H, Laskaris Z, et al. Cardiovascular disease hospitalizations in relation to exposure to the September 11, 2001 World Trade Center disaster and posttraumatic stress disorder. J Am Heart Assoc. 2013;2(5):e000431.

48. Kong SY, Song KJ, Shin SD, Ro YS. Cardiovascular Events after the Sewol Ferry Disaster, South Korea. Prehospital Disaster Med. 2019;34(2):142–8.

49. Joseph NT, Matthews KA, Myers HF. Conceptualizing health consequences of Hurricane Katrina from the perspective of socioeconomic status decline. Health Psychol Off J Div Health Psychol Am Psychol Assoc. 2014;33(2):139–46.

50. McKinney N, Houser C, Meyer-Arendt K. Direct and indirect mortality in Florida during the 2004 hurricane season. Int J Biometeorol. 2011;55(4):533–46.

51. Dirkzwager AJE, van der Velden PG, Grievink L, Yzermans CJ. Disaster-related posttraumatic stress disorder and physical health. Psychosom Med. 2007;69(5):435–40.

52. Jiao Z, Kakoulides SV, Moscona J, Whittier J, Srivastav S, Delafontaine P, et al. Effect of Hurricane Katrina on incidence of acute myocardial infarction in New Orleans three years after the storm. Am J Cardiol. 2012;109(4):502–5.

53. Gautam S, Menachem J, Srivastav SK, Delafontaine P, Irimpen A. Effect of Hurricane Katrina on the incidence of acute coronary syndrome at a primary angioplasty center in New Orleans. Disaster Med Public Health Prep. 2009;3(3):144–50.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

54. Nagayoshi Y, Yumoto S, Sakaguchi K, Shudo C, Takino S, Hashiyama M, et al. Heart attacks triggered by huge mud slides in mountain regions and severe flooding in inhabited areas. J Cardiol. 2015;65(2):117–20.

55. Jordan HT, Miller-Archie SA, Cone JE, Morabia A, Stellman SD. Heart disease among adults exposed to the September 11, 2001 World Trade Center disaster: results from the World Trade Center Health Registry. Prev Med. 2011;53(6):370–6.

56. Kim S, Kulkarni PA, Rajan M, Thomas P, Tsai S, Tan C, et al. Hurricane Sandy (New Jersey): Mortality Rates in the Following Month and Quarter. Am J Public Health. 2017;107(8):1304–7.

57. Fonseca VA, Smith H, Kuhadiya N, Leger SM, Yau CL, Reynolds K, et al. Impact of a natural disaster on diabetes: exacerbation of disparities and long-term consequences. Diabetes Care. 2009;32(9):1632–8.

58. Silva-Palacios F, Casanegra AI, Shapiro A, Phan M, Hawkins B, Li J, et al. Impact of tornadoes on hospital admissions for acute cardiovascular events. Thromb Res. 2015 Nov;136(5):907–10.

59. Bich TH, Quang LN, Ha LTT, Hanh TTD, Guha-Sapir D. Impacts of flood on health: epidemiologic evidence from Hanoi, Vietnam. Glob Health Action. 2011;4(101496665):6356.

60. Huizink AC, Slottje P, Witteveen AB, Bijlsma JA, Twisk JWR, Smidt N, et al. Long term health complaints following the Amsterdam Air Disaster in police officers and fire-fighters. Occup Environ Med. 2006;63(10):657–62.

61. Jordan HT, Brackbill RM, Cone JE, Debchoudhury I, Farfel MR, Greene CM, et al. Mortality among survivors of the Sept 11, 2001, World Trade Center disaster: results from the World Trade Center Health Registry cohort. Lancet Lond Engl. 2011;378(9794):879–87.

62. Hendrickson LA, Vogt RL. Mortality of Kauai Residents in the 12-Month Period following Hurricane Iniki. Am J Epidemiol. 1996 Jul 15;144(2):188–91.

63. Peters MN, Katz MJ, Moscona JC, Alkadri ME, Khazi Syed RH, Turnage TA, et al. Effect of Hurricane Katrina on chronobiology at onset of acute myocardial infarction during the subsequent three years. Am J Cardiol. 2013;111(6):800–3.

64. Koroma IB, Javadi D, Hann K, Harries AD, Smart F, Samba T. Non-communicable diseases in the Western Area District, Sierra Leone, following the Ebola outbreak. F1000Research [Internet]. 2019 Nov 19 [cited 2020 Jul 30];8. Available from: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6652098/

65. An K, Salyer J, Kao H-FS. Psychological strains, salivary biomarkers, and risks for coronary heart disease among hurricane survivors. Biol Res Nurs. 2015;17(3):311–20.

66. Lin S, Gomez MI, Gensburg L, Liu W, Hwang S-A. Respiratory and cardiovascular hospitalizations after the World Trade Center disaster. Arch Environ Occup Health. 2010;65(1):12–20.

67. Yu S, Alper HE, Nguyen A-M, Brackbill RM. Risk of Stroke Among Survivors of the September 11, 2001, World Trade Center Disaster. J Occup Environ Med. 2018;60(8):e371–6.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

68. Brackbill RM, Thorpe LE, DiGrande L, Perrin M, Sapp JH 2nd, Wu D, et al. Surveillance for World Trade Center disaster health effects among survivors of collapsed and damaged buildings. Morb Mortal Wkly Rep Surveill Summ Wash DC 2002. 2006;55(2):1–18.

69. Gerin W, Chaplin W, Schwartz JE, Holland J, Alter R, Wheeler R, et al. Sustained blood pressure increase after an acute stressor: the effects of the 11 September 2001 attack on the New York City World Trade Center: J Hypertens. 2005 Feb;23(2):279–84.

70. Husarewycz MN, El-Gabalawy R, Logsetty S, Sareen J. The association between number and type of traumatic life experiences and physical conditions in a nationally representative sample. Gen Hosp Psychiatry. 2014 Jan;36(1):26–32.

71. Ng J, Atkin SL, Rigby AS, Walton C, Kilpatrick ES. The effect of extensive flooding in Hull on the glycaemic control of patients with diabetes: Impact of flooding on glycaemic control. Diabet Med. 2011 May;28(5):519–24.

72. Rey G, Jougla E, Fouillet A, Pavillon G, Bessemoulin P, Frayssinet P, et al. The impact of major heat waves on all-cause and cause-specific mortality in France from 1971 to 2003. Int Arch Occup Environ Health. 2007 May 2;80(7):615–26.

73. Moscona JC, Peters MN, Maini R, Katigbak P, Deere B, Gonzales H, et al. The Incidence, Risk Factors, and Chronobiology of Acute Myocardial Infarction Ten Years After Hurricane Katrina. Disaster Med Public Health Prep. 2019 Apr;13(02):217–22.

74. Thethi TK, Yau CL, Shi L, Leger S, Nagireddy P, Waddadar J, et al. Time to recovery in diabetes and comorbidities following Hurricane Katrina. Disaster Med Public Health Prep. 2010;4 Suppl 1(101297401):S33-8.

75. Obrova J, Sovova E, Ivanova K, Taborsky M, Loyka S. Effects of the july 1997 floods in the Czech Republic on cardiac mortality. Disaster Med Public Health Prep. 2014;8(6):492–6.

84. Schultz WM, Kelli HM, Lisko JC, Varghese T, Shen J, Sandesara P, et al. Socioeconomic Status and Cardiovascular Outcomes: Challenges and Interventions. Circulation. 2018 May 15;137(20):2166–78.

85. Mohammed SH, Habtewold TD, Birhanu MM, Sissay TA, Tegegne BS, Abuzerr S, et al. Neighbourhood socioeconomic status and overweight/obesity: a systematic review and meta-analysis of epidemiological studies. BMJ Open [Internet]. 2019 Nov 14 [cited 2020 Nov 17];9(11). Available from: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6886990/

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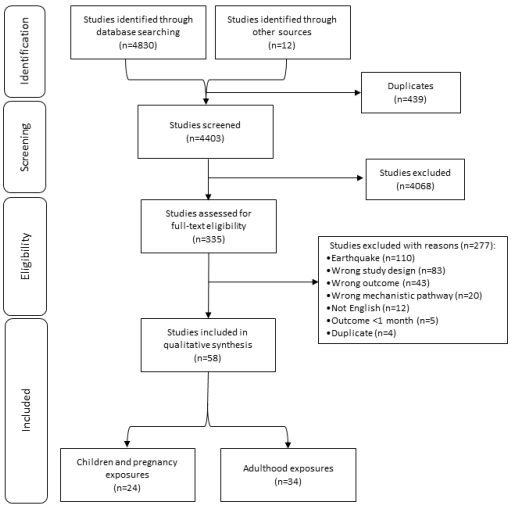


Figure 1. PRISMA flow chart

338x190mm (96 x 96 DPI)

APPENDIX

Figure A1. Search strategy for Medline

1	"social isolation.mp." or Social isolation/	1479
2	quaratine.mp. or exp Quarantine/	5589
3	Pandemics/	7247
4	Epidemics/	10146
5	Coronavirus or coronavirus.mp.	19310
6	humanitarian crises.mp.	232
7	exp mass casualty incidents/ or exp natural disasters/	18883
8	public health emergency.mp.	1636
9	cardiovasucalr.mp. or exp Cardiovascular Diseases/	2605399
10	hypertension.mp. or exp Hypertension/	484321
11	exp Obesity/ or obesity.mp.	321847
12	body mass index.mp. or exp Body Mass Index/	236149
13	stroke.mp, or exp Stroke/	310109
14	myocardial infarction.mp. or exp Myocardial Infarction/	247417
15	angina.mp. or exp Angina Pectoris/	698186
16	diabetes.mp. or Diabetes Mellitus/	646582
17	9 or 10 or 11 or 12 or 13 or 14 or 15 or 16	3552066
18	Disasters/ or disaster.mp.	39131
19	1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 or 18	10917
20	17 and 19	3688
21	limit 20 to humans	2998

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Table A2. Critical appraisal using Joanna Briggs Institute (JBI) checklist for cohort studies by exposure period (n=41)

Author, Year	Were the two groups similar and recruited from the same population?	Were the exposures measured similarly to assign people to both exposed and unexposed groups?	Was the exposure measured in a valid and reliable way?	Were confounding factors identified?	Were strategies to deal with confounding factors stated?	Were the groups/participants free of the outcome at the start of the study (or at the moment of exposure)?	Were the outcomes measured in a valid and reliable way?	Was the follow up time reported and sufficient to be long enough for outcomes to occur?	Was follow up complete, and if not, were the reasons to loss to follow up described and explored?	Were strategies to address incomplete follow up utilized?	Was appropriate statistical analysis used?
Perinatal											
Dancause, 2012	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Dancause, 2013	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No	Yes
Dancause, 2015	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Unclear	Yes
de Rooij, 2007	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Dorn, 2007	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Unclear	Yes
Ekamper, 2015	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Goudet, 2011	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Unclear	Unclear	Yes
Huang, 2010	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Unclear	Unclear	Yes
Hult, 2010	Yes	Yes	Yes	Yes	Yes	Unclear	Yes	Yes	Unclear	Unclear	Yes
Kroska, 2018	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Unclear	Yes
Lei, 2016	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Unclear	Yes
Liu, 2016	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Unclear	Yes
Lumey, 2012	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mazumder, 2010	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Unclear	Unclear	Yes
Mryskyla, 2013	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Unclear	Unclear	Yes

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Painter, 2006	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Ravelli, 1999	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Roseboom, 2001	Yes	Yes	Yes	Yes	Yes	Yes	Unclear	Yes	Yes	Yes	Yes
Schrier, 2010	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Unclear	Unclear	Yes
Sotomayer, 2013	Yes	Yes	Yes	Yes	Yes	Unclear	Yes	Yes	Unclear	Unclear	Yes
Stein, 2006	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adult											
Baum, 2019	Yes	Yes	Yes	Yes	Yes		Yes	Yes	Yes	Yes	Yes
Brackbill, 2006	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Unclear	Yes	Yes
Dirkzwager, 2007	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Unclear	Unclear	Yes
Fonseca, 2009	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Not applicable	Yes
Gautam, 2009	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Unclear	Unclear	Yes
Huizink, 2006	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes
Jiao, 2012	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Unclear	Unclear	Yes
Jordan, 2011 (A)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No	Yes
Jordan, 2011 (B)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Unclear	Yes	Yes
Jordan, 2013	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Unclear	Yes	Yes
Joseph, 2014	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Unclear	Unclear	Yes
Lawrence, 2019	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Unclear	Not applicable	Yes
Moscona, 2019	Yes	Yes	Yes	Yes	No	Unclear	Yes	Yes	Unclear	Unclear	Yes
Ng, 2011	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Unclear	Unclear	Yes
Peters, 2013	Yes	Yes	Yes	No	No	No	Yes	Yes	Unclear	Unclear	Yes
Rey, 2007	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Unclear	Unclear	Yes

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Thethi, 2010	Yes	Yes	Yes	No	No	No	Yes	Yes	Yes	Unclear	Yes
Trasande, 2018	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Unclear	Not applicable	Yes
Vanasse, 2016	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Unclear	Not applicable	
Yu, 2018	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No	Yes

Table A3. Critical appraisal using Joanna Briggs Institute (JBI) checklist for cross-sectional studies by exposure period (n=7)

Author, Year	Were the criteria for inclusion in the sample clearly defined?	Were the study subjects and the setting described in detail?	Was the exposure measured in a valid and reliable way?	Were objective, standard criteria used for measurement of the condition?	Were confounding factors identified?	Were strategies to deal with confounding factors stated?	Were the outcomes measured in a valid and reliable way?	Was appropriate statistical analysis used?
Perinatal								
Bercovich, 2014	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes
Oni, 2015	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adult								
An, 2015	Yes	Yes	Yes	Yes	No	No	Yes	Yes
Bich, 2011	No	Yes	Yes	Yes	No	No	No	Yes
Husarewycz, 2014	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Katratzias, 2015	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Koroma, 2019	No	No	Yes	Yes	No	No	Yes	Yes

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Table A4. Critical appraisal using Joanna Briggs Institute (JBI) checklist for quasi-experimental studies by exposure period (n=10)

Author, Year	Is it clear in the study what is the 'cause' and what is the 'effect' (i.e. there is no confusion about which variable comes first)?	Were the participants included in any comparisons similar?	Were the participants included in any comparisons receiving similar treatment/care, other than the exposure or intervention of interest?	Was there a control group?	Were there multiple measurements of the outcome both pre and post the intervention/exposure?	Was follow up complete and if not, were differences between groups in terms of their follow up adequately described and analyzed?	Were the outcomes of participants included in any comparisons measured in the same way?	Were outcomes measured in a reliable way?	Was appropriate statistical analysis used?
Perinatal									
Xiao, 2019	Yes	Unclear	Not applicable	Yes	No	Yes	Yes	Yes	Yes
Adult									
Becquart, 2018	Yes	Unclear	Not applicable	No	No	Yes	Yes	Yes	Yes
Gerin, 2005	Yes	Yes	Not applicable	Yes	Yes	Yes	Yes	Yes	Yes
Hendrickson, 1996	Yes	Unclear	Not applicable	No	Yes	Yes	Yes	Yes	Yes
Kim, 2017	Yes	Unclear	Not applicable	No	Yes	Yes	Yes	Yes	Yes
Kong, 2019	Yes	Unclear	Not applicable	Yes	No	Yes	Yes	Yes	Yes
Lin, 2010	Yes	Unclear	Not applicable	Yes	No	Yes	Yes	Yes	Yes
McKinney, 2011	Yes	Unclear	Not applicable	No	Yes	Yes	Yes	Yes	Yes
Nagayoshi, 2015	Yes	Unclear	Not applicable	Yes	No	Yes	Yes	Yes	Yes
Silva-Palacios, 2015	Yes	Unclear	Not applicable	No	No	Yes	Yes	Yes	Yes

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	3
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	4
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	4
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	4
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	4
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	11 (Table 1)
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	5
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	5
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	5
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	5
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	5
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	5

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	N/A
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	N/A
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	6
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	6
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	9
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	6-8
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	N/A
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	N/A
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	N/A
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	9
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	9-10
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	10
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	1

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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Secondary Subject Heading:	Epidemiology
Keywords:	PUBLIC HEALTH, EPIDEMIOLOGY, DIABETES & ENDOCRINOLOGY

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Impact of disasters, including pandemics, on cardiometabolic outcomes across the life-course: A systematic review

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ABSTRACT

Background: Disasters are events that disrupt the daily functioning of a community or society, and may increase long-term risk of adverse cardiometabolic outcomes, including cardiovascular disease, obesity, and diabetes. The objectives of this study were to conduct a systematic review to determine the impact of disasters, including pandemics, on cardiometabolic outcomes across the life-course.

Design: A systematic search was conducted in May 2020 using two electronic databases, EMBASE and Medline. All studies were screened in duplicate at title and abstract, and full-text level. Studies were eligible for inclusion if they assessed the association between a population-level or community disaster and cardiometabolic outcomes ≥ 1 month following the disaster. There were no restrictions on age, year of publication, country, or population. Data were extracted on study characteristics, exposure (e.g., type of disaster, region, year), cardiometabolic outcomes, and measures of effect. Study quality was evaluated using the Joanna Briggs Institute critical appraisal tools.

Results: A total of 58 studies were included, with 24 studies reporting the effects of exposure to disaster during pregnancy/childhood and 34 studies reporting the effects of exposure during adulthood. Studies included exposure to natural (n=35; 60%) and human-made (n=23; 40%) disasters, with only 3 (5%) of these studies evaluating previous pandemics. Most studies reported increased cardiometabolic risk, including increased cardiovascular disease incidence or mortality, diabetes, and obesity, but not all. Few studies evaluated the biologic mechanisms or high-risk subgroups that may be at a greater risk of negative health outcomes following disasters.

Conclusions: The findings from this study suggest that the burden of disasters extend beyond the known direct harm, and attention is needed on the detrimental indirect long-term effects on cardiometabolic health. Given the current COVID-19 pandemic, these findings may inform public health prevention strategies to mitigate the impact of future cardiometabolic risk.

PROSPERO registration: CRD – 42020186074

Strengths and limitations of this study:

- This systematic review is one of the first to review the literature on disasters, including pandemics, and subsequent cardiometabolic outcomes throughout the life-course.
- A total of 58 studies were included and they covered a range of disasters (e.g., famine, war, terrorism, natural disasters, and infectious disease epidemics), periods of exposure from pregnancy, childhood to older adulthood, and a wide breadth of cardiometabolic outcomes were evaluated.
- This review contributes a synthesis of the literature on the impact of disasters and cardiometabolic outcomes, that may inform public health interventions to mitigate the long-term impact of current COVID-19 pandemic.

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BACKGROUND

Disasters, as defined by the World Health Organization (WHO), are events that disrupt the daily functioning of a community or society causing material, economic or environmental losses, overwhelming local capacity (1). Disasters can be categorized into natural disasters, human-made disasters, and hybrid disasters (2). Natural disasters include natural phenomenon above and beneath the earth's surface (e.g., tsunamis or landslides), meteorological phenomenon (e.g., tornadoes or floods) or biological phenomenon (e.g., epidemics and pandemics) (2). Human-made disasters include adverse transportation incidents, technological events (e.g., fire or toxic leaks), terrorism, warfare or conflict (2). A hybrid disaster results from both human and natural forces, such as the clearing of a jungle that results in a landslide (2). All types of disasters can result in public health emergencies as they typically impact a significant proportion of people (3). Epidemics, defined as a greater than expected increase in cases of a disease, and pandemics, which cross countries and continents, are types of natural disasters with far-reaching global disruption (4). The COVID-19 pandemic is a present-day example of a global disaster that is unlike any disaster in recent history. Understanding the potential long-term health implications of the current COVID-19 pandemic and resulting public health mitigation strategies is urgently needed.

Previous systematic reviews have focused on acute outcomes, specifically on the psychological impact of quarantine during pandemics (5), the impact on health outcomes after disasters in older adults (6), medically unexplained physical symptoms following disasters (7), and chronic medical interventions following a natural disaster (8). It is biologically plausible that exposure to a disaster may lead to long-term or chronic outcomes that could arise many years later and this may be modified by the time of exposure across the life-course. Consistent with established models of life-course epidemiology there may be critical periods of exposure (e.g., during development in childhood), where exposure to a disaster substantially increases later life disease risk, or exposure to a disaster may contribute to a chain of risk or accumulation of risks across the life course (9,10). There is currently no review on the long-term impacts of disasters, or more specifically, epidemics and pandemics on cardiometabolic outcomes across the life-course. Noncommunicable diseases (NCDs), including cardiovascular disease (CVD), obesity, and diabetes, are the leading cause of morbidity and mortality worldwide (11,12). NCDs are attributed to 71% of all global deaths annually, with approximately 14 million CVD-related deaths and 1.6 million diabetes-related deaths (12). Findings from the Global Burden of Diseases Study indicate that CVD and diabetes account for over 20% of the global burden of disability, with diabetes recently emerging as the fourth leading cause of disability globally (11). Exposure to disasters may cause

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cardiometabolic outcomes to emerge or worsen through several different mechanistic pathways including stress exposure (13), lack of access to health services (14), food security and behavioural changes such as alterations in physical activity, sleep, and diet (15). It is important to understand the impact of disasters on the incidence of new cardiometabolic diseases and changes in disease status in all populations and age groups. Particular subgroups of a population may be more or less susceptible to cardiometabolic outcomes and understanding this can inform targeted interventions. The primary objective of this review was to determine the impact of disasters, including pandemics on risk of cardiometabolic outcomes across the life-course. The secondary objectives were to determine how to reduce the impact of chronic disease outcomes following a disaster and to identify populations at highest risk of cardiometabolic outcomes following a disaster.

METHODS

A systematic review was conducted following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) (16). This review was registered on PROSPERO (CRD – 42020186074).

Search strategy

A systematic search was conducted in May 2020 using the electronic databases EMBASE and MEDLINE. The health research librarians at McMaster University assisted in developing the search strategy. The search broadly captured two concepts: disasters and cardiometabolic outcomes (e.g., diabetes, obesity, hypertension). The complete search strategy for EMBASE can be found in Table 1. The search strategy for MEDLINE can be found in the Appendix (Table A1). Reference lists of eligible studies and relevant systematic reviews were hand searched to identify additional articles.

Eligibility criteria

Studies were eligible for inclusion if they assessed the relationship between a population-level or community disaster and the risk of future cardiometabolic outcomes including CVD, diabetes or obesity or cardiometabolic risk scores (17). CVD included myocardial infarction, stroke, hypertension and angina. There were no restrictions on year of publication, country of disaster, or population. Only studies evaluating the impact of real-world disasters in humans were included. Due to the research team's capacity, only studies published in English were included. Observational and quasi-experimental study designs, including case-control, cohort and other longitudinal study designs or natural

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experiments were included. Outcomes that were not cardiometabolic-related or acute cardiometabolic events, such as an immediate complication (defined as <1 month after disaster), were excluded. Studies that assessed the exposure to a chemical as a result of the disaster, were excluded, as we were not interested in outcomes resulting from chemical exposure. Earthquake studies were also excluded since a systematic review was published in 2018 that assessed the impact of earthquakes on cardiometabolic outcomes (18).

Study selection

After running the search, all identified studies were imported into Covidence and duplicates were removed (19). Studies were screened at title and abstract-level, and then at full-text by any two of the following independent reviewers VD, JL, MSA, YYM, ATA, ES, SI, JDM, RR LNA. Conflicts were resolved by a third reviewer who made the final decision regarding eligibility for inclusion.

Data extraction

A data extraction template was created, and pilot tested prior to data extraction. Data were then extracted from all studies by any two of the following independent abstractors VD, JL, SMA, YYM, ATA, RR, ES and conflicts were resolved by a third independent abstractor. Study characteristics including year of publication, study design, country of disaster, sample size, and length of study were extracted where reported. Specific information on the exposure and outcome in each study were extracted including the type and name of the disaster, country and year of the disaster, the outcome of interest, and how the exposure and outcome were measured. Finally, any information on subgroups including age, population, sex and disaster type were also extracted, if applicable.

Critical appraisal

Critical appraisal was conducted using the Joanna Briggs Institute (JBI) Critical Appraisal Tools (20). This tool was chosen due to the availability of checklists for a wide range of study designs, including cohort, cross-sectional and quasi-experimental designs (20). The quasi-experimental study design checklist was used for natural experiments including time-series studies and pre/post designs, as it was decided this was the most appropriate tool. All studies were critically appraised independently by any two of the following individuals VD, JL, MA, YYM, ATA, ES, SI and a third individual was consulted for any discrepancies.

Data analysis

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Data from the included studies were narratively synthesized. Results are presented by exposure period (perinatal/childhood versus adulthood) and by cardiometabolic outcome (obesity, CVD, and diabetes). Characteristics of studies are presented as frequencies and percentages. Due to the heterogeneity of studies, a meta-analysis was not conducted.

RESULTS

A total of 4830 studies were identified through the electronic database search. An additional 12 studies were identified through manual searching of the reference lists of relevant studies. After removing duplicates (n=439), 4403 studies were screened at title and abstract level. After applying inclusion and exclusion criteria, 4068 studies were excluded, leaving 335 studies screened for full-text eligibility. A total of 58 studies were eligible for inclusion into the review. The complete screening process is described in Figure 1.

Description of studies

Table 2 provides a summary of included studies. Of 58 included studies, 24 studies (15,21–43) investigated exposure to disasters during pregnancy or childhood while the remaining 34 studies (14,44–76) investigated exposure to disaster during adulthood. Almost all studies (n=49) assessed cardiometabolic outcomes during adulthood, only two studies assessed outcomes during pregnancy (27,28) and seven studies assessed outcomes during childhood and adolescence (15,21–26). The length of studies, including prospective follow-up and retrospective assessment, ranged from one month to 95 years. Most studies (n=36) focused on disasters that occurred in North America (14,21–28,38,39,42,45–49,51,52,54,55,57–60,63–65,67–72,75,76), followed by Europe (n=13) (29–33,35,41,43,53,62,73,74). The remaining disasters occurred in Asia (n=7) (15,37,40,44,50,56,61) and Africa (n=2) (36,66). The characteristics of included studies and key findings are provided in Table 3 for disaster exposure during the perinatal period and childhood, and Table 4 for exposure during adulthood.

Exposure to disaster in the perinatal and childhood period

Of the 24 studies that evaluated perinatal and childhood exposure to disaster, 12 studies evaluated human-made disasters (29–35,40,41,47) and the remaining 12 evaluated natural disasters

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(15,21–28,38,39,42) of which two were pandemics (38,39). Most studies (n=15) assessed the disaster as the main exposure of interest (24,28–33,35,36,39–42). The remaining studies evaluated stress (e.g., maternal stress, disaster-related PTSD, subjective stress, objective hardship) (21–25,27,47), maternal weight and maternal nutrition status (15,34), cognitive appraisal (26), and coping strategies (27) that were the result of the disaster as the exposure variable. The age when cardiometabolic outcomes were assessed varied across studies, with two studies evaluating pregnancy outcomes (27,28), 8 studies evaluating outcomes among children and youth (15,21–26,47), four studies assessed outcomes during young to mid adulthood (>18-40 years of age) (27,29,36,37), and 10 during later adulthood (≥50 years of age) (30,32–35,38–42). One study did not specify the exact age, rather evaluated outcomes throughout adulthood, from 18-63 years of age (31). Detailed characteristics and findings of all studies that assessed perinatal and childhood exposures to disasters can be found in Table 3.

Within the 10 studies that evaluated perinatal or childhood disaster exposure in relation to pregnancy, childhood, or youth outcomes, one study evaluated a human-made disaster (the World Trade Center Attacks) and the other nine studies evaluated natural disasters (including ice storms, floods and hurricanes) and the findings were mixed. The one study that evaluated a human made disaster found limited evidence of any increased cardiometabolic risk and a small decrease in both BMI and zBMI was observed for children exposed to the World Trade Centre attacks compared to those who were not but found no differences in triglycerides or lipids (47). Within the eight studies that evaluated exposure to a natural disaster during the perinatal and childhood period and cardiometabolic outcomes in later childhood, there were six studies that evaluated measures of child growth and four of these studies reported increased BMI or adiposity in later childhood (21,23,25,26), one study was null (22) and one study reported increased wasting or malnutrition following exposure to a flood (15). Importantly, the one study where increased wasting was observed was in Bangladesh, whereas all of the studies that observed increased risk of obesity were in North America. Of the studies that evaluated childhood cardiometabolic outcomes other than growth, one study found increased insulin concentrations at age 13 (24). There were two studies that evaluated exposure to a natural disaster (Hurricane Katrina and Hurricane Sandy) during pregnancy and both found increased incidence or hospital visits for gestational hypertension and diabetes (27,28).

There were 14 studies that evaluated exposure to a disaster in pregnancy or childhood in relation to the subsequent onset of adult cardiometabolic conditions; 11 of these studies evaluated

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exposure to human made disasters and three evaluated natural disasters. Within the 11 studies that evaluated human made disasters, one study found no association between exposure to the Dutch famine and coronary artery disease in older adulthood (32). While the remaining 10 studies all found some evidence of increased cardiometabolic outcomes in adulthood following perinatal or childhood exposure, the results were mixed with many null results depending on outcome or exposure. For example, prenatal exposure to famine was associated with higher LDL and CHD approximately 28 years later, however no difference was found for glucose, insulin, BMI or other lipids (29). The results were not consistent across outcomes, for example, increased risk of hypertension was found in three studies (29,36,40) but not in two studies (35,37). Within the three studies that evaluated adult cardiometabolic outcomes following exposure to a natural disaster, all three studies found increased risk of cardiovascular diseases or mortality following prenatal exposure to famine or the 1918 influenza pandemic (38,39,42).

Studies on adult exposure to disaster and subsequent cardiometabolic outcomes

Thirty-four studies investigated the effects of exposure to disasters during adulthood on cardiometabolic outcomes. The length of follow-up ranged from one month to 13 years. There were 23 studies that examined natural disasters (14,44–46,48,51,52,54–56,58–61,64–67,72–76), and 11 studies that examined human-made disasters (43,49,50,53,57,62,63,68–71). Of these studies, only one evaluated the impact of an infectious disease epidemic (66). Most studies (n=27) considered the disaster as the main exposure of interest (14,43–46,48,50,52,54–56,58–66,68,71–76). The remaining seven studies assessed disaster-related stress (53,57,67,69), including post-traumatic stress disorder (PTSD) and psychological strain, unemployment rates as a result of the disaster (51) and exposure to damaged or collapsed buildings during the World Trade Center disaster (70). Detailed characteristics and findings of all studies that assessed adult exposures to disasters are included in Table 4.

The studies that assessed exposure to human-made disasters (n=11) during adulthood reported mixed results in terms of associations with outcomes and statistical significance. Three studies assessed PTSD related to disasters and found an increased association with stoke (69), heart disease (57) and cardiovascular/vascular problems (53), two of which were exposure to the World Trade Centre disaster and the third was a Fireworks depot explosion. Two studies assessing exposure to the World Trade Centre disaster and Amsterdam Air disaster found an increased association with cardiovascular

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hospitalizations (49) and cardiovascular symptom complaints (62) in rescue workers compared to non-rescue workers. Of the remaining six studies, three studies reported an increased association with hypertension (43), systolic blood pressure (71) and CVD mortality (63), however, the exact exposure varied across studies. For instance, one study explored the level of exposure, defined as low, intermediate, or high to the World Trade Centre disaster (63), whereas another study evaluated exposure to the Volendam Pub Fire among parents who had children who were injured or died (43). The final three studies assessing exposure to human-made disasters (World Trade Centre disaster and Sewol Ferry disaster) reported mixed results with some showing a decreased association or null findings (53,68,70).

Among studies that evaluated the impact of exposure to natural disasters (n=23), six studies that evaluated exposure to Hurricanes Sandy, Katrina, Iniki, and the Flood of Saint-Jean-sur-Richelieu reported an increased association with cardiometabolic outcomes(45,46,51,58,64,72). One of these studies specifically investigated unemployment as a result of Hurricane Katrina and found those who were unemployed, compared to those who remained employed were 5.65 times more likely to have a cardiometabolic event ($p<0.05$) (51). Nine studies reported a statistically significant increase in outcomes following exposure disaster (52,54–56,61,67,73–75). For instance, one study found those who reported higher levels of psychological strain after surviving Hurricane Ike, compared to those with lower levels of psychological strain had higher mean blood glucose and obesity four months after the disaster (67). Whereas, another study found a higher proportion of people experiencing worse hypertension who were living in households affected by the 2008 Hanoi flood compared to those who lived in an unaffected households in both rural and urban areas (61). Two of these studies reported an increase in incidence of AMI and AMI hospital admission pre-Hurricane Katrina, compared to post-Hurricane Katrina (54,55). The remaining six studies reported a mix of findings. Three studies found varying associations across outcomes reported within their study. For example, Fonseca et al., found an increased mean difference pre- and post-Hurricane Katrina for glycemic control, systolic blood pressure, diastolic blood pressure, and HDL, but not for LDL and triglycerides (59). Nine studies reported mixed findings across outcomes within the study. Four of these studies found both an increase and decrease in outcomes when comparing mean difference or proportion pre- and post-disaster (14,48,66,76). One study found those with higher reports of exposure to natural disaster throughout the life course were significantly different from those with lower reports (44). The final two studies found a decreased

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proportion of AMI following Hurricane Katrina (65) and no significant association between exposure to the Oklahoma tornado and hospital admission for cardiovascular events (60).

Mediation and modification of cardiometabolic outcomes

Across all studies, few evaluated effect modification or subgroups of a population that may be at a greater risk of negative health outcomes following disasters. Eight studies stratified by sex (30,34,36,41,45,50,57,64), gestational timing of exposure (22,29,31–33,35,38), year of birth or age at outcome (42,64,69), urban or rural area (37), race (45,48) and socioeconomic status (67), however, results varied greatly due to the differences in exposure period, disaster type, cardiometabolic outcome and age at outcome. One study explored the possible mediators between cognitive appraisal following the Quebec Ice Storm and obesity. It was noted that negative cognitive appraisal was found to predict obesity via DNA methylation of diabetes-related genes (26). No studies evaluated or discussed possible interventions to mitigate risk of cardiometabolic disease following a disaster.

Critical appraisal

The critical appraisal assessment for all study designs can be found in Tables A2-A4. Among the cohort studies, most studies met all criteria included in the checklist indicating high study quality. For instance, almost all cohort studies had comparable populations that were recruited in a similar way and exposures that were assessed in the same way across populations. However, across almost all cohort studies, information on follow-up or strategies to address incomplete follow-up were unclear or not addressed. The critical appraisal results for cross-sectional studies were inconsistent with a small number of studies meeting only some checklist requirements. For quasi-experimental studies, the checklist requirement for within person comparisons were not applicable for all studies, however, all studies clearly defined the cause and effect within the study.

DISCUSSION

Principal findings

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A total of 58 studies were identified and they covered a wide breadth of exposures to both natural and human-made disasters, including famine, war, terrorism, natural disasters, and infectious disease epidemics. Exposures were investigated in pregnancy and childhood exposure through to adulthood with outcomes measured 1 month to 95 years later. The reviewed studies reflect a true-life course body of literature with exposures at multiple ages and long-term exposures. A range of cardiometabolic outcomes including obesity, hypertension, myocardial infarction, diabetes, and cardiac mortality were investigated. Given the varied nature of the studies, it was difficult to draw overall conclusions, but the vast majority of studies provided some evidence of increased cardiometabolic risk following disaster exposure. There were only 11 studies that reported no increased risk or had unclear findings. Across these studies, there was a variety of disaster exposure, outcomes, and follow-up periods, however seven of these studies did not report adjustment or consideration of any confounders.

Relation to other studies

To the best of our knowledge, this is the first review to systematically review the literature on a broad range of disasters and cardiometabolic health outcomes across the life-course. Other reviews have focused on a specific population, such as older adults, specific disaster types (e.g., natural disasters only) or other health conditions (e.g., mental health) or acute outcomes (5,6,8). However, across most reviews it was apparent the heterogenous nature of included studies makes it difficult to summarize findings and make overall conclusions and recommendations. For instance, Chan et al. found exposure to natural disasters negatively affected those with chronic conditions, although authors noted limitations due to limited literature (8). Another systematic review found very heterogenous results when reviewing the literature on health outcomes after disasters for older adults with chronic disease (5). The studies included here were from multiple disciplines and utilized a variety of studies designs, assessed several different outcomes, and applied different statistical approaches. Overall the results suggested increased risk of adverse cardiometabolic outcomes following disasters, although this was not apparent across all included studies (6). The unexpected nature of disasters, uniqueness of population or region affected, and scale of damage lead to research studies that vary greatly. Although previous reviews and the current review have identified quite heterogenous studies, overall conclusions suggest risk of disease increases after exposure to disasters.

Biologic mechanisms

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Several potential mechanisms were discussed in the included studies that may contribute to the observed associations between disaster exposure and increased cardiometabolic outcomes, include the role of both objective and subjective stress, nutritional changes, and reduced access to healthcare. One study that explored mediators in the association between stress and obesity measures identified the role of DNA methylation in this association (26). It is well postulated that the activation of a stress response following a stressful event leads to changes in the nervous, cardiovascular, endocrine and immune systems (77). Exposure to disasters including famine, war, terrorism, natural disasters, and infectious disease epidemics may activate a stress response, altering the progression of disease development (77). The repeated or prolonged exposure to various disasters, such as a pandemic spanning over months, may lead to worse health outcomes. Reduction in health services is another possible mechanism leading to worse health outcomes. Healthcare services may be directed toward the immediate response to health-related consequences caused by the disaster (e.g., illness from a pandemic, injuries associated with a terrorist attack or natural disaster), limiting access to primary care (78). This interruption to services may decrease screening or early treatment ultimately leading to the rise in chronic diseases. Lastly, social determinants of health are known to be important risk factors for cardiometabolic conditions (79). At least one study investigated whether the observed associations were due to changes in educational attainment (42) and unemployment (51) . More investigation of the role of social determinants as modifiers or mediators of the associations between disasters and long-term cardiometabolic outcomes may be warranted. Despite numerous proposed biologic mechanisms and well-established life course frameworks, relatively few studies actually evaluated potential causal pathways using a life course framework, and this may contribute to some of the observed heterogeneity in results.

Strengths and limitations

This review had several strengths including the comprehensive evaluation of the impact of a wide range of disaster exposures on various cardiometabolic outcomes at different periods throughout the life-course. The search strategy was developed in consultation with health science librarians at McMaster University to ensure the most comprehensive search was developed and relevant literature was identified. The timely findings of this synthesis are a strength of this review, given the current COVID-19 pandemic, which is affecting millions of people worldwide. While only a small proportion of the identified studies focused on pandemics and epidemics, the findings may serve to guide our

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understanding of expected outcomes, and to develop future research to study the effects of COVID-19 on cardiometabolic outcomes.

Although this review had several strengths, interpretation of findings should be done with caution due to limitations. First, the heterogeneity across studies restricted the ability to conduct a meta-analysis. Studies varied in terms of study design, reported measures of effect, the comparison group (e.g., some studies did not include a comparator group), length of follow-up, timing and measurement of exposure, and primary outcomes and how they were measured. Given the multi-disciplinary nature of the identified studies, a wide range of analytic approaches were used, and measures of effect varied. These differences in addition to the lack of statistical significance across studies make it difficult to draw overall conclusions. Many of the studies used a retrospective cohort study design and relied on administrative data sources as such many studies were unable to comprehensively adjust for confounders, including social determinants of health. Measurement error and misclassification of exposure status is also possible since many studies did not objectively measure disaster exposure or degree of impact, and instead used proxy measures of disaster exposure based on time and geography.

Very few studies have evaluated the long-term impacts of pandemics and epidemics on cardiometabolic outcomes, identifying a current gap in the literature. This made it difficult to truly assess if exposure to disasters at sensitive periods of development had lasting effects much later in life. Studies also reported insufficient data on subgroups that were at increased risk of worse cardiometabolic health outcomes and interventions that were implemented to mitigate risk of cardiometabolic outcomes. In addition, results were not often explored by sex and gender, or did not apply an equity lens. It has been noted that those of different levels of socioeconomic status experience differential cardiometabolic outcomes (80,81). This signifies the importance of exploring associations between exposure to disasters and cardiometabolic outcomes stratified by these factors. Understanding how these associations differ will also help to identify groups of people who will experience worse outcomes following a disaster.

Study implications

To the best of our knowledge, this is the first study to comprehensively explore the impact of several different types of disasters on cardiometabolic outcomes at different periods throughout the life-course. The results suggest that increased risk is observed for disaster exposure at any period over the life-course from the perinatal child and adult periods. These findings emphasize that the burden of

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disasters extends beyond the known direct harms they cause, and attention is needed on the detrimental indirect long-term effects on cardiometabolic health and chronic disease. Given the current COVID-19 pandemic, this review may be helpful in raising awareness of the potential increase in cardiometabolic health outcomes post-pandemic, to ensure appropriate public health mitigation measures are developed and implemented to prevent long term cardiometabolic outcomes at the population level.

Unanswered questions and future research

Future research should evaluate the impact of pandemics, such as COVID-19, on future cardiometabolic health throughout the life-course. It may also be of interest for future research to explore the impact of implementing public health measures, such as physical distancing to reduce transmission of a virus, and the implications following a disaster with access to healthcare on health outcomes. This information would be helpful in planning public health responses to different disasters. In addition, further investigation of possible mechanisms, such as disruptions to healthcare or medication access, and changes in dietary intake or physical activity, is needed. This would help to develop preventative strategies targeted at these mechanisms to help reduce the possible cardiometabolic consequences after experiencing a disaster. This review found insufficient evidence identifying subgroups of the population who are at the greatest risk or specific disaster related risk factors that that increase cardiometabolic disease development following a pandemic. This is an important gap that needs to be addressed by future research.

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Table 1. Search strategy for EMBASE

1	social isolation.mp. or social isolation/	24963
2	quarantine.mp. or quarantine/	4752
3	*epidemic/	32686
4	*pandemic/	4387
5	disease outbreak.mp.	2321
6	disaster/	13321
7	*natural disaster/	968
8	humanitarian crisis.mp.	257
9	mass casuality.mp. or mass disaster/	3654
10	coronavirus.mp. or coronaviridae/	23106
11	cardiovascular disease.mp. or *cardiovascular disease/	357319
12	*diabetes mellitus/	210248
13	*cerebrovascular accident/	78444
14	*heart infarction/	99072
15	*angina pectoris/	22631
16	*obesity/	178134
17	public health emergency.mp.	1752
18	*body mass/	31459
19	*hypertension/	198593
20	1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 or 9 or 10 or 17	109105
21	11 or 12 or 13 or 14 or 15 or 16 or 18 or 19	1087681
22	20 and 21	2047
23	limit 22 to human	1832

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Table 2. Characteristics of included studies (n=58)

Characteristics	N (%)
Continent	
North America	36 (62%)
Europe	13 (22%)
Asia	7 (12%)
Africa	2 (3%)
Year of Publication	
2010-2020	44 (76%)
2000-2009	12 (21%)
1996-1999	2 (3%)
Study design	
Cohort/longitudinal	41 (71%)
Quasi-experimental[1]	10 (17%)
Cross-sectional	7 (12%)
Sample size	
≥10,000	19 (33%)
1,000- <10,000	10 (17%)
≤1,000	24 (41%)
Not specified	5 (9%)
Exposure life stage	
Pregnancy/Childhood	24 (41%)
Adulthood	34 (57%)
Outcome life stage	
Pregnancy	2 (3%)
Childhood [2]	8 (13%)
Adult	47 (81%)
Disaster	
Human-made	23 (40%)
Natural	35 (60%)
Cardiometabolic outcome [3]	
Cardiovascular disease [4]	41 (71%)
Diabetes [5]	11 (19%)
Obesity or BMI	12 (21%)
Mortality from cardiovascular disease	9 (16%)
Cardiometabolic risk during pregnancy [6]	2 (3%)

1. Includes pre/post study design, time series and natural experiments
2. Children defined as ≤18 years of age
3. Does not equal to 100% as studies report multiple cardiometabolic outcomes
4. Includes hypertension, coronary artery disease/heart disease, angina, heart attack/myocardial infarction, metabolic syndrome, cardiac disease related blood markers, stroke
5. Diabetes, blood glucose, metabolic syndrome
6. Gestational diabetes, gestational hypertension, pre-eclampsia

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Table 3. Characteristics of included studies investigating the association between exposure to a disaster during the perinatal and childhood periods and cardiometabolic outcomes across the life-course, by disaster type (n=24)

Study	Study design	Country	Name of disaster	Year	Sample size	Primary exposure and comparator	Average follow-up	Outcomes	Primary results[1]
Human-made disaster with child/youth outcome									
Trasande, 2018 (47)	Prospective cohort study	USA	World Trade Center attacks (9/11)	2001	402	New York children and youth enrolled in the World Trade Center Health Registry (birthdates: Sept 11, 1993 to September 10, 2001) compared to individuals born during the same time period who were ineligible for enrollment in the WTCHR	2 years	Youth outcomes: 1) BMI (kg/m2) 2) zBMI 3) Triglycerides (mg/dL) 4) Cholesterol (mg/dL) 5) LDL (mg/dL) 6) HDL (mg/dL)	Regression coefficient and 95% CI: 1) BMI: -1.12 (-2.11, -0.12) 2) zBMI: -0.24 (95% CI: -0.49. -0.002) 3) logTrig: 0.02 (95% CI: -0.07, 0.12) 4) logChol (0.02 (95% CI: -0.02, 0.06) 5) log LDL: 0.06 (95% CI: -0.001, 0.12) 6) logHDL: -0.04 (95% CI: -0.10, 0.03)
Human-made disaster with adult outcome									
Bercovich, 2014 (40)	Cross-sectional	Israel	Holocaust	1941-1945	300	European Jews born in 1940-1945 with exposure to the holocaust compared to European Jews during the same time period born	N/A	Adult outcomes: 1) Hypertension 2) Diabetes 3) Dyslipidemia 4) Any cardiovascular disease	1) Adjusted OR: 2.2, 95%CI: 1.2-3.8 2) Adjusted OR: 2.2, 95%CI: 1.2-4.2 3) Adjusted OR: 3.1, 95%CI: 1.7-5.7 4) Adjusted OR: 2.6, 95%CI: 1.4-4.7
de Rooij, 2007 (30)	Cohort	Netherlands	Dutch Famine	1944-1945	783	Prenatal exposure to Dutch famine defined as people born between January 7, 1945 and December 8 1945 compared to people born before Jan 7 1945 or conceived after Dec 8 1945	58 years	Metabolic syndrome at age 58	Metabolic syndrome OR: 1.2; 95% CI: 0.9, 1.7
Ekamper, 2015 (31)	Cohort	Netherlands	Dutch Famine	1944-1945	41096	Male military conscripts born between Jan 1944 and 1946 and compared to military conscripts born before 1944 or after 1946	63 years	Adult outcomes: 1) Heart disease mortality 2) Cerebrovascular disease mortality 3) Diabetes mellitus mortality	Hazard ratio (HR): 1) HR: 0.94; 95% CI: 0.77, 1.15 2) HR: 1.55, 95% CI: 0.95, 2.51 3) HR: 1.61, 95% CI: 0.91, 2.86
Huang, 2010 (37)	Cohort	China	1959-1961 Chinese Famine	1959-1961	35,025	County level famine intensity for women born during 1957-1962 compared to women born post-famine in 1963	32 years	Adult outcomes at age 32: 1) BMI among rural sample 2) BMI among urban sample 3) Hypertension among rural sample 4) Hypertension among urban sample	1) Average effect = 0.92, 95% CI: 0.32, 1.51 2) Average effect = 0.03, 95% CI: -2.82, 2.87 3) Log odds = 1.23, 95% CI: -0.38, 2.84 4) Log odds = 0.37 95% CI: -2.07, 2.80
Hult, 2010 (36)	Cohort	Nigeria	Biafran Famine	1967-1970	1,339	Individuals exposed to famine during early childhood (born 1965-1967) or exposed to famine in fetal life and infancy (born 1968-Jan 1970) compared to people born between 1971 and 1973	~40 years	Adult outcomes at age ~40 years: 1) Hypertension 2) Diabetes 3) Overweight (BMI>25 kg/m^2) 4) Obesity (BMI>30kg/m^2)	Adjusted OR (95% CI) 1) Childhood exposure OR: 1.42, 95% CI: 0.63, 3.13; Fetal-infant exposure: OR: 2.50, 95% CI: 1.19, 5.26 2) Childhood exposure OR: 1.81, 95% CI: 0.64, 5.15; Fetal-infant exposure OR: 2.56, 95% CI: 0.92, 7.17 3) Childhood exposure OR: 1.02, 95% CI: 0.77, 1.34; Fetal-infant exposure: OR: 1.41, 95% CI: 1.03, 1.93 4) Childhood exposure OR: 1.20, 95% CI:

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									0.87, 1.67; Fetal-infant exposure: OR: 1.30, 95% CI: 0.92, 1.85
Lumey , 2012[1] (32)	Cohort	Netherlands	Dutch Famine	1944-1945	1075	Infants whose mothers were exposed to famine during or immediately preceding pregnancy (born Feb 1, 1945, March 31, 1946) compared to individuals born in the same hospital before or after famine	~56-62 years	Adult outcomes at 56-62 years Coronary artery disease	Early gestation HR: 1.26, 95%CI: 0.59, 2.70 Late gestation HR: 1.31, 95%CI: 0.67, 2.57
Painter, 2006 (33)	Cohort	Netherlands	The Dutch Famine	1944-1945	975	Infants who were born between January 1945 and December 1945 who were exposed to famine in utero compared to infants born before the famine (November 1943 and January 1945) and after the famine (December 1945 and February 1947)	~50-58 years	Adult outcomes at 50-58 years: Coronary artery disease	HR: 1.9, 95% CI: 1.0, 3.8
Ravelli, 1999[2] (34)	Cohort	Netherlands	The Dutch Famine	1944-1945	741	Infants exposed to famine during different periods of gestation (late, mid and early) whose maternal daily ration was <1000 kcal (born between January 1945 and December 1945) compared to those born not during the famine	50 years	Obesity adult outcomes at 50 years stratified by sex: 1) Weight (kg) 2) BMI (kg/m2) 3) Waist circumference (cm)	Mean difference (95% CI) between exposure during late or early gestation versus nonexposed: Men: 1) late: 0.8 (-3.1, 4.7); early: 1.5 (-3.5, 6.6) 2) late: 0.4 (-3.5, 4.5); early: 0.5 (-4.6, 6.0) 3) late: 1.8 (-1.4, 4.9); early: (1.8 (-2.4, 6.0) Women: 1) Late: -1.8 (-6.1, 2.5); early: 7.9 (2.5, 13.2) 2) late: -2.1 (-7.0, 3.1); early 7.4 (0.7, 14.5) 3) late: -0.7 (-4.4, 3.0); early: 5.7 (1.1, 10.3)
Roseboom, 2001[2] (29)	Cohort	Netherlands	The Dutch Famine	1944-1945	2414	Infants who were exposed to famine in utero whose mother had a daily ration <1000 calories during any 13-week period of gestion compared to infants who were born before or conceived after the famine period (before November 1943 or after February 1947)	~28 years	Adult outcomes at 28 years; 1) Plasma glucose (mmol/l) 2) Plasma insulin (pmol/l) 3) Total cholesterol (mmol/l) 4) HDL (mmmol/l) 5) LDL (mmol/l) 6) LDL/HDL cholesterol 7) BMI (kg/m) 8) CHD 9) Systolic BP (mmHg) 10) Diastolic BP (mmHg)	Mean values of outcomes for late gestation and early gestation: 1) Late: 6.3; early: 6.1 2) Late: 200; early: 207 3) Late: 5.83; early: 6.13 4) Late: 5.83; early: 6.13 5) Late: 1.32; early: 1.26* 6) Late: 3.87; early: 3.26* 7) Late: 26.7; early: 28.1 8) Late: 2.5; early: 8.8* 9) Late: 127.4; early: 123.4 10) Late: 86.4; early: 84.8 * p<0.05
Schrier, 2011 (41)	Cohort	Finland	Winter War and Continuation War	1939-1940 (Winter War), 1941-1944	13,039	Individuals in utero who were exposed to bombings that occurred for 48 days between 1934-1944 compared to those who were not exposed in utero	~60-70	Adult outcomes: 1) Coronary heart disease 2) Cerebrovascular disease	Results are shown graphically Higher CHD survival rates among women 64+ and among men aged 50-54 exposed while in utero

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				(Continuation War)					
Stein, 2006 (35)	Cohort	Netherlands	The Dutch Famine	1944-1945	971	Prenatal exposure to famine defined as the weeks post-last menstrual period that mother was exposed to an official ration of <900 kcal/week (gestation weeks: 1-10, 11-20, 21-30, or 31-delivery)	59	Adult outcomes: 1) systolic BP (mmHg) 2) Diastolic BP (mmHg) 3) Hypertension	Adjusted regression coefficients 1) 1-10 weeks: 1.20 (95% CI: -3.28, 5.69); 11-20 weeks: -1.19 (95% CI: -4.92, 2,55); 21-30 weeks: 1.33 (95% CI: -2.24, 4.90); 31-delivery: 2.02 (95% CI: -1.53, 5.57) 2) 1-10 weeks: 1.10 (95% CI: -1.36, 3.57); 11-20 weeks: -1.26 (95% CI: -3.32, 0.80); 21-30 weeks: 1.19 (95% CI: -0.78, 3.15); 31-delivery: 0.71 (95% CI: -1.24, 2.66) 3) 1-10 weeks: 1.14 (95% CI: 0.62, 2.11); 11-20 weeks: 0.98 (95% CI: 0.59, 1.65); 21-30 weeks: 1.23 (95% CI: 0.74, 1.05); 31-delivery: 1.42 (95% CI: 0.86, 2.35)
Natural disaster with pregnancy outcomes									
Oni, 2015 (27)	Cross-sectional	USA	Hurricane Katrina	2005	146	Women who were pregnant during Hurricane Katrina or became pregnant immediately after hurricane compared to those who were not exposed to the hurricane; Women who experienced prenatal stress caused by Hurricane Katrina compared to those who did not experience stress	9 months	Pregnancy related outcomes: 1) Pregnancy induced hypertension 2) Gestational diabetes	1) Hurricane exposure: adjusted OR: 1.22 (95% CI: 0.81, 1.84); perceived stress: adjusted OR: 1.16 (95% CI: 1.05, 1.30); 2) Hurricane exposure: adjusted OR: 1.04 (95% CI: 0.69,1.57); perceived stress: adjusted OR: 1.13 (95% CI: 1.02, 1.25)
Xiao, 2019 (28)	Time series/Quasi-experimental	USA	Hurricane Sandy	2012	Not reported	Exposure to Hurricane Sandy lasting impacts defined as the following 12 months after Sandy (November 2012-October 2013) compared to the November to October in other years during November 2005 to October 2014 among women who were pregnant	12 months	Outcomes in adults: 1) Emergency department visits for gestational hypertension 2) Emergency department visits for diabetes or abnormal glucose	1) Increased at 7 months: 7.3% (95% CI: 1.0%, 13.9%) 2) Increased at 8 months:26.3% (95% CI: 3.9%, 53.6%) *results for 12 months reported graphically
Natural disaster with child/youth outcomes									
Cao-Lei, 2016 (26)	Cohort	Canada	Quebec Ice Storm	1998	31	Negative cognitive appraisal of the impact of the ice storm among pregnant women compared to neutral or positive appraisal	13 years	Outcomes among children at age 13: 1) Central adiposity (waist to height ratio) 2) BMI (kg/m2)	Mean (standard deviation) 1) Exposed: 20.86 (3.73); unexposed: 22.84 (5.19) 2) Exposed: 0.43 (0.04); unexposed: 0.45 (0.06)
Dancause, 2012 (23)	Cohort	Canada	Quebec Ice Storm	1998	111	Higher objective PNMS scores compared to lower scores among women who were pregnant or conceived within 3 months of the storm	5.5 years	Childhood obesity at 5.5 years of age	OR: 1.37, 95% CI: 1.06-1.77
Dancause, 2013 (24)	Cohort	Canada	Quebec Ice Storm	1998	32	Higher objective hardship compared to lower hardship scores reported among pregnant women exposed to the storm	13.4 years	Childhood insulin secretion at 13 years of age	Insulin secretion: Adjusted linear regression standardized coefficient = 0.52, p<0.01
Dancause, 2015 (22)	Cohort	USA	Iowa Flood	2008	106	Higher reported measures of objective hardship and subjective distress compared to lower scores among pregnant women during the floods	2.5-4 years	Childhood outcomes: 1) Child BMI z-scores at age 2.5 2) Child BMI z-scores at age 4 3) Difference in BMI from age 2.5 to 4 4) Child adiposity	Beta coefficient (p-value) 1) -0.07 (p=0.56) 2) -0.22 (p=0.07) 3) 0.11 (p=0.41) 4) 0.00 (p=0.97) 5) -0.06 (p=0.72) 6) 0.03 (p=0.82)

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								(skinfolds) at age 2.5 5) Child adiposity (skinfolds) at age 4 6) Difference in adiposity from age 2.5 to 4	
Goudet, 2011 (15)	Cohort	Bangladesh	1998 Bangladesh Flood	1998	220	Maternal malnutrition among mothers of infants and young children following flood exposure defined as underweight (BMI<18.5 kg/m2) compared to normal (BMI>=18.5)	12 months	Child outcomes at 12-36 months of age: 1) Underweight (weight for age z-score <-2) 2) Stunted (height for age z-score < -2) 3) Wasted (weight for height z-score < -2)	1) Adjusted OR = 3.509, 95%CI: 1.022,12.048) 2) Adjusted OR: 4.447, 95%CI: 1.044,18.943 3) Adjusted OR: 2.097, 95%CI: 0.507, 8.671
Kroska, 2018 (21)	Longitudinal study	USA	Iowa Flood	2008	103	Levels of maternal stress among those exposed to Iowa floods	2.5 years	Children outcomes at 2.5 years: BMI (kg/m2)	Standardized coefficient: 0.2071 (p=0.0322)
Liu, 2016 (25)	Longitudinal study	Canada	Quebec Ice Storm	1998	52-111 at different time points	Levels of maternal stress (objective hardship and subjective stress) among those exposed to Iowa floods	5.5-15.5 years	Children outcomes at 5.5-15.5 years: 1) BMI (kg/m2) 2) Waist to height ratio	Correlation R (p-value) Age 8.5 1) Objective hardship: 0.21 (p=0.05) 2) Objective hardship: 0.23 (p=0.03) Age 15.5 1) Objective hardship: 0.34 (p=0.02) 2) Objective hardship: 0.44 (p<0.01)
Natural disaster with adult outcomes									
Mazumder, 2010[23] (38)	Cohort	USA	1918 Influenza Pandemic	1918-1919	101,068	Infants who were born during the Influenza pandemic (third and fourth quarter of 1918, and the first, second and third quarter of 1919) compared to those born in the last quarter of 1919	~ 60-82 years	Adult outcomes at 60-82 years: 1) Diabetes 2) Heart disease	Excess cases of diabetes/heart disease: 1) 1918 Q4: 7.7% excess (95% CI: -10.6, 25.9); 1919 Q1: -5.2 (95% CI: -22.9, 12.5); 1919 Q2: 36.7% excess (95% CI: 18.9, 54.4); 2) 1918 Q4: 4.6% excess (95% CI: -4.3, 13.5); 1919 Q1: 10.9% excess (95% CI: 2.3, 19.6); 1919 Q2: 6.4% excess (95% CI: -2.2, 15.1);
Mryskyla, 2013[3] (39)	Cohort	USA	1918 Influenza Pandemic	1918-1919	81,571	Infants who were born during the Influenza pandemic (born during different quarters of 1917, 1918 and 1919) compared to those born in 1920-1924	~ 63-95 years	Adult outcomes at 63-95 years: Cardiovascular mortality	1918 Q1: HR: 1.05 (95% Ci: 0.94, 1.17); 1918 Q2: HR: 1.02 (95% Ci: 0.91, 1.14); 1918 Q3 HR: 0.99 (95% CI: 0.89, 1.10); 1918 Q4 HR: 0.97 (95% Ci: 0.87, 1.09); 1919 Q1 HR 1.07 (95% CI: 0.96, 1.19); 1919 Q2: HR:1.06 (95% CI: 0.95, 1.19);
Sotomayer, 2013 (42)	Cohort Natural experiment	Puerto Rico	Hurricane San Felipe and San Cipiran	1928 and 1932	11990	Those born during 1929 and 1933 were defined as exposed to the hurricanes compared to individuals born outside of these years between 1920 and 1940	Not reported (average age = 70)	Outcomes at ~70 years of age: 1) Diabetes 2) Hypertension 3) High cholesterol 4) CVD 5) AMI 6) Coronary/angina 7) Stroke	Linear regression estimates (p-value) 1) San Felipe: 5.94 (p<0.01); San Ciprian: 5.43 (p<0.01) 2) San Felipe: 4.73 (p<0.01); San Ciprian: 6.39 (p<0.01) 3) San Felipe: 8.85 (p<0.01); San Ciprian: 5.28 (p<0.01) 4) San Felipe: -1.48; San Ciprian: 1.33 5) San Felipe: 0.81; San Ciprian: 3.26 (p<0.01

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									6) San Felipe: 0.40; San Ciprian: -0.60 7) San Felipe: -0.25; San Ciprian: 0.58

1. Results are numbered to correspond with the numbered outcomes in the outcomes column
2. Only presenting results for early and late gestation; results for mid gestation are not included in summary table but can be found in studies
3. Not all results presented for different exposure groups

Abbreviations: AMI: Acute myocardial infarction; BMI: Body mass index; BP: Blood pressure; CI: Confidence Interval; CHD: Coronary heart disease; Chol: Cholesterol; CVD: Cardiovascular disease; HDL: High-density lipoproteins; HR: Hazard ratio; LDL: Low-density lipoproteins OR: Odds Ratio; PMNS: Prenatal Maternal Stress; Q1-Q4: quarter; SE: Standard error; Trig; Triglycerides.

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Table 4. Description of studies investigating the association between exposure to a disaster during adulthood and cardiometabolic outcomes across the life-course, by disaster type (n=34)

Study	Study design	Country	Name of disaster	Year	Sample size	Primary exposure and comparator	Average follow-up	Outcomes	Primary results[1]
Human-made disaster									
Brackbill, 2006 (70)	Cohort	USA	World Trade Center attacks (9/11)	2001	8,418	Adult survivors of 9/11 present at time of first airplane impact in a structure that was damaged compared to those that collapsed; time of evacuation before compared to after damage	1 year	1) Hypertension 2) Coronary heart disease 3) Angina 4) Heart attack 5) Diabetes 6) Stroke	1) Building type: adjusted OR: 1.2 (p<0.05); time of evacuation: adjusted OR: 0.9 (0.6, 1.3) 2) Building type: adjusted OR: 0.8 (0.4, 1.6); time of evacuation: adjusted OR: 0.5 (0.1, 2.2) 3) Building type: adjusted OR: 0.8 (0.4, 1.6); time of evacuation: adjusted OR: 0.7 (0.2, 3.1) 4) Building type: adjusted OR: 2.1 (0.9, 4.9); 5) Time of evacuation: adjusted OR: 0.7 (0.3, 1.7) 6) Building type: adjusted OR: 1.5 (0.6, 4.0)
Dirkzwager, 2007 (53)	Cohort	Netherlands	Fireworks depot explosion	2000	896	PTSD among those exposed to the fireworks disaster 19 months following the disaster compared to those with no PTSD exposed to the fireworks explosion	18 months	1) Cardiovascular 2) Vascular problems	1) Physical health problems OR: 1.23; 95% CI: 0.78, 1.94; New health problems (not present pre disaster): 1.11; 0.65, 1.89 2) Physical health problems OR: 2.12 95% CI: 1.23, 3.68; 1.92; New health problems (not present pre disaster) OR: 1.92, 95% CI: 1.04, 3.55.
Dorn, 2007 (43)	Cohort	Netherlands	Volendam Pub Fire	2001	2255	Parents of children with burns from fire parents of children without burns, bereaved parents compared to community controls who were not trapped in fire	4 years	Incidence of hypertension	Bereaved parents: OR: 2.42, 95%CI: 0.90, 6.55); parents of victims with burns: OR: 1.43, 95%CI: 0.97, 2.11; parents of victims without burns: OR:1.44, 95%CI: 0.92, 2.26)
Gerin, 2005 (71)	Pre/post design/quasi experimental	USA	World Trade Center attacks (9/11)	2001	528	Adults 2 months before 9/11 compared to 2 months after 9/11 across 4 cities (Chicago, Washington DC, New York, and Mississippi)	4 months	Systolic blood pressure	Difference (SE) New York: 1.58 (0.82) p<0.05 Chicago: 2.15 (0.32) p<0.001 Mississippi: 2.92 (0.67) p<0.001 Washington DC: 8.67 (1.16) p<0.001
Huizink, 2006 (62)	Cohort	Netherlands	Amsterdam Air Disaster	1992	1996	Police officers and firefighters who performed at least one disaster-related task compared to professional colleagues who did not perform any disaster-related tasks	8.5 years	Cardiovascular complaints	Adulthood outcomes Police officers: OR: 1.76 (95% CI: 1.35, 2.29) Firefighters: OR: 3.3 (95% CI: 1.70, 6.41)
Jordan, 2011A (57)	Prospective cohort	USA	World Trade Center attacks (9/11)	2001	39,324	9/11-related PTSD compared to no PTSD	2.9 years	Heart disease	Women adjusted OR: 1.68 (95% CI: 1.33, 2.12) Men adjusted OR: 1.62 (95% CI: 1.34, 1.96)
Jordan, 2011B (63)	Prospective cohort study	USA	World Trade Center attacks (9/11)	2001	39324	Low, intermediate and high exposure to 9/11	2.9 years	Heart disease mortality	Intermediate exposure: HR: 1.21 (95% CI: 80, 1.83) High exposure: HR: 2.06 (95% CI: 1.10, 3.86)
Jordan, 2013[2] (49)	Cohort	USA	World Trade Center attacks (9/11)	2001	46,346	Low, intermediate and high exposure to 9/11	7 years	CVD hospitalizations	Rescue/recovery workers: women: high: adjusted HR: 3.29 (95% CI: 0.85, 12.69); men: high: 1.82 (95% CI: 1.06, 3.13) Non-rescue/recovery workers: women: high: adjusted HR: 0.88 (95% CI: 0.54, 1.43); men: high: adjusted HR: 0.94 (95% CI: 0.60, 1.47)

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Kong, 2019 (50)	Pre/post design/quasi experimental	South Korea	Sewol Ferry Disaster	2014	73,632	Exposure to the Sewol Ferry Disaster in one-week periods from May 21 through June 17, 2014 compared to the reference period (March 2015-April 2015)	8 weeks	Adulthood outcomes 1) Acute MI 2) Angina	1) 8 weeks after Sewol: IRR: 0.91 (95% CI: 0.81, 1.02) 2) 8 weeks after Seowl: IRR: 0.93 (95% CI: 0.85, 1.01)
Lin, 2010 (68)	Pre/post design/quasi experimental	USA	World Trade Center attacks (9/11)	2001	Not reported	Areas affected by 9/11 compared to areas not affected by 9/11	10 years	Adulthood outcomes for Cardiovascular disease hospitalizations	Prevalence ratio (95% CI): 08/14–09/10: 0.51 (95% CI: 0.26, 1.00) 09/11–09/17: 0.56 (95% CI: 0.28, 1.11); 09/18-09/24: 0.77 (95% CI: 0.44, 1.32); 09/25-10/01: 0.49 (95% CI: 0.24, 1.00); 10/02-10/08: 0.98 (95% CI: 0.53, 1.87); 10/09-10/15: 1.09 (95% CI: 0.60, 1.98); 10/16–10/22: 0.50 (95% CI: 0.26, 0.95); 10/23–10/29: : 0.45 (95% CI: 0.20, 0.98); 10/30–11/05: 0.48 (95% CI: 0.23, 0.97)
Yu, 2018 (69)	Cohort Study	USA	World Trade Center attacks (9/11)	2001	42,527	9/11-related PTSD compared to no PTSD	13 years	Stroke	Adjusted HR: 1.69 (95% CI: 1.42, 2.02)
Natural disaster									
An, 2015 (67)	Cross-sectional	USA	Hurricane Ike	2008	19	Psychological strains among Hurricane Ike survivors	3 months	1) Blood glucose (mg/dL) 2) Obesity (BMI; kg/m2)	Mean (high vs low) and standard deviation: PTSD symptom: 22.44 (4.93) vs.12.86 (10.48); p=0.014; perceived stress:23.00 (5.03) vs. 28.11 (5.07) p=0.048 2) 28.43 kg/m^2 (3.92) vs. 20.83kg/m^2 (3.92) p=0.018
Baum, 2019 (14)	Cohort study	USA	Hurricane Sandy	2012	81 544	Veterans who used Manhattan VA Medical Center before Hurricane Sandy and experienced decreased access to health care services compared to veterans who used the VA Bronx, Brooklyn or West Haven medical centers	2 years	1) Uncontrolled hypertension 2) Systolic BP (mmHg) 3) Diastolic BP (mmHg) 4) Uncontrolled diabetes 5) Uncontrolled cholesterol (mg/dL) 6) Weight (lbs)	% differential change (95% CI): 1) 6 months: 19.3 (4.5, 8.7); 12 months: 4.5 (3.1, 5.9); 18 months: 5.0 (3.5, 6.5); 24 months: 2.1 (0.5, 3.6) 2) 6 months: 3.8 (3.1, 4.5); 12 months: 2.3 (1.7, 2.9); 18 months: 3.1 (2.5, 3.7); 1.5 (0.9, 2.1) 3) 6 months: 2.7 (2.3, 3.1); 12 months: 2.2 (1.9, 2.6); 18 months 2.9 (2.5, 3.3), 24 months: 2.0 (1.7, 2.4) 4) 6 months: 1.9 (-0.1, 4.0); 12 months: 1.7 (-0.3, 3.6); 18 months: 0.8 (-1.2, 2.8); 24 months: -0.2 (-2.2, 1.8) 5) 6 months: 1.3 (-0.1, 2.6); 12 months: 0.6 (-0.6, 1.8); 18 months: -0.7 (-2.0, 0.6); 24 months: -0.2 (-1.4, 1.0) 6) 6 months: -0.1 (-0.5, 0.2); 12 months: 0.2 (-0.2, 0.5); 18 months: -0.2 (-0.5, 0.2); 24 months: 0.5 (0.1, 0.9)
Becquart, 2018 (48)	Time series/quasi experimental	USA	Hurricane Katrina	2005	383 552	Exposure to hurricane before, during and after among older adults in Louisiana in the affected counties	1 year	Hospitalizations due to CVD	Mean (SD) Orleans: T1: 7.25 (2.44); T2: 3.91 (1.45)*; T3; 18.47 (17.3)*; T4:13.76 (6.51)*; T5: 9.54 (2.78); T6: 4.69 (2.08) Jefferson: T1: 5.90 (1.90); T2: 5.01(1.52); T3: 8.118 (3.70)*; T4: 7.25 (2.15)*; T5: 5.26 (1.53); T6: 4.65 (1.57)* East BR: T1: 8.69 (2.74); T2: 9.11 (2.69); T3: 6.52 (2.58); T4: 6.55 (1.70)*; T5: 6.69 (2.42)*; T6: 7.39

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									(2.37)* * p<0.05
Bich, 2011 (61)	Cross-sectional	Vietnam	Historic flood in 2008	2008	781	Individuals who resided in households affected by flood in Hanoi in 2005 compared to non-affected households	1 month	Worsening hypertension after rain/flood	Rural: non flooded 33.3%; flooded: 51.2%; Urban: non flooded 20.3% flooded: 42.9%* * p<0.05
Fonseca, 2009 (59)	Cohort	USA	Hurricane Katrina	2005	1795	Adults with diabetes who were in the databases from 3 health care systems 6 months before the hurricane (Feb 28, 2005-Aug 27, 2005) compared to 6-16 months after the hurricane (March 1, 2006-December 31, 2006)	22 months	1) Glycemic control/A1C 2) Systolic BP (mmHg) 3) Diastolic BP (mmHg) 4) HDL (mg/dL) 5) LDL (mg/dL) 6) Triglycerides (mg/dL)	Difference in mean (SD) 1) 0.1 (1.6) (p<0.01) 2) 10.5 (20.4) (p<0.01) 3) 3.9 (13.1) (p<0.01) 4) 6.0 (35.5) (p<0.01) 5) -2.4 (9.2) (p<0.01) 6) -2.1 (137.5) (p=0.60)
Gautam, 2009 (55)	Retrospective cohort	USA	Hurricane Katrina	2005	396	Exposure to Hurricane Katrina compared to period before hurricane	4 years	Incidence of AMI admission	Pre-Katrina group: 150 admissions for AMI (0.71%) Post-Katrina group: 246 admission for AMI (2.18%) p<0.0001
Hendrickson, 2996 (64)	Pre/post design/quasi experimental	USA	Hurricane Iniki	1992	Not reported	Mortality data for residents of Kauai for 5-year period 1987-1991 prior disaster compared to the year immediately following the hurricane (Oct 1 1992-Sept 30, 1993)	6 years total	Mortality by: 1) Heart disease 2) Stroke 3) Diabetes mellitus	1) RR: 0.96 (95% CI: 0.79-1.17) 2) RR: 1.20 (95% CI: 0.81-1.78) 3) RR: 2.61 (95% CI: 1.44-4.74)
Husarewycz, 2014 (72)	Cross-sectional	USA	Natural disaster/terrorism	Lifetime disaster experience	34,653	Number of times directly experienced natural disaster/terrorism compared to no experiences	1 year	1) Cardiovascular disease 2) Hypertension/arteriosclerosis 3) Diabetes 4) Obesity	1) OR: 1.28 (95% CI: 1.10, 1.49) 2) OR: 1.08 (95% CI: 0.95, 1.24) 3) OR: 1.10 (95% CI: 0.94, 1,29) 4) OR: 1.01 (95% CI: 0.90, 1.14)
Jiao, 2012 (54)	Retrospective cohort observational study	USA	Hurricane Katrina	2005	Not reported	2 years prior to the hurricane (August 29, 2003 - August 28, 2005) compared to the 3-year period post-Hurricane Katrina (February 14, 2006 - February 13, 2009)	5 years	Incidence of AMI	Pre-Katrina: 0.7% compared to post-Katrina: 2% (p<0.001)
Joseph, 2014 (51)	Cohort/longitudinal	USA	Hurricane Katrina	2005	215	African Americans who experienced acute unemployment due to Hurricane compared to those who remained employed	4 years	Cardiometabolic event	Adjusted OR = 5.65, p < .05
Karatzias , 2015 (44)	Cross-sectional	Hong Kong	Natural disaster	Not specified	1147	Experience of natural disaster across life-course compared to less or no experiences	Survey done from August to December 2012	1) Hypertension 2) Heart disease 3) Diabetes	Chi square (p-value) 1) X^2: 3.3 p=0.047 2) X^2: 3.6 p=0.056 3) X^2: 2.5 p=0.088
Kim, 2017 (58)	Pre/post design/quasi experimental	USA	Hurricane Sandy	2012	Not reported	The month of Hurricane Sandy (October 28, 2012-November 27, 2012) compared to the same month in 2009-2011; Sandy quarter (October 28, 2012-January 27, 2013) compared to the same period in 2009-2011 among elderly people	Sandy month: Oct 28, 2012-Nov 27, 2012 Sandy quarter: Oct 28, 2012-Jan 27, 2013	Cardiovascular disease-related death	Sandy quarter: adjusted RR: 1.06; 95% CI: 1.02, 1.10 Sandy month: adjusted RR: 1.10; 95% CI: 1.02, 1.18
Koroma, 2019 (66)	Cross-sectional	Sierra-Leone	Ebola	2014-2015	10011	District facilities for six-month periods before Ebola (June-December 2012), during Ebola (June-December 2014) and post-Ebola (June-December 2015)	June-December 2012,2013,2014	1) Cardiovascular disease 2) Hypertension 3) Diabetes	Number of people with non-communicable diseases 1) Pre-Ebola: 355, Ebola: 300, Post-Ebola: 196 2) Pre-Ebola: 282, Ebola: 230, Post-Ebola: 457 3) Pre-Ebola: 3716, Ebola: 1851, Post-Ebola: 2463

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Lawrence, 2019 (45)	Prospective Cohort Study	USA	Superstorm Sandy	2012	651858	Residing in counties affected by Superstorm Sandy compared to non-affected counties; Superstorm period compared to reference periods (short-term and long-term (4 and 12 months))	1 year	Emergency department visits, outpatient visits, and hospital admissions for cardiovascular disease	4 months: Superstorm sandy period: RR: 2.10 (95% CI: 2.10, 2,10); Affected counties RR: 2.62 (95% CI: 2.62, 2.63) 12 months: Superstorm sandy period: RR: 2.01 (95% CI: 2.00, 2.01); Affected counties RR: 2.64 (95% CI: 2.64, 2.65)
McKinney, 2011 (52)	Time-series/quasi experimental	USA	Hurricane Charley, Frances, Ivan and Jeanne and Tropical Storm Bonnie	2004	Not reported	Counties in 2004 directly impacted by the hurricanes, ordered evacuated regardless of the level of damage that occurred and adjacent to the impact zone where direct deaths were reported compared to compared to the same areas in 2001-2006	5 years	Heart-related mortality	Results shown graphically Significantly elevated heart-related deaths
Moscona, 2019 (75)	Retrospective cohort study	USA	Hurricane Katrina	2005	2-year pre-Katrina - 21,079 10-year post Katrina - 84,751	Individuals who lived in New Orleans who went to the Tulane University Health Sciences Center compared to the two months prior to the Hurricane	12 years	1) Hospital admission for Incidence of AMI 2) Changes in CAD 3) Changes in diabetes mellitus 4) Changes in hypertension 5) Changes in hyperlipidemia	Pre-Katrina versus Post-Katrina 1) 0.7% vs 2.8% (p<0.001) 2) 36.4% vs. 47.9%, (p= 0.01) 3) 31.3% vs. 39.9% (p= 0.04) 4) 71.1% vs 80.6%, (p=0.12) 5) 45.4% vs. 59.3% (p = 0.005)
Nagayoshi, 2015 (56)	Pre/post design/quasi experimental	Japan	July 12, 2012 heavy rain and mudslides "mountain tsunamis"	2012	583	Individuals who were admitted at Aso Central Hospital from July 12 to August 31, 2012 compared to the 3-year period before flooding	3 years	1) Hospital admission for cardiovascular outcomes 2) Cardiovascular events	1) 4.5/month before compared to 16.8/month after; p < 0.01 2) 5.1/month before compared to 16.8/month after; p < 0.01).
Ng, 2011 (73)	Cohort	United Kingdom	Flood	June, 2007	1,743	Diabetics affected by floods compared to diabetics not affected by floods	2 years	Glycemic control/HbA1c levels	Mean HbA1c before 7.6% (7.5–7.7) vs. After 7.9% (7.7–8.0); p = 0.002
Peters, 2013 (65)	Retrospective Cohort	USA	Hurricane Katrina	2005	698	Admission to Tulane University Health Sciences Centre in the 3-year period post-Katrina compared to the 6-year period pre-Katrina	9 years	Chronobiology of AMI onset	Pre-Katrina: 45% vs post-Katrina: 30.9%, p=0.002
Rey, 2007 (74)	Longitudinal	France	6 Heat Waves	1971-2003	Not reported	Time of heat wave compared to the expected mortality during the 3 years prior to the heat wave	N/A	Excess cardiovascular disease death	41% in 1975 to 23% in 2003
Sliva-Palacios, 2015 (60)	Pre/post design/quasi experimental	USA	Oklahoma Tornado	2013	22,607	Victims of the Oklahoma Tornado Outbreaks compared to the same people pre-tornado and same period one year prior	6 Months	Hospital admissions for CVE	One year prior: PR = 1.05 95% CI: 0.91 to 1.21, p = 0.50; 3-month pre-tornado: PR = 0.96, 95% CI: 0.83 to 1.21, p = 0.63
Thethi, 2010 (76)	Cohort	USA	Hurricane Katrina	2005	1523	Individuals exposed to Hurricane Katrina compared to 6-16 months pre-Hurricane Katrina (February 28, 2005-August 27, 2005)	6 months before Katrina and 6-16 months after Katrina and follow up 1 year after the first post-Katrina visit	1) LDL (mg/dL) 2) HDL (mg/dL) 3) Triglycerides (mg/dL) 4) Cholesterol (mg/dL) 5) Diastolic blood pressure (mm/Hg) 6) Systolic blood pressure (mm/Hg)	Mean Pre-Katrina vs Post-Katrina: 1) 101.34 vs 107.44 2) 43.53 vs 41.08 3) 160.8 vs 158.65 4) 181.9vs 181.39 5) 70.99 vs 74.88 6) 130.73 vs 141.27

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Vanasse, 2016 (46)	Population-based retrospective cohort study with a time series design	Canada	Flood of Saint-Jean-sur-Richelieu	2011	111,317	Exposure to flood in spring 2011 and exposure to flooded area (Area 1) compared to same period in Spring 2010 and 2012 and non-flooded areas in the same town (Areas 2, 3 and 4)	4 months	Acute cardiovascular event	Spring 2010: aOR 1.25 (95%CI: 0.81, 1.92); spring 2012 aOR: 1.27 (95% CI: 0.82, 1.92); Non-flooded areas 2: aOR: 1.11 (95%CI: 0.79, 1.59), Non-flooded area 3: aOR: 0.94 (95% CI: 0.68, 1.32); Non-flooded area 4: aOR 1.08 (95%CI: 0.78, 1.47)

1. 1. Results are numbered to correspond with the numbered outcomes in the outcomes column
2. Only results for extreme outcomes are reported in table, remaining results can be found in the study.

Abbreviations: aOR: Adjusted odds ratio; AMI: Acute myocardial infarction BMI: Body mass index; BP: Blood pressure; CI: Confidence Interval; CAD: Coronary artery disease; CHD: Coronary heart disease; CVD: Cardiovascular disease; CVE: Cardiovascular events; HbA1c ; Hemoglobin A1C; HDL: High-density lipoproteins; HR: Hazard ratio; LDL: Low-density lipoproteins OR: Odds Ratio; PR: Prevalence Ratio; PTSD: Post-traumatic stress disorder; RR: Relative risk; SD: Standard deviation; SE: Standard Error; Trig; Triglycerides.; T1-T6; Time.

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Figure 1. PRISMA flow diagram

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1. World Health Organization. WHO | Definitions: emergencies [Internet]. WHO. World Health Organization; 2020 [cited 2020 May 12]. Available from: https://www.who.int/hac/about/definitions/en/

2. Mohamed SI. An overview on disasters. Disaster Prevention and Management: An International Journal. 2007 Jan 1;16(5):687–703.

3. Giorgadze T, Maisuradze I, Japaridze A, Utiashvili Z, Abesadze G. Disasters and their consequences for public health. Georgian Med News. 2011 May;(194):59–63.

4. Ordway D-M, February 1 JR, 2016. Global trends in human infectious disease: Rising number of outbreaks, fewer per-capita cases [Internet]. Journalist's Resource. 2016 [cited 2020 May 8]. Available from: https://journalistsresource.org/studies/society/public-health/global-rise-human-infectious-disease-outbreaks/

5. Brooks SK, Webster RK, Smith LE, Woodland L, Wessely S, Greenberg N, et al. The psychological impact of quarantine and how to reduce it: rapid review of the evidence. The Lancet. 2020 Mar 14;395(10227):912–20.

6. Bell SA, Horowitz J, Iwashyna TJ. Health Outcomes After Disaster for Older Adults With Chronic Disease: A Systematic Review. Gerontologist [Internet]. [cited 2020 May 8]; Available from: https://academic.oup.com/gerontologist/advance-article/doi/10.1093/geront/gnz123/5565119

7. Jackson JL, George S, Hinchey S. Medically Unexplained Physical Symptoms. J Gen Intern Med. 2009 Apr;24(4):540–2.

8. Chan EYY, Sondorp E. Medical interventions following natural disasters: missing out on chronic medical needs. Asia Pac J Public Health. 2007;19 Spec No:45–51.

9. life course approach to chronic disease epidemiology: conceptual models, empirical challenges and interdisciplinary perspectives | International Journal of Epidemiology | Oxford Academic [Internet]. [cited 2021 Feb 28]. Available from: https://academic.oup.com/ije/article/31/2/285/617688

10. Kuh D, Ben-Shlomo Y, Lynch J, Hallqvist J, Power C. Life course epidemiology. Journal of Epidemiology & Community Health. 2003 Oct 1;57(10):778–83.

11. James SL, Abate D, Abate KH, Abay SM, Abbafati C, Abbasi N, et al. Global, regional, and national incidence, prevalence, and years lived with disability for 354 diseases and injuries for 195 countries and territories, 1990–2017: a systematic analysis for the Global Burden of Disease Study 2017. The Lancet. 2018 Nov;392(10159):1789–858.

12. Forouzanfar MH, Afshin A, Alexander LT, Anderson HR, Bhutta ZA, Biryukov S, et al. Global, regional, and national comparative risk assessment of 79 behavioural, environmental and occupational, and metabolic risks or clusters of risks, 1990–2015: a systematic analysis for the Global Burden of Disease Study 2015. The Lancet. 2016 Oct;388(10053):1659–724.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

13. Tan SY, Yip A. Hans Selye (1907–1982): Founder of the stress theory. Singapore Med J. 2018 Apr;59(4):170–1.

14. Baum A., Barnett M.L., Wisnivesky J., Schwartz M.D. Association between a Temporary Reduction in Access to Health Care and Long-term Changes in Hypertension Control among Veterans after a Natural Disaster. JAMA Network Open. 2019;2(11):e1915111.

15. Goudet S, Griffths P, Bogin BA. Mother's body mass index as a predictor of infant's nutritional status in the post-emergency phase of a flood. Disasters. 2011 Oct;35(4):701–19.

16. Moher D, Liberati A, Tetzlaff J, Altman DG, PRISMA Group. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. Ann Intern Med. 2009 Aug 18;151(4):264–9, W64.

17. Kamel M, Smith BT, Wahi G, Carsley S, Birken CS, Anderson LN. Continuous cardiometabolic risk score definitions in early childhood: a scoping review. Obesity Reviews. 2018;19(12):1688–99.

18. G B, G T, Kk N, V K, K V, A S, et al. Impact of major earthquakes on the incidence of acute coronary syndromes - A systematic review of the literature [Internet]. Vol. 59, Hellenic journal of cardiology : HJC = Hellenike kardiologike epitheorese. Hellenic J Cardiol; 2018 [cited 2020 Sep 10]. Available from: https://pubmed.ncbi.nlm.nih.gov/29807192/

19. Covidence - Better systematic review management [Internet]. Covidence. [cited 2020 Sep 10]. Available from: https://www.covidence.org/

20. Aromataris E, Munn Z, editors. JBI Manual for Evidence Synthesis [Internet]. JBI; 2020 [cited 2020 Sep 10]. Available from: https://wiki.jbi.global/display/MANUAL

21. Kroska EB, O'Hara MW, Elgbeili G, Hart KJ, Laplante DP, Dancause KN, et al. The impact of maternal flood-related stress and social support on offspring weight in early childhood. Arch Womens Ment Health. 2018 Apr 1;21(2):225–33.

22. Dancause KN, Laplante DP, Hart KJ, O'Hara MW, Elgbeili G, Brunet A, et al. Prenatal stress due to a natural disaster predicts adiposity in childhood: the Iowa Flood Study. Journal of obesity. 2015;2015(101526295):570541.

23. Dancause KN, Laplante DP, Fraser S, Brunet A, Ciampi A, Schmitz N, et al. Prenatal exposure to a natural disaster increases risk for obesity in 51/2-year-old children. Pediatric research. 2012;71(1):126–31.

24. Dancause KN, Veru F, Andersen RE, Laplante DP, King S. Prenatal stress due to a natural disaster predicts insulin secretion in adolescence. Early Hum Dev. 2013 Sep;89(9):773–6.

25. Liu GT, Dancause KN, Elgbeili G, Laplante DP, King S. Disaster-related prenatal maternal stress explains increasing amounts of variance in body composition through childhood and adolescence: Project Ice Storm. Environ Res. 2016;150:1–7.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

26. Cao-Lei L, Dancause KN, Elgbeili G, Laplante DP, Szyf M, King S. Pregnant women's cognitive appraisal of a natural disaster affects their children's BMI and central adiposity via DNA methylation: Project Ice Storm. Early Human Development. 2016 Dec 1;103:189–92.

27. Oni O, Harville E, Xiong X, Buekens P. Relationships among stress coping styles and pregnancy complications among women exposed to Hurricane Katrina. Journal of obstetric, gynecologic, and neonatal nursing : JOGNN. 2015;44(2):256–67.

28. Xiao J, Huang M, Zhang W, Rosenblum A, Ma W, Meng X, et al. The immediate and lasting impact of Hurricane Sandy on pregnancy complications in eight affected counties of New York State. The Science of the total environment. 2019;678(uj0, 0330500):755–60.

29. Roseboom TJ, van der Meulen JH, Ravelli AC, Osmond C, Barker DJ, Bleker OP. Effects of prenatal exposure to the Dutch famine on adult disease in later life: an overview. Mol Cell Endocrinol. 2001 Dec 20;185(1–2):93–8.

30. de Rooij SR, Painter RC, Holleman F, Bossuyt PM, Roseboom TJ. The metabolic syndrome in adults prenatally exposed to the Dutch famine. Am J Clin Nutr. 2007 Oct 1;86(4):1219–24.

31. Ekamper P, van Poppel F, Stein AD, Bijwaard GE, Lumey LH. Prenatal Famine Exposure and Adult Mortality From Cancer, Cardiovascular Disease, and Other Causes Through Age 63 Years. Am J Epidemiol. 2015 Feb 15;181(4):271–9.

32. Lumey LH, Martini LH, Myerson M, Stein AD, Prineas RJ. No relation between coronary artery disease or electrocardiographic markers of disease in middle age and prenatal exposure to the Dutch famine of 1944-5. Heart. 2012 Nov;98(22):1653–9.

33. Painter RC, de Rooij SR, Bossuyt PM, Simmers TA, Osmond C, Barker DJ, et al. Early onset of coronary artery disease after prenatal exposure to the Dutch famine. Am J Clin Nutr. 2006 Aug;84(2):322–7; quiz 466–7.

34. Ravelli AC, van Der Meulen JH, Osmond C, Barker DJ, Bleker OP. Obesity at the age of 50 y in men and women exposed to famine prenatally. Am J Clin Nutr. 1999 Nov;70(5):811–6.

35. Stein AD, Zybert PA, van der Pal-de Bruin K, Lumey LH. Exposure to famine during gestation, size at birth, and blood pressure at age 59 y: evidence from the Dutch Famine. Eur J Epidemiol. 2006;21(10):759–65.

36. Hult M, Tornhammar P, Ueda P, Chima C, Bonamy A-KE, Ozumba B, et al. Hypertension, diabetes and overweight: looming legacies of the Biafran famine. PLoS ONE. 2010 Oct 22;5(10):e13582.

37. Huang C, Li Z, Wang M, Martorell R. Early life exposure to the 1959-1961 Chinese famine has long-term health consequences. J Nutr. 2010 Oct;140(10):1874–8.

38. Mazumder B, Almond D, Park K, Crimmins EM, Finch CE. Lingering prenatal effects of the 1918 influenza pandemic on cardiovascular disease. J Dev Orig Health Dis. 2010 Feb;1(1):26–34.

39. Myrskyla M, Mehta NK, Chang VW. Early life exposure to the 1918 influenza pandemic and old-age mortality by cause of death. American journal of public health. 2013;103(7):e83-90.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

40. Bercovich E, Keinan-Boker L, Shasha SM. Long-Term Health Effects in Adults Born during the Holocaust. 2014;16:5.

41. Schreier NK, Moltchanova EV, Blomstedt PA, Kajantie E, Eriksson JG. Prenatal exposure to wartime stress – long-term effect on coronary heart disease in later life. Annals of Medicine. 2011 Nov 1;43(7):555–61.

42. Sotomayor O. Fetal and infant origins of diabetes and ill health: evidence from Puerto Rico's 1928 and 1932 hurricanes. Econ Hum Biol. 2013 Jul;11(3):281–93.

43. Dorn T, Yzermans CJ, Guijt H, van der Zee J. Disaster-related stress as a prospective risk factor for hypertension in parents of adolescent fire victims. American journal of epidemiology. 2007;165(4):410–7.

44. Karatzias T, Yan E, Jowett S. Adverse life events and health: A population study in Hong Kong. Journal of Psychosomatic Research. 2015 Feb;78(2):173–7.

45. Lawrence WR, Lin Z, Lipton EA, Birkhead G, Primeau M, Dong G-H, et al. After the Storm: Short-term and Long-term Health Effects Following Superstorm Sandy among the Elderly. Disaster med public health prep. 2019 Feb;13(1):28–32.

46. Vanasse A, Cohen A, Courteau J, Bergeron P, Dault R, Gosselin P, et al. Association between Floods and Acute Cardiovascular Diseases: A Population-Based Cohort Study Using a Geographic Information System Approach. International journal of environmental research and public health. 2016;13(2):168.

47. Trasande L, Koshy TT, Gilbert J, Burdine LK, Marmor M, Han X, et al. Cardiometabolic profiles of adolescents and young adults exposed to the World Trade Center Disaster. Environmental research. 2018;160(ei2, 0147621):107–14.

48. Becquart NA, Naumova EN, Singh G, Chui KKH. Cardiovascular Disease Hospitalizations in Louisiana Parishes' Elderly before, during and after Hurricane Katrina. International journal of environmental research and public health. 2018;16(1).

49. Jordan HT, Stellman SD, Morabia A, Miller-Archie SA, Alper H, Laskaris Z, et al. Cardiovascular disease hospitalizations in relation to exposure to the September 11, 2001 World Trade Center disaster and posttraumatic stress disorder. Journal of the American Heart Association. 2013;2(5):e000431.

50. Kong SY, Song KJ, Shin SD, Ro YS. Cardiovascular Events after the Sewol Ferry Disaster, South Korea. Prehospital and disaster medicine. 2019;34(2):142–8.

51. Joseph NT, Matthews KA, Myers HF. Conceptualizing health consequences of Hurricane Katrina from the perspective of socioeconomic status decline. Health psychology : official journal of the Division of Health Psychology, American Psychological Association. 2014;33(2):139–46.

52. McKinney N, Houser C, Meyer-Arendt K. Direct and indirect mortality in Florida during the 2004 hurricane season. International journal of biometeorology. 2011;55(4):533–46.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

53. Dirkzwager AJE, van der Velden PG, Grievink L, Yzermans CJ. Disaster-related posttraumatic stress disorder and physical health. Psychosomatic medicine. 2007;69(5):435–40.

54. Jiao Z, Kakoulides SV, Moscona J, Whittier J, Srivastav S, Delafontaine P, et al. Effect of Hurricane Katrina on incidence of acute myocardial infarction in New Orleans three years after the storm. The American journal of cardiology. 2012;109(4):502–5.

55. Gautam S, Menachem J, Srivastav SK, Delafontaine P, Irimpen A. Effect of Hurricane Katrina on the incidence of acute coronary syndrome at a primary angioplasty center in New Orleans. Disaster medicine and public health preparedness. 2009;3(3):144–50.

56. Nagayoshi Y, Yumoto S, Sakaguchi K, Shudo C, Takino S, Hashiyama M, et al. Heart attacks triggered by huge mud slides in mountain regions and severe flooding in inhabited areas. Journal of cardiology. 2015;65(2):117–20.

57. Jordan HT, Miller-Archie SA, Cone JE, Morabia A, Stellman SD. Heart disease among adults exposed to the September 11, 2001 World Trade Center disaster: results from the World Trade Center Health Registry. Preventive medicine. 2011;53(6):370–6.

58. Kim S, Kulkarni PA, Rajan M, Thomas P, Tsai S, Tan C, et al. Hurricane Sandy (New Jersey): Mortality Rates in the Following Month and Quarter. American journal of public health. 2017;107(8):1304–7.

59. Fonseca VA, Smith H, Kuhadiya N, Leger SM, Yau CL, Reynolds K, et al. Impact of a natural disaster on diabetes: exacerbation of disparities and long-term consequences. Diabetes care. 2009;32(9):1632–8.

60. Silva-Palacios F, Casanegra AI, Shapiro A, Phan M, Hawkins B, Li J, et al. Impact of tornadoes on hospital admissions for acute cardiovascular events. Thrombosis Research. 2015 Nov;136(5):907–10.

61. Bich TH, Quang LN, Ha LTT, Hanh TTD, Guha-Sapir D. Impacts of flood on health: epidemiologic evidence from Hanoi, Vietnam. Global health action. 2011;4(101496665):6356.

62. Huizink AC, Slottje P, Witteveen AB, Bijlsma JA, Twisk JWR, Smidt N, et al. Long term health complaints following the Amsterdam Air Disaster in police officers and fire-fighters. Occupational and environmental medicine. 2006;63(10):657–62.

63. Jordan HT, Brackbill RM, Cone JE, Debchoudhury I, Farfel MR, Greene CM, et al. Mortality among survivors of the Sept 11, 2001, World Trade Center disaster: results from the World Trade Center Health Registry cohort. Lancet (London, England). 2011;378(9794):879–87.

64. Hendrickson LA, Vogt RL. Mortality of Kauai Residents in the 12-Month Period following Hurricane Iniki. American Journal of Epidemiology. 1996 Jul 15;144(2):188–91.

65. Peters MN, Katz MJ, Moscona JC, Alkadri ME, Khazi Syed RH, Turnage TA, et al. Effect of Hurricane Katrina on chronobiology at onset of acute myocardial infarction during the subsequent three years. The American journal of cardiology. 2013;111(6):800–3.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

66. Koroma IB, Javadi D, Hann K, Harries AD, Smart F, Samba T. Non-communicable diseases in the Western Area District, Sierra Leone, following the Ebola outbreak. F1000Res [Internet]. 2019 Nov 19 [cited 2020 Jul 30];8. Available from: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6652098/

67. An K, Salyer J, Kao H-FS. Psychological strains, salivary biomarkers, and risks for coronary heart disease among hurricane survivors. Biological research for nursing. 2015;17(3):311–20.

68. Lin S, Gomez MI, Gensburg L, Liu W, Hwang S-A. Respiratory and cardiovascular hospitalizations after the World Trade Center disaster. Archives of environmental & occupational health. 2010;65(1):12–20.

69. Yu S, Alper HE, Nguyen A-M, Brackbill RM. Risk of Stroke Among Survivors of the September 11, 2001, World Trade Center Disaster. Journal of occupational and environmental medicine. 2018;60(8):e371–6.

70. Brackbill RM, Thorpe LE, DiGrande L, Perrin M, Sapp JH 2nd, Wu D, et al. Surveillance for World Trade Center disaster health effects among survivors of collapsed and damaged buildings. Morbidity and mortality weekly report Surveillance summaries (Washington, DC : 2002). 2006;55(2):1–18.

71. Gerin W, Chaplin W, Schwartz JE, Holland J, Alter R, Wheeler R, et al. Sustained blood pressure increase after an acute stressor: the effects of the 11 September 2001 attack on the New York City World Trade Center: Journal of Hypertension. 2005 Feb;23(2):279–84.

72. Husarewycz MN, El-Gabalawy R, Logsetty S, Sareen J. The association between number and type of traumatic life experiences and physical conditions in a nationally representative sample. General Hospital Psychiatry. 2014 Jan;36(1):26–32.

73. Ng J, Atkin SL, Rigby AS, Walton C, Kilpatrick ES. The effect of extensive flooding in Hull on the glycaemic control of patients with diabetes: Impact of flooding on glycaemic control. Diabetic Medicine. 2011 May;28(5):519–24.

74. Rey G, Jougla E, Fouillet A, Pavillon G, Bessemoulin P, Frayssinet P, et al. The impact of major heat waves on all-cause and cause-specific mortality in France from 1971 to 2003. Int Arch Occup Environ Health. 2007 May 2;80(7):615–26.

75. Moscona JC, Peters MN, Maini R, Katigbak P, Deere B, Gonzales H, et al. The Incidence, Risk Factors, and Chronobiology of Acute Myocardial Infarction Ten Years After Hurricane Katrina. Disaster med public health prep. 2019 Apr;13(02):217–22.

76. Thethi TK, Yau CL, Shi L, Leger S, Nagireddy P, Waddadar J, et al. Time to recovery in diabetes and comorbidities following Hurricane Katrina. Disaster medicine and public health preparedness. 2010;4 Suppl 1(101297401):S33-8.

77. Schneiderman N, Ironson G, Siegel SD. STRESS AND HEALTH: Psychological, Behavioral, and Biological Determinants. Annu Rev Clin Psychol. 2005;1:607–28.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

78. Committee on Post-Disaster Recovery of a Community's Public Health M, Policy B on HS, Medicine I of. Health Care [Internet]. Healthy, Resilient, and Sustainable Communities After Disasters: Strategies, Opportunities, and Planning for Recovery. National Academies Press (US); 2015 [cited 2021 Feb 16]. Available from: https://www.ncbi.nlm.nih.gov/books/NBK316524/

79. Puckrein GA, Egan BM, Howard G. Social and Medical Determinants of Cardiometabolic Health: The Big Picture. Ethn Dis. 25(4):521–4.

80. Schultz WM, Kelli HM, Lisko JC, Varghese T, Shen J, Sandesara P, et al. Socioeconomic Status and Cardiovascular Outcomes: Challenges and Interventions. Circulation. 2018 May 15;137(20):2166–78.

81. Mohammed SH, Habtewold TD, Birhanu MM, Sissay TA, Tegegne BS, Abuzerr S, et al. Neighbourhood socioeconomic status and overweight/obesity: a systematic review and meta-analysis of epidemiological studies. BMJ Open [Internet]. 2019 Nov 14 [cited 2020 Nov 17];9(11). Available from: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6886990/

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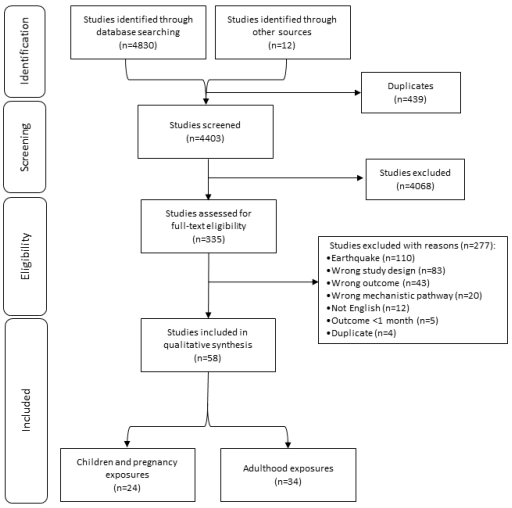


Figure 1. PRISMA flow chart

338x190mm (96 x 96 DPI)

APPENDIX

Figure A1. Search strategy for Medline

1	"social isolation.mp." or Social isolation/	1479
2	quaratine.mp. or exp Quarantine/	5589
3	Pandemics/	7247
4	Epidemics/	10146
5	Coronavirus or coronavirus.mp.	19310
6	humanitarian crises.mp.	232
7	exp mass casualty incidents/ or exp natural disasters/	18883
8	public health emergency.mp.	1636
9	cardiovasucalr.mp. or exp Cardiovascular Diseases/	2605399
10	hypertension.mp. or exp Hypertension/	484321
11	exp Obesity/ or obesity.mp.	321847
12	body mass index.mp. or exp Body Mass Index/	236149
13	stroke.mp, or exp Stroke/	310109
14	myocardial infarction.mp. or exp Myocardial Infarction/	247417
15	angina.mp. or exp Angina Pectoris/	698186
16	diabetes.mp. or Diabetes Mellitus/	646582
17	9 or 10 or 11 or 12 or 13 or 14 or 15 or 16	3552066
18	Disasters/ or disaster.mp.	39131
19	1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 or 18	10917
20	17 and 19	3688
21	limit 20 to humans	2998

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Table A2. Critical appraisal using Joanna Briggs Institute (JBI) checklist for cohort studies by exposure period (n=41)

Author, Year	Were the two groups similar and recruited from the same population?	Were the exposures measured similarly to assign people to both exposed and unexposed groups?	Was the exposure measured in a valid and reliable way?	Were confounding factors identified?	Were strategies to deal with confounding factors stated?	Were the groups/participants free of the outcome at the start of the study (or at the moment of exposure)?	Were the outcomes measured in a valid and reliable way?	Was the follow up time reported and sufficient to be long enough for outcomes to occur?	Was follow up complete, and if not, were the reasons to loss to follow up described and explored?	Were strategies to address incomplete follow up utilized?	Was appropriate statistical analysis used?
Perinatal											
Dancause, 2012	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Dancause, 2013	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No	Yes
Dancause, 2015	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Unclear	Yes
de Rooij, 2007	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Dorn, 2007	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Unclear	Yes
Ekamper, 2015	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Goudet, 2011	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Unclear	Unclear	Yes
Huang, 2010	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Unclear	Unclear	Yes
Hult, 2010	Yes	Yes	Yes	Yes	Yes	Unclear	Yes	Yes	Unclear	Unclear	Yes
Kroska, 2018	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Unclear	Yes
Lei, 2016	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Unclear	Yes
Liu, 2016	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Unclear	Yes
Lumey, 2012	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mazumder, 2010	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Unclear	Unclear	Yes
Mryskyla, 2013	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Unclear	Unclear	Yes

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Painter, 2006	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Ravelli, 1999	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Roseboom, 2001	Yes	Yes	Yes	Yes	Yes	Yes	Unclear	Yes	Yes	Yes	Yes
Schrier, 2010	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Unclear	Unclear	Yes
Sotomayer, 2013	Yes	Yes	Yes	Yes	Yes	Unclear	Yes	Yes	Unclear	Unclear	Yes
Stein, 2006	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adult											
Baum, 2019	Yes	Yes	Yes	Yes	Yes		Yes	Yes	Yes	Yes	Yes
Brackbill, 2006	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Unclear	Yes	Yes
Dirkzwager, 2007	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Unclear	Unclear	Yes
Fonseca, 2009	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Not applicable	Yes
Gautam, 2009	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Unclear	Unclear	Yes
Huizink, 2006	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes
Jiao, 2012	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Unclear	Unclear	Yes
Jordan, 2011 (A)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No	Yes
Jordan, 2011 (B)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Unclear	Yes	Yes
Jordan, 2013	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Unclear	Yes	Yes
Joseph, 2014	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Unclear	Unclear	Yes
Lawrence, 2019	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Unclear	Not applicable	Yes
Moscona, 2019	Yes	Yes	Yes	Yes	No	Unclear	Yes	Yes	Unclear	Unclear	Yes
Ng, 2011	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Unclear	Unclear	Yes
Peters, 2013	Yes	Yes	Yes	No	No	No	Yes	Yes	Unclear	Unclear	Yes
Rey, 2007	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Unclear	Unclear	Yes

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Thethi, 2010	Yes	Yes	Yes	No	No	No	Yes	Yes	Yes	Unclear	Yes
Trasande, 2018	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Unclear	Not applicable	Yes
Vanasse, 2016	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Unclear	Not applicable	
Yu, 2018	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No	Yes

Table A3. Critical appraisal using Joanna Briggs Institute (JBI) checklist for cross-sectional studies by exposure period (n=7)

Author, Year	Were the criteria for inclusion in the sample clearly defined?	Were the study subjects and the setting described in detail?	Was the exposure measured in a valid and reliable way?	Were objective, standard criteria used for measurement of the condition?	Were confounding factors identified?	Were strategies to deal with confounding factors stated?	Were the outcomes measured in a valid and reliable way?	Was appropriate statistical analysis used?
Perinatal								
Bercovich, 2014	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes
Oni, 2015	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adult								
An, 2015	Yes	Yes	Yes	Yes	No	No	Yes	Yes
Bich, 2011	No	Yes	Yes	Yes	No	No	No	Yes
Husarewycz, 2014	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Katratzias, 2015	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Koroma, 2019	No	No	Yes	Yes	No	No	Yes	Yes

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Table A4. Critical appraisal using Joanna Briggs Institute (JBI) checklist for quasi-experimental studies by exposure period (n=10)

Author, Year	Is it clear in the study what is the 'cause' and what is the 'effect' (i.e. there is no confusion about which variable comes first)?	Were the participants included in any comparisons similar?	Were the participants included in any comparisons receiving similar treatment/care, other than the exposure or intervention of interest?	Was there a control group?	Were there multiple measurements of the outcome both pre and post the intervention/exposure?	Was follow up complete and if not, were differences between groups in terms of their follow up adequately described and analyzed?	Were the outcomes of participants included in any comparisons measured in the same way?	Were outcomes measured in a reliable way?	Was appropriate statistical analysis used?
Perinatal									
Xiao, 2019	Yes	Unclear	Not applicable	Yes	No	Yes	Yes	Yes	Yes
Adult									
Becquart, 2018	Yes	Unclear	Not applicable	No	No	Yes	Yes	Yes	Yes
Gerin, 2005	Yes	Yes	Not applicable	Yes	Yes	Yes	Yes	Yes	Yes
Hendrickson, 1996	Yes	Unclear	Not applicable	No	Yes	Yes	Yes	Yes	Yes
Kim, 2017	Yes	Unclear	Not applicable	No	Yes	Yes	Yes	Yes	Yes
Kong, 2019	Yes	Unclear	Not applicable	Yes	No	Yes	Yes	Yes	Yes
Lin, 2010	Yes	Unclear	Not applicable	Yes	No	Yes	Yes	Yes	Yes
McKinney, 2011	Yes	Unclear	Not applicable	No	Yes	Yes	Yes	Yes	Yes
Nagayoshi, 2015	Yes	Unclear	Not applicable	Yes	No	Yes	Yes	Yes	Yes
Silva-Palacios, 2015	Yes	Unclear	Not applicable	No	No	Yes	Yes	Yes	Yes

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	3
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	4
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	4
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	4
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	4
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	11 (Table 1)
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	5
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	5
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	5
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	5
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	5
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	5

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	N/A
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	N/A
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	6
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	6
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	10
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	6-10
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	N/A
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	N/A
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	N/A
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	10
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	12-13
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	14
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	1

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Impact of disasters, including pandemics, on cardiometabolic outcomes across the life-course: A systematic review

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ABSTRACT

Background: Disasters are events that disrupt the daily functioning of a community or society, and may increase long-term risk of adverse cardiometabolic outcomes, including cardiovascular disease, obesity, and diabetes. The objectives of this study were to conduct a systematic review to determine the impact of disasters, including pandemics, on cardiometabolic outcomes across the life-course.

Design: A systematic search was conducted in May 2020 using two electronic databases, EMBASE and Medline. All studies were screened in duplicate at title and abstract, and full-text level. Studies were eligible for inclusion if they assessed the association between a population-level or community disaster and cardiometabolic outcomes ≥ 1 month following the disaster. There were no restrictions on age, year of publication, country, or population. Data were extracted on study characteristics, exposure (e.g., type of disaster, region, year), cardiometabolic outcomes, and measures of effect. Study quality was evaluated using the Joanna Briggs Institute critical appraisal tools.

Results: A total of 58 studies were included, with 24 studies reporting the effects of exposure to disaster during pregnancy/childhood and 34 studies reporting the effects of exposure during adulthood. Studies included exposure to natural (n=35; 60%) and human-made (n=23; 40%) disasters, with only 3 (5%) of these studies evaluating previous pandemics. Most studies reported increased cardiometabolic risk, including increased cardiovascular disease incidence or mortality, diabetes, and obesity, but not all. Few studies evaluated the biologic mechanisms or high-risk subgroups that may be at a greater risk of negative health outcomes following disasters.

Conclusions: The findings from this study suggest that the burden of disasters extend beyond the known direct harm, and attention is needed on the detrimental indirect long-term effects on cardiometabolic health. Given the current COVID-19 pandemic, these findings may inform public health prevention strategies to mitigate the impact of future cardiometabolic risk.

PROSPERO registration: CRD – 42020186074

Strengths and limitations of this study:

- This systematic review is one of the first to review the literature on disasters, including pandemics, and subsequent cardiometabolic outcomes throughout the life-course.
- A comprehensive search strategy was used to identify studies that covered a range of disasters (e.g., famine, war, terrorism, natural disasters, and infectious disease epidemics), periods of exposure from pregnancy, childhood to older adulthood, and a wide breadth of cardiometabolic outcomes.
- Only studies published in English were included and a search of the grey literature was not conducted.
- Due to the heterogeneity of studies, a meta-analysis could not be conducted, and results were only synthesized narratively.
- Limited evidence was available on the impact of pandemics specifically, and few studies evaluated proposed mechanisms or risk modification across subgroups of the populations.

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BACKGROUND

Disasters, as defined by the World Health Organization (WHO), are events that disrupt the daily functioning of a community or society causing material, economic or environmental losses, overwhelming local capacity (1). Disasters can be categorized into natural disasters, human-made disasters, and hybrid disasters (2). Natural disasters include natural phenomenon above and beneath the earth's surface (e.g., tsunamis or landslides), meteorological phenomenon (e.g., tornadoes or floods) or biological phenomenon (e.g., epidemics and pandemics) (2). Human-made disasters include adverse transportation incidents, technological events (e.g., fire or toxic leaks), terrorism, warfare or conflict (2). A hybrid disaster results from both human and natural forces, such as the clearing of a jungle that results in a landslide (2). All types of disasters can result in public health emergencies as they typically impact a significant proportion of people (3). Epidemics, defined as a greater than expected increase in cases of a disease, and pandemics, which cross countries and continents, are types of natural disasters with far-reaching global disruption (4). The COVID-19 pandemic is a present-day example of a global disaster that is unlike any disaster in recent history. Understanding the potential long-term health implications of the current COVID-19 pandemic and resulting public health mitigation strategies is urgently needed.

Previous systematic reviews have focused on acute outcomes, specifically on the psychological impact of quarantine during pandemics (5), the impact on health outcomes after disasters in older adults (6), medically unexplained physical symptoms following disasters (7), and chronic medical interventions following a natural disaster (8). It is biologically plausible that exposure to a disaster may lead to long-term or chronic outcomes that could arise many years later and this may be modified by the time of exposure across the life-course. Consistent with established models of life-course epidemiology there may be critical periods of exposure (e.g., during development in childhood), where exposure to a disaster substantially increases later life disease risk, or exposure to a disaster may contribute to a chain of risk or accumulation of risks across the life course (9,10). There is currently no review on the long-term impacts of disasters, or more specifically, epidemics and pandemics on cardiometabolic outcomes across the life-course. Noncommunicable diseases (NCDs), including cardiovascular disease (CVD), obesity, and diabetes, are the leading cause of morbidity and mortality worldwide (11,12). NCDs are attributed to 71% of all global deaths annually, with approximately 14 million CVD-related deaths and 1.6 million diabetes-related deaths (12). Findings from the Global Burden of Diseases Study indicate that CVD and diabetes account for over 20% of the global burden of disability, with diabetes recently emerging as the fourth leading cause of disability globally (11). Exposure to disasters may cause

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cardiometabolic outcomes to emerge or worsen through several different mechanistic pathways including stress exposure (13), lack of access to health services (14), food security and behavioural changes such as alterations in physical activity, sleep, and diet (15). It is important to understand the impact of disasters on the incidence of new cardiometabolic diseases and changes in disease status in all populations and age groups. Particular subgroups of a population may be more or less susceptible to cardiometabolic outcomes and understanding this can inform targeted interventions. The primary objective of this review was to determine the impact of disasters, including pandemics on risk of cardiometabolic outcomes across the life-course. The secondary objectives were to determine how to reduce the impact of chronic disease outcomes following a disaster and to identify populations at highest risk of cardiometabolic outcomes following a disaster.

METHODS

A systematic review was conducted following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) (16). This review was registered on PROSPERO (CRD – 42020186074).

Search strategy

A systematic search was conducted in May 2020 using the electronic databases EMBASE and MEDLINE. The health research librarians at McMaster University assisted in developing the search strategy. The search broadly captured two concepts: disasters and cardiometabolic outcomes (e.g., diabetes, obesity, hypertension). The complete search strategy for EMBASE can be found in Table 1. The search strategy for MEDLINE can be found in the Appendix (Table A1). Reference lists of eligible studies and relevant systematic reviews were hand searched to identify additional articles.

Eligibility criteria

Studies were eligible for inclusion if they assessed the relationship between a population-level or community disaster and the risk of future cardiometabolic outcomes including CVD, diabetes or obesity or cardiometabolic risk scores (17). CVD included myocardial infarction, stroke, hypertension and angina. There were no restrictions on year of publication, country of disaster, or population. Only studies evaluating the impact of real-world disasters in humans were included. Due to the research team's capacity, only studies published in English were included. Observational and quasi-experimental study designs, including case-control, cohort and other longitudinal study designs or natural

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experiments were included. Outcomes that were not cardiometabolic-related or acute cardiometabolic events, such as an immediate complication (defined as <1 month after disaster), were excluded. Studies that assessed the exposure to a chemical as a result of the disaster, were excluded, as we were not interested in outcomes resulting from chemical exposure. Earthquake studies were also excluded since a systematic review was published in 2018 that assessed the impact of earthquakes on cardiometabolic outcomes (18).

Study selection

After running the search, all identified studies were imported into Covidence and duplicates were removed (19). Studies were screened at title and abstract-level, and then at full-text by any two of the following independent reviewers VD, JL, MSA, YYM, ATA, ES, SI, JDM, RR LNA. Conflicts were resolved by a third reviewer who made the final decision regarding eligibility for inclusion.

Data extraction

A data extraction template was created, and pilot tested prior to data extraction. Data were then extracted from all studies by any two of the following independent abstractors VD, JL, SMA, YYM, ATA, RR, ES and conflicts were resolved by a third independent abstractor. Study characteristics including year of publication, study design, country of disaster, sample size, and length of study were extracted where reported. Specific information on the exposure and outcome in each study were extracted including the type and name of the disaster, country and year of the disaster, the outcome of interest, and how the exposure and outcome were measured. Finally, any information on subgroups including age, population, sex and disaster type were also extracted, if applicable.

Critical appraisal

Critical appraisal was conducted using the Joanna Briggs Institute (JBI) Critical Appraisal Tools (20). This tool was chosen due to the availability of checklists for a wide range of study designs, including cohort, cross-sectional and quasi-experimental designs (20). The quasi-experimental study design checklist was used for natural experiments including time-series studies and pre/post designs, as it was decided this was the most appropriate tool. All studies were critically appraised independently by any two of the following individuals VD, JL, MA, YYM, ATA, ES, SI and a third individual was consulted for any discrepancies.

Data analysis

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Data from the included studies were narratively synthesized. Results are presented by exposure period (perinatal/childhood versus adulthood) and by cardiometabolic outcome (obesity, CVD, and diabetes). Characteristics of studies are presented as frequencies and percentages. Due to the heterogeneity of studies, a meta-analysis was not conducted.

RESULTS

A total of 4830 studies were identified through the electronic database search. An additional 12 studies were identified through manual searching of the reference lists of relevant studies. After removing duplicates (n=439), 4403 studies were screened at title and abstract level. After applying inclusion and exclusion criteria, 4068 studies were excluded, leaving 335 studies screened for full-text eligibility. A total of 58 studies were eligible for inclusion into the review. The complete screening process is described in Figure 1.

Description of studies

Table 2 provides a summary of included studies. Of 58 included studies, 24 studies (15,21–43) investigated exposure to disasters during pregnancy or childhood while the remaining 34 studies (14,44–76) investigated exposure to disaster during adulthood. Almost all studies (n=49) assessed cardiometabolic outcomes during adulthood, only two studies assessed outcomes during pregnancy (27,28) and seven studies assessed outcomes during childhood and adolescence (15,21–26). The length of studies, including prospective follow-up and retrospective assessment, ranged from one month to 95 years. Most studies (n=36) focused on disasters that occurred in North America (14,21–28,38,39,42,45–49,51,52,54,55,57–60,63–65,67–72,75,76), followed by Europe (n=13) (29–33,35,41,43,53,62,73,74). The remaining disasters occurred in Asia (n=7) (15,37,40,44,50,56,61) and Africa (n=2) (36,66). The characteristics of included studies and key findings are provided in Table 3 for disaster exposure during the perinatal period and childhood, and Table 4 for exposure during adulthood.

Exposure to disaster in the perinatal and childhood period

Of the 24 studies that evaluated perinatal and childhood exposure to disaster, 12 studies evaluated human-made disasters (29–35,40,41,47) and the remaining 12 evaluated natural disasters

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(15,21–28,38,39,42) of which two were pandemics (38,39). Most studies (n=15) assessed the disaster as the main exposure of interest (24,28–33,35,36,39–42). The remaining studies evaluated stress (e.g., maternal stress, disaster-related PTSD, subjective stress, objective hardship) (21–25,27,47), maternal weight and maternal nutrition status (15,34), cognitive appraisal (26), and coping strategies (27) that were the result of the disaster as the exposure variable. The age when cardiometabolic outcomes were assessed varied across studies, with two studies evaluating pregnancy outcomes (27,28), 8 studies evaluating outcomes among children and youth (15,21–26,47), four studies assessed outcomes during young to mid adulthood (>18-40 years of age) (27,29,36,37), and 10 during later adulthood (≥50 years of age) (30,32–35,38–42). One study did not specify the exact age, rather evaluated outcomes throughout adulthood, from 18-63 years of age (31). Detailed characteristics and findings of all studies that assessed perinatal and childhood exposures to disasters can be found in Table 3.

Within the 10 studies that evaluated perinatal or childhood disaster exposure in relation to pregnancy, childhood, or youth outcomes, one study evaluated a human-made disaster (the World Trade Center Attacks) and the other nine studies evaluated natural disasters (including ice storms, floods and hurricanes) and the findings were mixed. The one study that evaluated a human made disaster found limited evidence of any increased cardiometabolic risk and a small decrease in both BMI and zBMI was observed for children exposed to the World Trade Centre attacks compared to those who were not but found no differences in triglycerides or lipids (47). Within the eight studies that evaluated exposure to a natural disaster during the perinatal and childhood period and cardiometabolic outcomes in later childhood, there were six studies that evaluated measures of child growth and four of these studies reported increased BMI or adiposity in later childhood (21,23,25,26), one study was null (22) and one study reported increased wasting or malnutrition following exposure to a flood (15). Importantly, the one study where increased wasting was observed was in Bangladesh, whereas all of the studies that observed increased risk of obesity were in North America. Of the studies that evaluated childhood cardiometabolic outcomes other than growth, one study found increased insulin concentrations at age 13 (24). There were two studies that evaluated exposure to a natural disaster (Hurricane Katrina and Hurricane Sandy) during pregnancy and both found increased incidence or hospital visits for gestational hypertension and diabetes (27,28).

There were 14 studies that evaluated exposure to a disaster in pregnancy or childhood in relation to the subsequent onset of adult cardiometabolic conditions; 11 of these studies evaluated

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exposure to human made disasters and three evaluated natural disasters. Within the 11 studies that evaluated human made disasters, one study found no association between exposure to the Dutch famine and coronary artery disease in older adulthood (32). While the remaining 10 studies all found some evidence of increased cardiometabolic outcomes in adulthood following perinatal or childhood exposure, the results were mixed with many null results depending on outcome or exposure. For example, prenatal exposure to famine was associated with higher LDL and CHD approximately 28 years later, however no difference was found for glucose, insulin, BMI or other lipids (29). The results were not consistent across outcomes, for example, increased risk of hypertension was found in three studies (29,36,40) but not in two studies (35,37). Within the three studies that evaluated adult cardiometabolic outcomes following exposure to a natural disaster, all three studies found increased risk of cardiovascular diseases or mortality following prenatal exposure to famine or the 1918 influenza pandemic (38,39,42).

Studies on adult exposure to disaster and subsequent cardiometabolic outcomes

Thirty-four studies investigated the effects of exposure to disasters during adulthood on cardiometabolic outcomes. The length of follow-up ranged from one month to 13 years. There were 23 studies that examined natural disasters (14,44–46,48,51,52,54–56,58–61,64–67,72–76), and 11 studies that examined human-made disasters (43,49,50,53,57,62,63,68–71). Of these studies, only one evaluated the impact of an infectious disease epidemic (66). Most studies (n=27) considered the disaster as the main exposure of interest (14,43–46,48,50,52,54–56,58–66,68,71–76). The remaining seven studies assessed disaster-related stress (53,57,67,69), including post-traumatic stress disorder (PTSD) and psychological strain, unemployment rates as a result of the disaster (51) and exposure to damaged or collapsed buildings during the World Trade Center disaster (70). Detailed characteristics and findings of all studies that assessed adult exposures to disasters are included in Table 4.

The studies that assessed exposure to human-made disasters (n=11) during adulthood reported mixed results in terms of associations with outcomes and statistical significance. Three studies assessed PTSD related to disasters and found an increased association with stoke (69), heart disease (57) and cardiovascular/vascular problems (53), two of which were exposure to the World Trade Centre disaster and the third was a Fireworks depot explosion. Two studies assessing exposure to the World Trade Centre disaster and Amsterdam Air disaster found an increased association with cardiovascular

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hospitalizations (49) and cardiovascular symptom complaints (62) in rescue workers compared to non-rescue workers. Of the remaining six studies, three studies reported an increased association with hypertension (43), systolic blood pressure (71) and CVD mortality (63), however, the exact exposure varied across studies. For instance, one study explored the level of exposure, defined as low, intermediate, or high to the World Trade Centre disaster (63), whereas another study evaluated exposure to the Volendam Pub Fire among parents who had children who were injured or died (43). The final three studies assessing exposure to human-made disasters (World Trade Centre disaster and Sewol Ferry disaster) reported mixed results with some showing a decreased association or null findings (53,68,70).

Among studies that evaluated the impact of exposure to natural disasters (n=23), six studies that evaluated exposure to Hurricanes Sandy, Katrina, Iniki, and the Flood of Saint-Jean-sur-Richelieu reported an increased association with cardiometabolic outcomes(45,46,51,58,64,72). One of these studies specifically investigated unemployment as a result of Hurricane Katrina and found those who were unemployed, compared to those who remained employed were 5.65 times more likely to have a cardiometabolic event ($p<0.05$) (51). Nine studies reported a statistically significant increase in outcomes following exposure disaster (52,54–56,61,67,73–75). For instance, one study found those who reported higher levels of psychological strain after surviving Hurricane Ike, compared to those with lower levels of psychological strain had higher mean blood glucose and obesity four months after the disaster (67). Whereas, another study found a higher proportion of people experiencing worse hypertension who were living in households affected by the 2008 Hanoi flood compared to those who lived in an unaffected households in both rural and urban areas (61). Two of these studies reported an increase in incidence of AMI and AMI hospital admission pre-Hurricane Katrina, compared to post-Hurricane Katrina (54,55). The remaining six studies reported a mix of findings. Three studies found varying associations across outcomes reported within their study. For example, Fonseca et al., found an increased mean difference pre- and post-Hurricane Katrina for glycemic control, systolic blood pressure, diastolic blood pressure, and HDL, but not for LDL and triglycerides (59). Nine studies reported mixed findings across outcomes within the study. Four of these studies found both an increase and decrease in outcomes when comparing mean difference or proportion pre- and post-disaster (14,48,66,76). One study found those with higher reports of exposure to natural disaster throughout the life course were significantly different from those with lower reports (44). The final two studies found a decreased

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proportion of AMI following Hurricane Katrina (65) and no significant association between exposure to the Oklahoma tornado and hospital admission for cardiovascular events (60).

Mediation and modification of cardiometabolic outcomes

Across all studies, few evaluated effect modification or subgroups of a population that may be at a greater risk of negative health outcomes following disasters. Eight studies stratified by sex (30,34,36,41,45,50,57,64), gestational timing of exposure (22,29,31–33,35,38), year of birth or age at outcome (42,64,69), urban or rural area (37), race (45,48) and socioeconomic status (67), however, results varied greatly due to the differences in exposure period, disaster type, cardiometabolic outcome and age at outcome. One study explored the possible mediators between cognitive appraisal following the Quebec Ice Storm and obesity. It was noted that negative cognitive appraisal was found to predict obesity via DNA methylation of diabetes-related genes (26). No studies evaluated or discussed possible interventions to mitigate risk of cardiometabolic disease following a disaster.

Critical appraisal

The critical appraisal assessment for all study designs can be found in the Appendix (Tables A2-A4). Among the cohort studies, most studies met all criteria included in the checklist indicating high study quality. For instance, almost all cohort studies had comparable populations that were recruited in a similar way and exposures that were assessed in the same way across populations. However, across almost all cohort studies, information on follow-up or strategies to address incomplete follow-up were unclear or not addressed. The critical appraisal results for cross-sectional studies were inconsistent with a small number of studies meeting only some checklist requirements. For quasi-experimental studies, the checklist requirement for within person comparisons were not applicable for all studies, however, all studies clearly defined the cause and effect within the study.

DISCUSSION

Principal findings

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A total of 58 studies were identified and they covered a wide breadth of exposures to both natural and human-made disasters, including famine, war, terrorism, natural disasters, and infectious disease epidemics. Exposures were investigated in pregnancy and childhood exposure through to adulthood with outcomes measured 1 month to 95 years later. The reviewed studies reflect a true-life course body of literature with exposures at multiple ages and long-term exposures. A range of cardiometabolic outcomes including obesity, hypertension, myocardial infarction, diabetes, and cardiac mortality were investigated. Given the varied nature of the studies, it was difficult to draw overall conclusions, but the vast majority of studies provided some evidence of increased cardiometabolic risk following disaster exposure. There were only 11 studies that reported no increased risk or had unclear findings. Across these studies, there was a variety of disaster exposure, outcomes, and follow-up periods, however seven of these studies did not report adjustment or consideration of any confounders.

Relation to other studies

To the best of our knowledge, this is the first review to systematically review the literature on a broad range of disasters and cardiometabolic health outcomes across the life-course. Other reviews have focused on a specific population, such as older adults, specific disaster types (e.g., natural disasters only) or other health conditions (e.g., mental health) or acute outcomes (5,6,8). However, across most reviews it was apparent the heterogenous nature of included studies makes it difficult to summarize findings and make overall conclusions and recommendations. For instance, Chan et al. found exposure to natural disasters negatively affected those with chronic conditions, although authors noted limitations due to limited literature (8). Another systematic review found very heterogenous results when reviewing the literature on health outcomes after disasters for older adults with chronic disease (5). The studies included here were from multiple disciplines and utilized a variety of studies designs, assessed several different outcomes, and applied different statistical approaches. Overall the results suggested increased risk of adverse cardiometabolic outcomes following disasters, although this was not apparent across all included studies (6). The unexpected nature of disasters, uniqueness of population or region affected, and scale of damage lead to research studies that vary greatly. Although previous reviews and the current review have identified quite heterogenous studies, overall conclusions suggest risk of disease increases after exposure to disasters.

Biologic mechanisms

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Several potential mechanisms were discussed in the included studies that may contribute to the observed associations between disaster exposure and increased cardiometabolic outcomes, include the role of both objective and subjective stress, nutritional changes, and reduced access to healthcare. One study that explored mediators in the association between stress and obesity measures identified the role of DNA methylation in this association (26). It is well postulated that the activation of a stress response following a stressful event leads to changes in the nervous, cardiovascular, endocrine and immune systems (77). Exposure to disasters including famine, war, terrorism, natural disasters, and infectious disease epidemics may activate a stress response, altering the progression of disease development (77). The repeated or prolonged exposure to various disasters, such as a pandemic spanning over months, may lead to worse health outcomes. Reduction in health services is another possible mechanism leading to worse health outcomes. Healthcare services may be directed toward the immediate response to health-related consequences caused by the disaster (e.g., illness from a pandemic, injuries associated with a terrorist attack or natural disaster), limiting access to primary care (78). This interruption to services may decrease screening or early treatment ultimately leading to the rise in chronic diseases. Lastly, social determinants of health are known to be important risk factors for cardiometabolic conditions (79). At least one study investigated whether the observed associations were due to changes in educational attainment (42) and unemployment (51) . More investigation of the role of social determinants as modifiers or mediators of the associations between disasters and long-term cardiometabolic outcomes may be warranted. Despite numerous proposed biologic mechanisms and well-established life course frameworks, relatively few studies actually evaluated potential causal pathways using a life course framework, and this may contribute to some of the observed heterogeneity in results.

Strengths and limitations

This review had several strengths including the comprehensive evaluation of the impact of a wide range of disaster exposures on various cardiometabolic outcomes at different periods throughout the life-course. The search strategy was developed in consultation with health science librarians at McMaster University to ensure the most comprehensive search was developed and relevant literature was identified. The timely findings of this synthesis are a strength of this review, given the current COVID-19 pandemic, which is affecting millions of people worldwide. While only a small proportion of the identified studies focused on pandemics and epidemics, the findings may serve to guide our

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understanding of expected outcomes, and to develop future research to study the effects of COVID-19 on cardiometabolic outcomes.

Although this review had several strengths, interpretation of findings should be done with caution due to limitations. First, the heterogeneity across studies restricted the ability to conduct a meta-analysis. Studies varied in terms of study design, reported measures of effect, the comparison group (e.g., some studies did not include a comparator group), length of follow-up, timing and measurement of exposure, and primary outcomes and how they were measured. Given the multi-disciplinary nature of the identified studies, a wide range of analytic approaches were used, and measures of effect varied. These differences in addition to the lack of statistical significance across studies make it difficult to draw overall conclusions. Many of the studies used a retrospective cohort study design and relied on administrative data sources as such many studies were unable to comprehensively adjust for confounders, including social determinants of health. Measurement error and misclassification of exposure status is also possible since many studies did not objectively measure disaster exposure or degree of impact, and instead used proxy measures of disaster exposure based on time and geography.

Very few studies have evaluated the long-term impacts of pandemics and epidemics on cardiometabolic outcomes, identifying a current gap in the literature. This made it difficult to truly assess if exposure to disasters at sensitive periods of development had lasting effects much later in life. Studies also reported insufficient data on subgroups that were at increased risk of worse cardiometabolic health outcomes and interventions that were implemented to mitigate risk of cardiometabolic outcomes. In addition, results were not often explored by sex and gender, or did not apply an equity lens. It has been noted that those of different levels of socioeconomic status experience differential cardiometabolic outcomes (80,81). This signifies the importance of exploring associations between exposure to disasters and cardiometabolic outcomes stratified by these factors. Understanding how these associations differ will also help to identify groups of people who will experience worse outcomes following a disaster.

Study implications

To the best of our knowledge, this is the first study to comprehensively explore the impact of several different types of disasters on cardiometabolic outcomes at different periods throughout the life-course. The results suggest that increased risk is observed for disaster exposure at any period over the life-course from the perinatal child and adult periods. These findings emphasize that the burden of

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disasters extends beyond the known direct harms they cause, and attention is needed on the detrimental indirect long-term effects on cardiometabolic health and chronic disease. Given the current COVID-19 pandemic, this review may be helpful in raising awareness of the potential increase in cardiometabolic health outcomes post-pandemic, to ensure appropriate public health mitigation measures are developed and implemented to prevent long term cardiometabolic outcomes at the population level.

Unanswered questions and future research

Future research should evaluate the impact of pandemics, such as COVID-19, on future cardiometabolic health throughout the life-course. It may also be of interest for future research to explore the impact of implementing public health measures, such as physical distancing to reduce transmission of a virus, and the implications following a disaster with access to healthcare on health outcomes. This information would be helpful in planning public health responses to different disasters. In addition, further investigation of possible mechanisms, such as disruptions to healthcare or medication access, and changes in dietary intake or physical activity, is needed. This would help to develop preventative strategies targeted at these mechanisms to help reduce the possible cardiometabolic consequences after experiencing a disaster. This review found insufficient evidence identifying subgroups of the population who are at the greatest risk or specific disaster related risk factors that that increase cardiometabolic disease development following a pandemic. This is an important gap that needs to be addressed by future research.

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Table 1. Search strategy for EMBASE

1	social isolation.mp. or social isolation/	24963
2	quarantine.mp. or quarantine/	4752
3	*epidemic/	32686
4	*pandemic/	4387
5	disease outbreak.mp.	2321
6	disaster/	13321
7	*natural disaster/	968
8	humanitarian crisis.mp.	257
9	mass casuality.mp. or mass disaster/	3654
10	coronavirus.mp. or coronaviridae/	23106
11	cardiovascular disease.mp. or *cardiovascular disease/	357319
12	*diabetes mellitus/	210248
13	*cerebrovascular accident/	78444
14	*heart infarction/	99072
15	*angina pectoris/	22631
16	*obesity/	178134
17	public health emergency.mp.	1752
18	*body mass/	31459
19	*hypertension/	198593
20	1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 or 9 or 10 or 17	109105
21	11 or 12 or 13 or 14 or 15 or 16 or 18 or 19	1087681
22	20 and 21	2047
23	limit 22 to human	1832

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Table 2. Characteristics of included studies (n=58)

Characteristics	N (%)
Continent	
North America	36 (62%)
Europe	13 (22%)
Asia	7 (12%)
Africa	2 (3%)
Year of Publication	
2010-2020	44 (76%)
2000-2009	12 (21%)
1996-1999	2 (3%)
Study design	
Cohort/longitudinal	41 (71%)
Quasi-experimental[1]	10 (17%)
Cross-sectional	7 (12%)
Sample size	
≥10,000	19 (33%)
1,000- <10,000	10 (17%)
≤1,000	24 (41%)
Not specified	5 (9%)
Exposure life stage	
Pregnancy/Childhood	24 (41%)
Adulthood	34 (57%)
Outcome life stage	
Pregnancy	2 (3%)
Childhood [2]	8 (13%)
Adult	47 (81%)
Disaster	
Human-made	23 (40%)
Natural	35 (60%)
Cardiometabolic outcome [3]	
Cardiovascular disease [4]	41 (71%)
Diabetes [5]	11 (19%)
Obesity or BMI	12 (21%)
Mortality from cardiovascular disease	9 (16%)
Cardiometabolic risk during pregnancy [6]	2 (3%)

1. Includes pre/post study design, time series and natural experiments
2. Children defined as ≤18 years of age
3. Does not equal to 100% as studies report multiple cardiometabolic outcomes
4. Includes hypertension, coronary artery disease/heart disease, angina, heart attack/myocardial infarction, metabolic syndrome, cardiac disease related blood markers, stroke
5. Diabetes, blood glucose, metabolic syndrome
6. Gestational diabetes, gestational hypertension, pre-eclampsia

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Table 3. Characteristics of included studies investigating the association between exposure to a disaster during the perinatal and childhood periods and cardiometabolic outcomes across the life-course, by disaster type (n=24)

Study	Study design	Country	Name of disaster	Year	Sample size	Primary exposure and comparator	Average follow-up	Outcomes	Primary results[1]
Human-made disaster with child/youth outcome									
Trasande, 2018 (47)	Prospective cohort study	USA	World Trade Center attacks (9/11)	2001	402	New York children and youth enrolled in the World Trade Center Health Registry (birthdates: Sept 11, 1993 to September 10, 2001) compared to individuals born during the same time period who were ineligible for enrollment in the WTCHR	2 years	Youth outcomes: 1) BMI (kg/m2) 2) zBMI 3) Triglycerides (mg/dL) 4) Cholesterol (mg/dL) 5) LDL (mg/dL) 6) HDL (mg/dL)	Regression coefficient and 95% CI: 1) BMI: -1.12 (-2.11, -0.12) 2) zBMI: -0.24 (95% CI: -0.49. -0.002) 3) logTrig: 0.02 (95% CI: -0.07, 0.12) 4) logChol (0.02 (95% CI: -0.02, 0.06) 5) log LDL: 0.06 (95% CI: -0.001, 0.12) 6) logHDL: -0.04 (95% CI: -0.10, 0.03)
Human-made disaster with adult outcome									
Bercovich, 2014 (40)	Cross-sectional	Israel	Holocaust	1941-1945	300	European Jews born in 1940-1945 with exposure to the holocaust compared to European Jews during the same time period born	N/A	Adult outcomes: 1) Hypertension 2) Diabetes 3) Dyslipidemia 4) Any cardiovascular disease	1) Adjusted OR: 2.2, 95%CI: 1.2-3.8 2) Adjusted OR: 2.2, 95%CI: 1.2-4.2 3) Adjusted OR: 3.1, 95%CI: 1.7-5.7 4) Adjusted OR: 2.6, 95%CI: 1.4-4.7
de Rooij, 2007 (30)	Cohort	Netherlands	Dutch Famine	1944-1945	783	Prenatal exposure to Dutch famine defined as people born between January 7, 1945 and December 8 1945 compared to people born before Jan 7 1945 or conceived after Dec 8 1945	58 years	Metabolic syndrome at age 58	Metabolic syndrome OR: 1.2; 95% CI: 0.9, 1.7
Ekamper, 2015 (31)	Cohort	Netherlands	Dutch Famine	1944-1945	41096	Male military conscripts born between Jan 1944 and 1946 and compared to military conscripts born before 1944 or after 1946	63 years	Adult outcomes: 1) Heart disease mortality 2) Cerebrovascular disease mortality 3) Diabetes mellitus mortality	Hazard ratio (HR): 1) HR: 0.94; 95% CI: 0.77, 1.15 2) HR: 1.55, 95% CI: 0.95, 2.51 3) HR: 1.61, 95% CI: 0.91, 2.86
Huang, 2010 (37)	Cohort	China	1959-1961 Chinese Famine	1959-1961	35,025	County level famine intensity for women born during 1957-1962 compared to women born post-famine in 1963	32 years	Adult outcomes at age 32: 1) BMI among rural sample 2) BMI among urban sample 3) Hypertension among rural sample 4) Hypertension among urban sample	1) Average effect = 0.92, 95% CI: 0.32, 1.51 2) Average effect = 0.03, 95% CI: -2.82, 2.87 3) Log odds = 1.23, 95% CI: -0.38, 2.84 4) Log odds = 0.37 95% CI: -2.07, 2.80
Hult, 2010 (36)	Cohort	Nigeria	Biafran Famine	1967-1970	1,339	Individuals exposed to famine during early childhood (born 1965-1967) or exposed to famine in fetal life and infancy (born 1968-Jan 1970) compared to people born between 1971 and 1973	~40 years	Adult outcomes at age ~40 years: 1) Hypertension 2) Diabetes 3) Overweight (BMI>25 kg/m^2) 4) Obesity (BMI>30kg/m^2)	Adjusted OR (95% CI) 1) Childhood exposure OR: 1.42, 95% CI: 0.63, 3.13; Fetal-infant exposure: OR: 2.50, 95% CI: 1.19, 5.26 2) Childhood exposure OR: 1.81, 95% CI: 0.64, 5.15; Fetal-infant exposure OR: 2.56, 95% CI: 0.92, 7.17 3) Childhood exposure OR: 1.02, 95% CI: 0.77, 1.34; Fetal-infant exposure: OR: 1.41, 95% CI: 1.03, 1.93 4) Childhood exposure OR: 1.20, 95% CI:

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									0.87, 1.67; Fetal-infant exposure: OR: 1.30, 95% CI: 0.92, 1.85
Lumey , 2012[1] (32)	Cohort	Netherlands	Dutch Famine	1944-1945	1075	Infants whose mothers were exposed to famine during or immediately preceding pregnancy (born Feb 1, 1945, March 31, 1946) compared to individuals born in the same hospital before or after famine	~56-62 years	Adult outcomes at 56-62 years Coronary artery disease	Early gestation HR: 1.26, 95%CI: 0.59, 2.70 Late gestation HR: 1.31, 95%CI: 0.67, 2.57
Painter, 2006 (33)	Cohort	Netherlands	The Dutch Famine	1944-1945	975	Infants who were born between January 1945 and December 1945 who were exposed to famine in utero compared to infants born before the famine (November 1943 and January 1945) and after the famine (December 1945 and February 1947)	~50-58 years	Adult outcomes at 50-58 years: Coronary artery disease	HR: 1.9, 95% CI: 1.0, 3.8
Ravelli, 1999[2] (34)	Cohort	Netherlands	The Dutch Famine	1944-1945	741	Infants exposed to famine during different periods of gestation (late, mid and early) whose maternal daily ration was <1000 kcal (born between January 1945 and December 1945) compared to those born not during the famine	50 years	Obesity adult outcomes at 50 years stratified by sex: 1) Weight (kg) 2) BMI (kg/m2) 3) Waist circumference (cm)	Mean difference (95% CI) between exposure during late or early gestation versus nonexposed: Men: 1) late: 0.8 (-3.1, 4.7); early: 1.5 (-3.5, 6.6) 2) late: 0.4 (-3.5, 4.5); early: 0.5 (-4.6, 6.0) 3) late: 1.8 (-1.4, 4.9); early: (1.8 (-2.4, 6.0) Women: 1) Late: -1.8 (-6.1, 2.5); early: 7.9 (2.5, 13.2) 2) late: -2.1 (-7.0, 3.1); early 7.4 (0.7, 14.5) 3) late: -0.7 (-4.4, 3.0); early: 5.7 (1.1, 10.3)
Roseboom, 2001[2] (29)	Cohort	Netherlands	The Dutch Famine	1944-1945	2414	Infants who were exposed to famine in utero whose mother had a daily ration <1000 calories during any 13-week period of gestion compared to infants who were born before or conceived after the famine period (before November 1943 or after February 1947)	~28 years	Adult outcomes at 28 years; 1) Plasma glucose (mmol/l) 2) Plasma insulin (pmol/l) 3) Total cholesterol (mmol/l) 4) HDL (mmmol/l) 5) LDL (mmol/l) 6) LDL/HDL cholesterol 7) BMI (kg/m) 8) CHD 9) Systolic BP (mmHg) 10) Diastolic BP (mmHg)	Mean values of outcomes for late gestation and early gestation: 1) Late: 6.3; early: 6.1 2) Late: 200; early: 207 3) Late: 5.83; early: 6.13 4) Late: 5.83; early: 6.13 5) Late: 1.32; early: 1.26* 6) Late: 3.87; early: 3.26* 7) Late: 26.7; early: 28.1 8) Late: 2.5; early: 8.8* 9) Late: 127.4; early: 123.4 10) Late: 86.4; early: 84.8 * p<0.05
Schrier, 2011 (41)	Cohort	Finland	Winter War and Continuation War	1939-1940 (Winter War), 1941-1944	13,039	Individuals in utero who were exposed to bombings that occurred for 48 days between 1934-1944 compared to those who were not exposed in utero	~60-70	Adult outcomes: 1) Coronary heart disease 2) Cerebrovascular disease	Results are shown graphically Higher CHD survival rates among women 64+ and among men aged 50-54 exposed while in utero

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				(Continuation War)					
Stein, 2006 (35)	Cohort	Netherlands	The Dutch Famine	1944-1945	971	Prenatal exposure to famine defined as the weeks post-last menstrual period that mother was exposed to an official ration of <900 kcal/week (gestation weeks: 1-10, 11-20, 21-30, or 31-delivery)	59	Adult outcomes: 1) systolic BP (mmHg) 2) Diastolic BP (mmHg) 3) Hypertension	Adjusted regression coefficients 1) 1-10 weeks: 1.20 (95% CI: -3.28, 5.69); 11-20 weeks: -1.19 (95% CI: -4.92, 2,55); 21-30 weeks: 1.33 (95% CI: -2.24, 4.90); 31-delivery: 2.02 (95% CI: -1.53, 5.57) 2) 1-10 weeks: 1.10 (95% CI: -1.36, 3.57); 11-20 weeks: -1.26 (95% CI: -3.32, 0.80); 21-30 weeks: 1.19 (95% CI: -0.78, 3.15); 31-delivery: 0.71 (95% CI: -1.24, 2.66) 3) 1-10 weeks: 1.14 (95% CI: 0.62, 2.11); 11-20 weeks: 0.98 (95% CI: 0.59, 1.65); 21-30 weeks: 1.23 (95% CI: 0.74, 1.05); 31-delivery: 1.42 (95% CI: 0.86, 2.35)
Natural disaster with pregnancy outcomes									
Oni, 2015 (27)	Cross-sectional	USA	Hurricane Katrina	2005	146	Women who were pregnant during Hurricane Katrina or became pregnant immediately after hurricane compared to those who were not exposed to the hurricane; Women who experienced prenatal stress caused by Hurricane Katrina compared to those who did not experience stress	9 months	Pregnancy related outcomes: 1) Pregnancy induced hypertension 2) Gestational diabetes	1) Hurricane exposure: adjusted OR: 1.22 (95% CI: 0.81, 1.84); perceived stress: adjusted OR: 1.16 (95% CI: 1.05, 1.30); 2) Hurricane exposure: adjusted OR: 1.04 (95% CI: 0.69,1.57); perceived stress: adjusted OR: 1.13 (95% CI: 1.02, 1.25)
Xiao, 2019 (28)	Time series/Quasi-experimental	USA	Hurricane Sandy	2012	Not reported	Exposure to Hurricane Sandy lasting impacts defined as the following 12 months after Sandy (November 2012-October 2013) compared to the November to October in other years during November 2005 to October 2014 among women who were pregnant	12 months	Outcomes in adults: 1) Emergency department visits for gestational hypertension 2) Emergency department visits for diabetes or abnormal glucose	1) Increased at 7 months: 7.3% (95% CI: 1.0%, 13.9%) 2) Increased at 8 months:26.3% (95% CI: 3.9%, 53.6%) *results for 12 months reported graphically
Natural disaster with child/youth outcomes									
Cao-Lei, 2016 (26)	Cohort	Canada	Quebec Ice Storm	1998	31	Negative cognitive appraisal of the impact of the ice storm among pregnant women compared to neutral or positive appraisal	13 years	Outcomes among children at age 13: 1) Central adiposity (waist to height ratio) 2) BMI (kg/m2)	Mean (standard deviation) 1) Exposed: 20.86 (3.73); unexposed: 22.84 (5.19) 2) Exposed: 0.43 (0.04); unexposed: 0.45 (0.06)
Dancause, 2012 (23)	Cohort	Canada	Quebec Ice Storm	1998	111	Higher objective PNMS scores compared to lower scores among women who were pregnant or conceived within 3 months of the storm	5.5 years	Childhood obesity at 5.5 years of age	OR: 1.37, 95% CI: 1.06-1.77
Dancause, 2013 (24)	Cohort	Canada	Quebec Ice Storm	1998	32	Higher objective hardship compared to lower hardship scores reported among pregnant women exposed to the storm	13.4 years	Childhood insulin secretion at 13 years of age	Insulin secretion: Adjusted linear regression standardized coefficient = 0.52, p<0.01
Dancause, 2015 (22)	Cohort	USA	Iowa Flood	2008	106	Higher reported measures of objective hardship and subjective distress compared to lower scores among pregnant women during the floods	2.5-4 years	Childhood outcomes: 1) Child BMI z-scores at age 2.5 2) Child BMI z-scores at age 4 3) Difference in BMI from age 2.5 to 4 4) Child adiposity	Beta coefficient (p-value) 1) -0.07 (p=0.56) 2) -0.22 (p=0.07) 3) 0.11 (p=0.41) 4) 0.00 (p=0.97) 5) -0.06 (p=0.72) 6) 0.03 (p=0.82)

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								(skinfolds) at age 2.5 5) Child adiposity (skinfolds) at age 4 6) Difference in adiposity from age 2.5 to 4	
Goudet, 2011 (15)	Cohort	Bangladesh	1998 Bangladesh Flood	1998	220	Maternal malnutrition among mothers of infants and young children following flood exposure defined as underweight (BMI<18.5 kg/m2) compared to normal (BMI>=18.5)	12 months	Child outcomes at 12-36 months of age: 1) Underweight (weight for age z-score <-2) 2) Stunted (height for age z-score < -2) 3) Wasted (weight for height z-score < -2)	1) Adjusted OR = 3.509, 95%CI: 1.022,12.048) 2) Adjusted OR: 4.447, 95%CI: 1.044,18.943 3) Adjusted OR: 2.097, 95%CI: 0.507, 8.671
Kroska, 2018 (21)	Longitudinal study	USA	Iowa Flood	2008	103	Levels of maternal stress among those exposed to Iowa floods	2.5 years	Children outcomes at 2.5 years: BMI (kg/m2)	Standardized coefficient: 0.2071 (p=0.0322)
Liu, 2016 (25)	Longitudinal study	Canada	Quebec Ice Storm	1998	52-111 at different time points	Levels of maternal stress (objective hardship and subjective stress) among those exposed to Iowa floods	5.5-15.5 years	Children outcomes at 5.5-15.5 years: 1) BMI (kg/m2) 2) Waist to height ratio	Correlation R (p-value) Age 8.5 1) Objective hardship: 0.21 (p=0.05) 2) Objective hardship: 0.23 (p=0.03) Age 15.5 1) Objective hardship: 0.34 (p=0.02) 2) Objective hardship: 0.44 (p<0.01)
Natural disaster with adult outcomes									
Mazumder, 2010[23] (38)	Cohort	USA	1918 Influenza Pandemic	1918-1919	101,068	Infants who were born during the Influenza pandemic (third and fourth quarter of 1918, and the first, second and third quarter of 1919) compared to those born in the last quarter of 1919	~ 60-82 years	Adult outcomes at 60-82 years: 1) Diabetes 2) Heart disease	Excess cases of diabetes/heart disease: 1) 1918 Q4: 7.7% excess (95% CI: -10.6, 25.9); 1919 Q1: -5.2 (95% CI: -22.9, 12.5); 1919 Q2: 36.7% excess (95% CI: 18.9, 54.4); 2) 1918 Q4: 4.6% excess (95% CI: -4.3, 13.5); 1919 Q1: 10.9% excess (95% CI: 2.3, 19.6); 1919 Q2: 6.4% excess (95% CI: -2.2, 15.1);
Mryskyla, 2013[3] (39)	Cohort	USA	1918 Influenza Pandemic	1918-1919	81,571	Infants who were born during the Influenza pandemic (born during different quarters of 1917, 1918 and 1919) compared to those born in 1920-1924	~ 63-95 years	Adult outcomes at 63-95 years: Cardiovascular mortality	1918 Q1: HR: 1.05 (95% Ci: 0.94, 1.17); 1918 Q2: HR: 1.02 (95% Ci: 0.91, 1.14); 1918 Q3 HR: 0.99 (95% CI: 0.89, 1.10); 1918 Q4 HR: 0.97 (95% Ci: 0.87, 1.09); 1919 Q1 HR 1.07 (95% CI: 0.96, 1.19); 1919 Q2: HR:1.06 (95% CI: 0.95, 1.19);
Sotomayer, 2013 (42)	Cohort Natural experiment	Puerto Rico	Hurricane San Felipe and San Cipiran	1928 and 1932	11990	Those born during 1929 and 1933 were defined as exposed to the hurricanes compared to individuals born outside of these years between 1920 and 1940	Not reported (average age = 70)	Outcomes at ~70 years of age: 1) Diabetes 2) Hypertension 3) High cholesterol 4) CVD 5) AMI 6) Coronary/angina 7) Stroke	Linear regression estimates (p-value) 1) San Felipe: 5.94 (p<0.01); San Ciprian: 5.43 (p<0.01) 2) San Felipe: 4.73 (p<0.01); San Ciprian: 6.39 (p<0.01) 3) San Felipe: 8.85 (p<0.01); San Ciprian: 5.28 (p<0.01) 4) San Felipe: -1.48; San Ciprian: 1.33 5) San Felipe: 0.81; San Ciprian: 3.26 (p<0.01

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									6) San Felipe: 0.40; San Ciprian: -0.60 7) San Felipe: -0.25; San Ciprian: 0.58

1. Results are numbered to correspond with the numbered outcomes in the outcomes column
2. Only presenting results for early and late gestation; results for mid gestation are not included in summary table but can be found in studies
3. Not all results presented for different exposure groups

Abbreviations: AMI: Acute myocardial infarction; BMI: Body mass index; BP: Blood pressure; CI: Confidence Interval; CHD: Coronary heart disease; Chol: Cholesterol; CVD: Cardiovascular disease; HDL: High-density lipoproteins; HR: Hazard ratio; LDL: Low-density lipoproteins OR: Odds Ratio; PMNS: Prenatal Maternal Stress; Q1-Q4: quarter; SE: Standard error; Trig; Triglycerides.

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Table 4. Description of studies investigating the association between exposure to a disaster during adulthood and cardiometabolic outcomes across the life-course, by disaster type (n=34)

Study	Study design	Country	Name of disaster	Year	Sample size	Primary exposure and comparator	Average follow-up	Outcomes	Primary results[1]
Human-made disaster									
Brackbill, 2006 (70)	Cohort	USA	World Trade Center attacks (9/11)	2001	8,418	Adult survivors of 9/11 present at time of first airplane impact in a structure that was damaged compared to those that collapsed; time of evacuation before compared to after damage	1 year	1) Hypertension 2) Coronary heart disease 3) Angina 4) Heart attack 5) Diabetes 6) Stroke	1) Building type: adjusted OR: 1.2 (p<0.05); time of evacuation: adjusted OR: 0.9 (0.6, 1.3) 2) Building type: adjusted OR: 0.8 (0.4, 1.6); time of evacuation: adjusted OR: 0.5 (0.1, 2.2) 3) Building type: adjusted OR: 0.8 (0.4, 1.6); time of evacuation: adjusted OR: 0.7 (0.2, 3.1) 4) Building type: adjusted OR: 2.1 (0.9, 4.9); 5) Time of evacuation: adjusted OR: 0.7 (0.3, 1.7) 6) Building type: adjusted OR: 1.5 (0.6, 4.0)
Dirkzwager, 2007 (53)	Cohort	Netherlands	Fireworks depot explosion	2000	896	PTSD among those exposed to the fireworks disaster 19 months following the disaster compared to those with no PTSD exposed to the fireworks explosion	18 months	1) Cardiovascular 2) Vascular problems	1) Physical health problems OR: 1.23; 95% CI: 0.78, 1.94; New health problems (not present pre disaster): 1.11; 0.65, 1.89 2) Physical health problems OR: 2.12 95% CI: 1.23, 3.68; 1.92; New health problems (not present pre disaster) OR: 1.92, 95% CI: 1.04, 3.55.
Dorn, 2007 (43)	Cohort	Netherlands	Volendam Pub Fire	2001	2255	Parents of children with burns from fire parents of children without burns, bereaved parents compared to community controls who were not trapped in fire	4 years	Incidence of hypertension	Bereaved parents: OR: 2.42, 95%CI: 0.90, 6.55); parents of victims with burns: OR: 1.43, 95%CI: 0.97, 2.11; parents of victims without burns: OR:1.44, 95%CI: 0.92, 2.26)
Gerin, 2005 (71)	Pre/post design/quasi experimental	USA	World Trade Center attacks (9/11)	2001	528	Adults 2 months before 9/11 compared to 2 months after 9/11 across 4 cities (Chicago, Washington DC, New York, and Mississippi)	4 months	Systolic blood pressure	Difference (SE) New York: 1.58 (0.82) p<0.05 Chicago: 2.15 (0.32) p<0.001 Mississippi: 2.92 (0.67) p<0.001 Washington DC: 8.67 (1.16) p<0.001
Huizink, 2006 (62)	Cohort	Netherlands	Amsterdam Air Disaster	1992	1996	Police officers and firefighters who performed at least one disaster-related task compared to professional colleagues who did not perform any disaster-related tasks	8.5 years	Cardiovascular complaints	Adulthood outcomes Police officers: OR: 1.76 (95% CI: 1.35, 2.29) Firefighters: OR: 3.3 (95% CI: 1.70, 6.41)
Jordan, 2011A (57)	Prospective cohort	USA	World Trade Center attacks (9/11)	2001	39,324	9/11-related PTSD compared to no PTSD	2.9 years	Heart disease	Women adjusted OR: 1.68 (95% CI: 1.33, 2.12) Men adjusted OR: 1.62 (95% CI: 1.34, 1.96)
Jordan, 2011B (63)	Prospective cohort study	USA	World Trade Center attacks (9/11)	2001	39324	Low, intermediate and high exposure to 9/11	2.9 years	Heart disease mortality	Intermediate exposure: HR: 1.21 (95% CI: 80, 1.83) High exposure: HR: 2.06 (95% CI: 1.10, 3.86)
Jordan, 2013[2] (49)	Cohort	USA	World Trade Center attacks (9/11)	2001	46,346	Low, intermediate and high exposure to 9/11	7 years	CVD hospitalizations	Rescue/recovery workers: women: high: adjusted HR: 3.29 (95% CI: 0.85, 12.69); men: high: 1.82 (95% CI: 1.06, 3.13) Non-rescue/recovery workers: women: high: adjusted HR: 0.88 (95% CI: 0.54, 1.43); men: high: adjusted HR: 0.94 (95% CI: 0.60, 1.47)

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Kong, 2019 (50)	Pre/post design/quasi experimental	South Korea	Sewol Ferry Disaster	2014	73,632	Exposure to the Sewol Ferry Disaster in one-week periods from May 21 through June 17, 2014 compared to the reference period (March 2015-April 2015)	8 weeks	Adulthood outcomes 1) Acute MI 2) Angina	1) 8 weeks after Sewol: IRR: 0.91 (95% CI: 0.81, 1.02) 2) 8 weeks after Seowl: IRR: 0.93 (95% CI: 0.85, 1.01)
Lin, 2010 (68)	Pre/post design/quasi experimental	USA	World Trade Center attacks (9/11)	2001	Not reported	Areas affected by 9/11 compared to areas not affected by 9/11	10 years	Adulthood outcomes for Cardiovascular disease hospitalizations	Prevalence ratio (95% CI): 08/14–09/10: 0.51 (95% CI: 0.26, 1.00) 09/11–09/17: 0.56 (95% CI: 0.28, 1.11); 09/18-09/24: 0.77 (95% CI: 0.44, 1.32); 09/25-10/01: 0.49 (95% CI: 0.24, 1.00); 10/02-10/08: 0.98 (95% CI: 0.53, 1.87); 10/09-10/15: 1.09 (95% CI: 0.60, 1.98); 10/16–10/22: 0.50 (95% CI: 0.26, 0.95); 10/23–10/29: : 0.45 (95% CI: 0.20, 0.98); 10/30–11/05: 0.48 (95% CI: 0.23, 0.97)
Yu, 2018 (69)	Cohort Study	USA	World Trade Center attacks (9/11)	2001	42,527	9/11-related PTSD compared to no PTSD	13 years	Stroke	Adjusted HR: 1.69 (95% CI: 1.42, 2.02)
Natural disaster									
An, 2015 (67)	Cross-sectional	USA	Hurricane Ike	2008	19	Psychological strains among Hurricane Ike survivors	3 months	1) Blood glucose (mg/dL) 2) Obesity (BMI; kg/m2)	Mean (high vs low) and standard deviation: PTSD symptom: 22.44 (4.93) vs.12.86 (10.48); p=0.014; perceived stress:23.00 (5.03) vs. 28.11 (5.07) p=0.048 2) 28.43 kg/m^2 (3.92) vs. 20.83kg/m^2 (3.92) p=0.018
Baum, 2019 (14)	Cohort study	USA	Hurricane Sandy	2012	81 544	Veterans who used Manhattan VA Medical Center before Hurricane Sandy and experienced decreased access to health care services compared to veterans who used the VA Bronx, Brooklyn or West Haven medical centers	2 years	1) Uncontrolled hypertension 2) Systolic BP (mmHg) 3) Diastolic BP (mmHg) 4) Uncontrolled diabetes 5) Uncontrolled cholesterol (mg/dL) 6) Weight (lbs)	% differential change (95% CI): 1) 6 months: 19.3 (4.5, 8.7); 12 months: 4.5 (3.1, 5.9); 18 months: 5.0 (3.5, 6.5); 24 months: 2.1 (0.5, 3.6) 2) 6 months: 3.8 (3.1, 4.5); 12 months: 2.3 (1.7, 2.9); 18 months: 3.1 (2.5, 3.7); 1.5 (0.9, 2.1) 3) 6 months: 2.7 (2.3, 3.1); 12 months: 2.2 (1.9, 2.6); 18 months 2.9 (2.5, 3.3), 24 months: 2.0 (1.7, 2.4) 4) 6 months: 1.9 (-0.1, 4.0); 12 months: 1.7 (-0.3, 3.6); 18 months: 0.8 (-1.2, 2.8); 24 months: -0.2 (-2.2, 1.8) 5) 6 months: 1.3 (-0.1, 2.6); 12 months: 0.6 (-0.6, 1.8); 18 months: -0.7 (-2.0, 0.6); 24 months: -0.2 (-1.4, 1.0) 6) 6 months: -0.1 (-0.5, 0.2); 12 months: 0.2 (-0.2, 0.5); 18 months: -0.2 (-0.5, 0.2); 24 months: 0.5 (0.1, 0.9)
Becquart, 2018 (48)	Time series/quasi experimental	USA	Hurricane Katrina	2005	383 552	Exposure to hurricane before, during and after among older adults in Louisiana in the affected counties	1 year	Hospitalizations due to CVD	Mean (SD) Orleans: T1: 7.25 (2.44); T2: 3.91 (1.45)*; T3; 18.47 (17.3)*; T4:13.76 (6.51)*; T5: 9.54 (2.78); T6: 4.69 (2.08) Jefferson: T1: 5.90 (1.90); T2: 5.01(1.52); T3: 8.118 (3.70)*; T4: 7.25 (2.15)*; T5: 5.26 (1.53); T6: 4.65 (1.57)* East BR: T1: 8.69 (2.74); T2: 9.11 (2.69); T3: 6.52 (2.58); T4: 6.55 (1.70)*; T5: 6.69 (2.42)*; T6: 7.39

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									(2.37)* * p<0.05
Bich, 2011 (61)	Cross-sectional	Vietnam	Historic flood in 2008	2008	781	Individuals who resided in households affected by flood in Hanoi in 2005 compared to non-affected households	1 month	Worsening hypertension after rain/flood	Rural: non flooded 33.3%; flooded: 51.2%; Urban: non flooded 20.3% flooded: 42.9%* * p<0.05
Fonseca, 2009 (59)	Cohort	USA	Hurricane Katrina	2005	1795	Adults with diabetes who were in the databases from 3 health care systems 6 months before the hurricane (Feb 28, 2005-Aug 27, 2005) compared to 6-16 months after the hurricane (March 1, 2006-December 31, 2006)	22 months	1) Glycemic control/A1C 2) Systolic BP (mmHg) 3) Diastolic BP (mmHg) 4) HDL (mg/dL) 5) LDL (mg/dL) 6) Triglycerides (mg/dL)	Difference in mean (SD) 1) 0.1 (1.6) (p<0.01) 2) 10.5 (20.4) (p<0.01) 3) 3.9 (13.1) (p<0.01) 4) 6.0 (35.5) (p<0.01) 5) -2.4 (9.2) (p<0.01) 6) -2.1 (137.5) (p=0.60)
Gautam, 2009 (55)	Retrospective cohort	USA	Hurricane Katrina	2005	396	Exposure to Hurricane Katrina compared to period before hurricane	4 years	Incidence of AMI admission	Pre-Katrina group: 150 admissions for AMI (0.71%) Post-Katrina group: 246 admission for AMI (2.18%) p<0.0001
Hendrickson, 2996 (64)	Pre/post design/quasi experimental	USA	Hurricane Iniki	1992	Not reported	Mortality data for residents of Kauai for 5-year period 1987-1991 prior disaster compared to the year immediately following the hurricane (Oct 1 1992-Sept 30, 1993)	6 years total	Mortality by: 1) Heart disease 2) Stroke 3) Diabetes mellitus	1) RR: 0.96 (95% CI: 0.79-1.17) 2) RR: 1.20 (95% CI: 0.81-1.78) 3) RR: 2.61 (95% CI: 1.44-4.74)
Husarewycz, 2014 (72)	Cross-sectional	USA	Natural disaster/terrorism	Lifetime disaster experience	34,653	Number of times directly experienced natural disaster/terrorism compared to no experiences	1 year	1) Cardiovascular disease 2) Hypertension/arteriosclerosis 3) Diabetes 4) Obesity	1) OR: 1.28 (95% CI: 1.10, 1.49) 2) OR: 1.08 (95% CI: 0.95, 1.24) 3) OR: 1.10 (95% CI: 0.94, 1,29) 4) OR: 1.01 (95% CI: 0.90, 1.14)
Jiao, 2012 (54)	Retrospective cohort observational study	USA	Hurricane Katrina	2005	Not reported	2 years prior to the hurricane (August 29, 2003 - August 28, 2005) compared to the 3-year period post-Hurricane Katrina (February 14, 2006 - February 13, 2009)	5 years	Incidence of AMI	Pre-Katrina: 0.7% compared to post-Katrina: 2% (p<0.001)
Joseph, 2014 (51)	Cohort/longitudinal	USA	Hurricane Katrina	2005	215	African Americans who experienced acute unemployment due to Hurricane compared to those who remained employed	4 years	Cardiometabolic event	Adjusted OR = 5.65, p < .05
Karatzias , 2015 (44)	Cross-sectional	Hong Kong	Natural disaster	Not specified	1147	Experience of natural disaster across life-course compared to less or no experiences	Survey done from August to December 2012	1) Hypertension 2) Heart disease 3) Diabetes	Chi square (p-value) 1) X^2: 3.3 p=0.047 2) X^2: 3.6 p=0.056 3) X^2: 2.5 p=0.088
Kim, 2017 (58)	Pre/post design/quasi experimental	USA	Hurricane Sandy	2012	Not reported	The month of Hurricane Sandy (October 28, 2012-November 27, 2012) compared to the same month in 2009-2011; Sandy quarter (October 28, 2012-January 27, 2013) compared to the same period in 2009-2011 among elderly people	Sandy month: Oct 28, 2012-Nov 27, 2012 Sandy quarter: Oct 28, 2012-Jan 27, 2013	Cardiovascular disease-related death	Sandy quarter: adjusted RR: 1.06; 95% CI: 1.02, 1.10 Sandy month: adjusted RR: 1.10; 95% CI: 1.02, 1.18
Koroma, 2019 (66)	Cross-sectional	Sierra-Leone	Ebola	2014-2015	10011	District facilities for six-month periods before Ebola (June-December 2012), during Ebola (June-December 2014) and post-Ebola (June-December 2015)	June-December 2012,2013,2014	1) Cardiovascular disease 2) Hypertension 3) Diabetes	Number of people with non-communicable diseases 1) Pre-Ebola: 355, Ebola: 300, Post-Ebola: 196 2) Pre-Ebola: 282, Ebola: 230, Post-Ebola: 457 3) Pre-Ebola: 3716, Ebola: 1851, Post-Ebola: 2463

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Lawrence, 2019 (45)	Prospective Cohort Study	USA	Superstorm Sandy	2012	651858	Residing in counties affected by Superstorm Sandy compared to non-affected counties; Superstorm period compared to reference periods (short-term and long-term (4 and 12 months))	1 year	Emergency department visits, outpatient visits, and hospital admissions for cardiovascular disease	4 months: Superstorm sandy period: RR: 2.10 (95% CI: 2.10, 2,10); Affected counties RR: 2.62 (95% CI: 2.62, 2.63) 12 months: Superstorm sandy period: RR: 2.01 (95% CI: 2.00, 2.01); Affected counties RR: 2.64 (95% CI: 2.64, 2.65)
McKinney, 2011 (52)	Time-series/quasi experimental	USA	Hurricane Charley, Frances, Ivan and Jeanne and Tropical Storm Bonnie	2004	Not reported	Counties in 2004 directly impacted by the hurricanes, ordered evacuated regardless of the level of damage that occurred and adjacent to the impact zone where direct deaths were reported compared to compared to the same areas in 2001-2006	5 years	Heart-related mortality	Results shown graphically Significantly elevated heart-related deaths
Moscona, 2019 (75)	Retrospective cohort study	USA	Hurricane Katrina	2005	2-year pre-Katrina - 21,079 10-year post Katrina - 84,751	Individuals who lived in New Orleans who went to the Tulane University Health Sciences Center compared to the two months prior to the Hurricane	12 years	1) Hospital admission for Incidence of AMI 2) Changes in CAD 3) Changes in diabetes mellitus 4) Changes in hypertension 5) Changes in hyperlipidemia	Pre-Katrina versus Post-Katrina 1) 0.7% vs 2.8% (p<0.001) 2) 36.4% vs. 47.9%, (p= 0.01) 3) 31.3% vs. 39.9% (p= 0.04) 4) 71.1% vs 80.6%, (p=0.12) 5) 45.4% vs. 59.3% (p = 0.005)
Nagayoshi, 2015 (56)	Pre/post design/quasi experimental	Japan	July 12, 2012 heavy rain and mudslides "mountain tsunamis"	2012	583	Individuals who were admitted at Aso Central Hospital from July 12 to August 31, 2012 compared to the 3-year period before flooding	3 years	1) Hospital admission for cardiovascular outcomes 2) Cardiovascular events	1) 4.5/month before compared to 16.8/month after; p < 0.01 2) 5.1/month before compared to 16.8/month after; p < 0.01).
Ng, 2011 (73)	Cohort	United Kingdom	Flood	June, 2007	1,743	Diabetics affected by floods compared to diabetics not affected by floods	2 years	Glycemic control/HbA1c levels	Mean HbA1c before 7.6% (7.5–7.7) vs. After 7.9% (7.7–8.0); p = 0.002
Peters, 2013 (65)	Retrospective Cohort	USA	Hurricane Katrina	2005	698	Admission to Tulane University Health Sciences Centre in the 3-year period post-Katrina compared to the 6-year period pre-Katrina	9 years	Chronobiology of AMI onset	Pre-Katrina: 45% vs post-Katrina: 30.9%, p=0.002
Rey, 2007 (74)	Longitudinal	France	6 Heat Waves	1971-2003	Not reported	Time of heat wave compared to the expected mortality during the 3 years prior to the heat wave	N/A	Excess cardiovascular disease death	41% in 1975 to 23% in 2003
Sliva-Palacios, 2015 (60)	Pre/post design/quasi experimental	USA	Oklahoma Tornado	2013	22,607	Victims of the Oklahoma Tornado Outbreaks compared to the same people pre-tornado and same period one year prior	6 Months	Hospital admissions for CVE	One year prior: PR = 1.05 95% CI: 0.91 to 1.21, p = 0.50; 3-month pre-tornado: PR = 0.96, 95% CI: 0.83 to 1.21, p = 0.63
Thethi, 2010 (76)	Cohort	USA	Hurricane Katrina	2005	1523	Individuals exposed to Hurricane Katrina compared to 6-16 months pre-Hurricane Katrina (February 28, 2005-August 27, 2005)	6 months before Katrina and 6-16 months after Katrina and follow up 1 year after the first post-Katrina visit	1) LDL (mg/dL) 2) HDL (mg/dL) 3) Triglycerides (mg/dL) 4) Cholesterol (mg/dL) 5) Diastolic blood pressure (mm/Hg) 6) Systolic blood pressure (mm/Hg)	Mean Pre-Katrina vs Post-Katrina: 1) 101.34 vs 107.44 2) 43.53 vs 41.08 3) 160.8 vs 158.65 4) 181.9vs 181.39 5) 70.99 vs 74.88 6) 130.73 vs 141.27

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Vanasse, 2016 (46)	Population-based retrospective cohort study with a time series design	Canada	Flood of Saint-Jean-sur-Richelieu	2011	111,317	Exposure to flood in spring 2011 and exposure to flooded area (Area 1) compared to same period in Spring 2010 and 2012 and non-flooded areas in the same town (Areas 2, 3 and 4)	4 months	Acute cardiovascular event	Spring 2010: aOR 1.25 (95%CI: 0.81, 1.92); spring 2012 aOR: 1.27 (95% CI: 0.82, 1.92); Non-flooded areas 2: aOR: 1.11 (95%CI: 0.79, 1.59), Non-flooded area 3: aOR: 0.94 (95% CI: 0.68, 1.32); Non-flooded area 4: aOR 1.08 (95%CI: 0.78, 1.47)

1. 1. Results are numbered to correspond with the numbered outcomes in the outcomes column
2. Only results for extreme outcomes are reported in table, remaining results can be found in the study.

Abbreviations: aOR: Adjusted odds ratio; AMI: Acute myocardial infarction BMI: Body mass index; BP: Blood pressure; CI: Confidence Interval; CAD: Coronary artery disease; CHD: Coronary heart disease; CVD: Cardiovascular disease; CVE: Cardiovascular events; HbA1c ; Hemoglobin A1C; HDL: High-density lipoproteins; HR: Hazard ratio; LDL: Low-density lipoproteins OR: Odds Ratio; PR: Prevalence Ratio; PTSD: Post-traumatic stress disorder; RR: Relative risk; SD: Standard deviation; SE: Standard Error; Trig; Triglycerides.; T1-T6; Time.

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Figure 1. PRISMA flow diagram

Ethical Approval Statement

Not applicable

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References

1. World Health Organization. WHO | Definitions: emergencies [Internet]. WHO. World Health Organization; 2020 [cited 2020 May 12]. Available from: https://www.who.int/hac/about/definitions/en/

2. Mohamed SI. An overview on disasters. Disaster Prevention and Management: An International Journal. 2007 Jan 1;16(5):687–703.

3. Giorgadze T, Maisuradze I, Japaridze A, Utiashvili Z, Abesadze G. Disasters and their consequences for public health. Georgian Med News. 2011 May;(194):59–63.

4. Ordway D-M, February 1 JR, 2016. Global trends in human infectious disease: Rising number of outbreaks, fewer per-capita cases [Internet]. Journalist's Resource. 2016 [cited 2020 May 8]. Available from: https://journalistsresource.org/studies/society/public-health/global-rise-human-infectious-disease-outbreaks/

5. Brooks SK, Webster RK, Smith LE, Woodland L, Wessely S, Greenberg N, et al. The psychological impact of quarantine and how to reduce it: rapid review of the evidence. The Lancet. 2020 Mar 14;395(10227):912–20.

6. Bell SA, Horowitz J, Iwashyna TJ. Health Outcomes After Disaster for Older Adults With Chronic Disease: A Systematic Review. Gerontologist [Internet]. [cited 2020 May 8]; Available from: https://academic.oup.com/gerontologist/advance-article/doi/10.1093/geront/gnz123/5565119

7. Jackson JL, George S, Hinchey S. Medically Unexplained Physical Symptoms. J Gen Intern Med. 2009 Apr;24(4):540–2.

8. Chan EYY, Sondorp E. Medical interventions following natural disasters: missing out on chronic medical needs. Asia Pac J Public Health. 2007;19 Spec No:45–51.

9. life course approach to chronic disease epidemiology: conceptual models, empirical challenges and interdisciplinary perspectives | International Journal of Epidemiology | Oxford Academic [Internet]. [cited 2021 Feb 28]. Available from: https://academic.oup.com/ije/article/31/2/285/617688

10. Kuh D, Ben-Shlomo Y, Lynch J, Hallqvist J, Power C. Life course epidemiology. Journal of Epidemiology & Community Health. 2003 Oct 1;57(10):778–83.

11. James SL, Abate D, Abate KH, Abay SM, Abbafati C, Abbasi N, et al. Global, regional, and national incidence, prevalence, and years lived with disability for 354 diseases and injuries for 195 countries and territories, 1990–2017: a systematic analysis for the Global Burden of Disease Study 2017. The Lancet. 2018 Nov;392(10159):1789–858.

12. Forouzanfar MH, Afshin A, Alexander LT, Anderson HR, Bhutta ZA, Biryukov S, et al. Global, regional, and national comparative risk assessment of 79 behavioural, environmental and occupational, and metabolic risks or clusters of risks, 1990–2015: a systematic analysis for the Global Burden of Disease Study 2015. The Lancet. 2016 Oct;388(10053):1659–724.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

13. Tan SY, Yip A. Hans Selye (1907–1982): Founder of the stress theory. Singapore Med J. 2018 Apr;59(4):170–1.

14. Baum A., Barnett M.L., Wisnivesky J., Schwartz M.D. Association between a Temporary Reduction in Access to Health Care and Long-term Changes in Hypertension Control among Veterans after a Natural Disaster. JAMA Network Open. 2019;2(11):e1915111.

15. Goudet S, Griffths P, Bogin BA. Mother's body mass index as a predictor of infant's nutritional status in the post-emergency phase of a flood. Disasters. 2011 Oct;35(4):701–19.

16. Moher D, Liberati A, Tetzlaff J, Altman DG, PRISMA Group. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. Ann Intern Med. 2009 Aug 18;151(4):264–9, W64.

17. Kamel M, Smith BT, Wahi G, Carsley S, Birken CS, Anderson LN. Continuous cardiometabolic risk score definitions in early childhood: a scoping review. Obesity Reviews. 2018;19(12):1688–99.

18. G B, G T, Kk N, V K, K V, A S, et al. Impact of major earthquakes on the incidence of acute coronary syndromes - A systematic review of the literature [Internet]. Vol. 59, Hellenic journal of cardiology : HJC = Hellenike kardiologike epitheorese. Hellenic J Cardiol; 2018 [cited 2020 Sep 10]. Available from: https://pubmed.ncbi.nlm.nih.gov/29807192/

19. Covidence - Better systematic review management [Internet]. Covidence. [cited 2020 Sep 10]. Available from: https://www.covidence.org/

20. Aromataris E, Munn Z, editors. JBI Manual for Evidence Synthesis [Internet]. JBI; 2020 [cited 2020 Sep 10]. Available from: https://wiki.jbi.global/display/MANUAL

21. Kroska EB, O'Hara MW, Elgbeili G, Hart KJ, Laplante DP, Dancause KN, et al. The impact of maternal flood-related stress and social support on offspring weight in early childhood. Arch Womens Ment Health. 2018 Apr 1;21(2):225–33.

22. Dancause KN, Laplante DP, Hart KJ, O'Hara MW, Elgbeili G, Brunet A, et al. Prenatal stress due to a natural disaster predicts adiposity in childhood: the Iowa Flood Study. Journal of obesity. 2015;2015(101526295):570541.

23. Dancause KN, Laplante DP, Fraser S, Brunet A, Ciampi A, Schmitz N, et al. Prenatal exposure to a natural disaster increases risk for obesity in 51/2-year-old children. Pediatric research. 2012;71(1):126–31.

24. Dancause KN, Veru F, Andersen RE, Laplante DP, King S. Prenatal stress due to a natural disaster predicts insulin secretion in adolescence. Early Hum Dev. 2013 Sep;89(9):773–6.

25. Liu GT, Dancause KN, Elgbeili G, Laplante DP, King S. Disaster-related prenatal maternal stress explains increasing amounts of variance in body composition through childhood and adolescence: Project Ice Storm. Environ Res. 2016;150:1–7.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

26. Cao-Lei L, Dancause KN, Elgbeili G, Laplante DP, Szyf M, King S. Pregnant women's cognitive appraisal of a natural disaster affects their children's BMI and central adiposity via DNA methylation: Project Ice Storm. Early Human Development. 2016 Dec 1;103:189–92.

27. Oni O, Harville E, Xiong X, Buekens P. Relationships among stress coping styles and pregnancy complications among women exposed to Hurricane Katrina. Journal of obstetric, gynecologic, and neonatal nursing : JOGNN. 2015;44(2):256–67.

28. Xiao J, Huang M, Zhang W, Rosenblum A, Ma W, Meng X, et al. The immediate and lasting impact of Hurricane Sandy on pregnancy complications in eight affected counties of New York State. The Science of the total environment. 2019;678(uj0, 0330500):755–60.

29. Roseboom TJ, van der Meulen JH, Ravelli AC, Osmond C, Barker DJ, Bleker OP. Effects of prenatal exposure to the Dutch famine on adult disease in later life: an overview. Mol Cell Endocrinol. 2001 Dec 20;185(1–2):93–8.

30. de Rooij SR, Painter RC, Holleman F, Bossuyt PM, Roseboom TJ. The metabolic syndrome in adults prenatally exposed to the Dutch famine. Am J Clin Nutr. 2007 Oct 1;86(4):1219–24.

31. Ekamper P, van Poppel F, Stein AD, Bijwaard GE, Lumey LH. Prenatal Famine Exposure and Adult Mortality From Cancer, Cardiovascular Disease, and Other Causes Through Age 63 Years. Am J Epidemiol. 2015 Feb 15;181(4):271–9.

32. Lumey LH, Martini LH, Myerson M, Stein AD, Prineas RJ. No relation between coronary artery disease or electrocardiographic markers of disease in middle age and prenatal exposure to the Dutch famine of 1944-5. Heart. 2012 Nov;98(22):1653–9.

33. Painter RC, de Rooij SR, Bossuyt PM, Simmers TA, Osmond C, Barker DJ, et al. Early onset of coronary artery disease after prenatal exposure to the Dutch famine. Am J Clin Nutr. 2006 Aug;84(2):322–7; quiz 466–7.

34. Ravelli AC, van Der Meulen JH, Osmond C, Barker DJ, Bleker OP. Obesity at the age of 50 y in men and women exposed to famine prenatally. Am J Clin Nutr. 1999 Nov;70(5):811–6.

35. Stein AD, Zybert PA, van der Pal-de Bruin K, Lumey LH. Exposure to famine during gestation, size at birth, and blood pressure at age 59 y: evidence from the Dutch Famine. Eur J Epidemiol. 2006;21(10):759–65.

36. Hult M, Tornhammar P, Ueda P, Chima C, Bonamy A-KE, Ozumba B, et al. Hypertension, diabetes and overweight: looming legacies of the Biafran famine. PLoS ONE. 2010 Oct 22;5(10):e13582.

37. Huang C, Li Z, Wang M, Martorell R. Early life exposure to the 1959-1961 Chinese famine has long-term health consequences. J Nutr. 2010 Oct;140(10):1874–8.

38. Mazumder B, Almond D, Park K, Crimmins EM, Finch CE. Lingering prenatal effects of the 1918 influenza pandemic on cardiovascular disease. J Dev Orig Health Dis. 2010 Feb;1(1):26–34.

39. Myrskyla M, Mehta NK, Chang VW. Early life exposure to the 1918 influenza pandemic and old-age mortality by cause of death. American journal of public health. 2013;103(7):e83-90.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

40. Bercovich E, Keinan-Boker L, Shasha SM. Long-Term Health Effects in Adults Born during the Holocaust. 2014;16:5.

41. Schreier NK, Moltchanova EV, Blomstedt PA, Kajantie E, Eriksson JG. Prenatal exposure to wartime stress – long-term effect on coronary heart disease in later life. Annals of Medicine. 2011 Nov 1;43(7):555–61.

42. Sotomayor O. Fetal and infant origins of diabetes and ill health: evidence from Puerto Rico's 1928 and 1932 hurricanes. Econ Hum Biol. 2013 Jul;11(3):281–93.

43. Dorn T, Yzermans CJ, Guijt H, van der Zee J. Disaster-related stress as a prospective risk factor for hypertension in parents of adolescent fire victims. American journal of epidemiology. 2007;165(4):410–7.

44. Karatzias T, Yan E, Jowett S. Adverse life events and health: A population study in Hong Kong. Journal of Psychosomatic Research. 2015 Feb;78(2):173–7.

45. Lawrence WR, Lin Z, Lipton EA, Birkhead G, Primeau M, Dong G-H, et al. After the Storm: Short-term and Long-term Health Effects Following Superstorm Sandy among the Elderly. Disaster med public health prep. 2019 Feb;13(1):28–32.

46. Vanasse A, Cohen A, Courteau J, Bergeron P, Dault R, Gosselin P, et al. Association between Floods and Acute Cardiovascular Diseases: A Population-Based Cohort Study Using a Geographic Information System Approach. International journal of environmental research and public health. 2016;13(2):168.

47. Trasande L, Koshy TT, Gilbert J, Burdine LK, Marmor M, Han X, et al. Cardiometabolic profiles of adolescents and young adults exposed to the World Trade Center Disaster. Environmental research. 2018;160(ei2, 0147621):107–14.

48. Becquart NA, Naumova EN, Singh G, Chui KKH. Cardiovascular Disease Hospitalizations in Louisiana Parishes' Elderly before, during and after Hurricane Katrina. International journal of environmental research and public health. 2018;16(1).

49. Jordan HT, Stellman SD, Morabia A, Miller-Archie SA, Alper H, Laskaris Z, et al. Cardiovascular disease hospitalizations in relation to exposure to the September 11, 2001 World Trade Center disaster and posttraumatic stress disorder. Journal of the American Heart Association. 2013;2(5):e000431.

50. Kong SY, Song KJ, Shin SD, Ro YS. Cardiovascular Events after the Sewol Ferry Disaster, South Korea. Prehospital and disaster medicine. 2019;34(2):142–8.

51. Joseph NT, Matthews KA, Myers HF. Conceptualizing health consequences of Hurricane Katrina from the perspective of socioeconomic status decline. Health psychology : official journal of the Division of Health Psychology, American Psychological Association. 2014;33(2):139–46.

52. McKinney N, Houser C, Meyer-Arendt K. Direct and indirect mortality in Florida during the 2004 hurricane season. International journal of biometeorology. 2011;55(4):533–46.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

53. Dirkzwager AJE, van der Velden PG, Grievink L, Yzermans CJ. Disaster-related posttraumatic stress disorder and physical health. Psychosomatic medicine. 2007;69(5):435–40.

54. Jiao Z, Kakoulides SV, Moscona J, Whittier J, Srivastav S, Delafontaine P, et al. Effect of Hurricane Katrina on incidence of acute myocardial infarction in New Orleans three years after the storm. The American journal of cardiology. 2012;109(4):502–5.

55. Gautam S, Menachem J, Srivastav SK, Delafontaine P, Irimpen A. Effect of Hurricane Katrina on the incidence of acute coronary syndrome at a primary angioplasty center in New Orleans. Disaster medicine and public health preparedness. 2009;3(3):144–50.

56. Nagayoshi Y, Yumoto S, Sakaguchi K, Shudo C, Takino S, Hashiyama M, et al. Heart attacks triggered by huge mud slides in mountain regions and severe flooding in inhabited areas. Journal of cardiology. 2015;65(2):117–20.

57. Jordan HT, Miller-Archie SA, Cone JE, Morabia A, Stellman SD. Heart disease among adults exposed to the September 11, 2001 World Trade Center disaster: results from the World Trade Center Health Registry. Preventive medicine. 2011;53(6):370–6.

58. Kim S, Kulkarni PA, Rajan M, Thomas P, Tsai S, Tan C, et al. Hurricane Sandy (New Jersey): Mortality Rates in the Following Month and Quarter. American journal of public health. 2017;107(8):1304–7.

59. Fonseca VA, Smith H, Kuhadiya N, Leger SM, Yau CL, Reynolds K, et al. Impact of a natural disaster on diabetes: exacerbation of disparities and long-term consequences. Diabetes care. 2009;32(9):1632–8.

60. Silva-Palacios F, Casanegra AI, Shapiro A, Phan M, Hawkins B, Li J, et al. Impact of tornadoes on hospital admissions for acute cardiovascular events. Thrombosis Research. 2015 Nov;136(5):907–10.

61. Bich TH, Quang LN, Ha LTT, Hanh TTD, Guha-Sapir D. Impacts of flood on health: epidemiologic evidence from Hanoi, Vietnam. Global health action. 2011;4(101496665):6356.

62. Huizink AC, Slottje P, Witteveen AB, Bijlsma JA, Twisk JWR, Smidt N, et al. Long term health complaints following the Amsterdam Air Disaster in police officers and fire-fighters. Occupational and environmental medicine. 2006;63(10):657–62.

63. Jordan HT, Brackbill RM, Cone JE, Debchoudhury I, Farfel MR, Greene CM, et al. Mortality among survivors of the Sept 11, 2001, World Trade Center disaster: results from the World Trade Center Health Registry cohort. Lancet (London, England). 2011;378(9794):879–87.

64. Hendrickson LA, Vogt RL. Mortality of Kauai Residents in the 12-Month Period following Hurricane Iniki. American Journal of Epidemiology. 1996 Jul 15;144(2):188–91.

65. Peters MN, Katz MJ, Moscona JC, Alkadri ME, Khazi Syed RH, Turnage TA, et al. Effect of Hurricane Katrina on chronobiology at onset of acute myocardial infarction during the subsequent three years. The American journal of cardiology. 2013;111(6):800–3.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

66. Koroma IB, Javadi D, Hann K, Harries AD, Smart F, Samba T. Non-communicable diseases in the Western Area District, Sierra Leone, following the Ebola outbreak. F1000Res [Internet]. 2019 Nov 19 [cited 2020 Jul 30];8. Available from: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6652098/

67. An K, Salyer J, Kao H-FS. Psychological strains, salivary biomarkers, and risks for coronary heart disease among hurricane survivors. Biological research for nursing. 2015;17(3):311–20.

68. Lin S, Gomez MI, Gensburg L, Liu W, Hwang S-A. Respiratory and cardiovascular hospitalizations after the World Trade Center disaster. Archives of environmental & occupational health. 2010;65(1):12–20.

69. Yu S, Alper HE, Nguyen A-M, Brackbill RM. Risk of Stroke Among Survivors of the September 11, 2001, World Trade Center Disaster. Journal of occupational and environmental medicine. 2018;60(8):e371–6.

70. Brackbill RM, Thorpe LE, DiGrande L, Perrin M, Sapp JH 2nd, Wu D, et al. Surveillance for World Trade Center disaster health effects among survivors of collapsed and damaged buildings. Morbidity and mortality weekly report Surveillance summaries (Washington, DC : 2002). 2006;55(2):1–18.

71. Gerin W, Chaplin W, Schwartz JE, Holland J, Alter R, Wheeler R, et al. Sustained blood pressure increase after an acute stressor: the effects of the 11 September 2001 attack on the New York City World Trade Center: Journal of Hypertension. 2005 Feb;23(2):279–84.

72. Husarewycz MN, El-Gabalawy R, Logsetty S, Sareen J. The association between number and type of traumatic life experiences and physical conditions in a nationally representative sample. General Hospital Psychiatry. 2014 Jan;36(1):26–32.

73. Ng J, Atkin SL, Rigby AS, Walton C, Kilpatrick ES. The effect of extensive flooding in Hull on the glycaemic control of patients with diabetes: Impact of flooding on glycaemic control. Diabetic Medicine. 2011 May;28(5):519–24.

74. Rey G, Jougla E, Fouillet A, Pavillon G, Bessemoulin P, Frayssinet P, et al. The impact of major heat waves on all-cause and cause-specific mortality in France from 1971 to 2003. Int Arch Occup Environ Health. 2007 May 2;80(7):615–26.

75. Moscona JC, Peters MN, Maini R, Katigbak P, Deere B, Gonzales H, et al. The Incidence, Risk Factors, and Chronobiology of Acute Myocardial Infarction Ten Years After Hurricane Katrina. Disaster med public health prep. 2019 Apr;13(02):217–22.

76. Thethi TK, Yau CL, Shi L, Leger S, Nagireddy P, Waddadar J, et al. Time to recovery in diabetes and comorbidities following Hurricane Katrina. Disaster medicine and public health preparedness. 2010;4 Suppl 1(101297401):S33-8.

77. Schneiderman N, Ironson G, Siegel SD. STRESS AND HEALTH: Psychological, Behavioral, and Biological Determinants. Annu Rev Clin Psychol. 2005;1:607–28.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

78. Committee on Post-Disaster Recovery of a Community's Public Health M, Policy B on HS, Medicine I of. Health Care [Internet]. Healthy, Resilient, and Sustainable Communities After Disasters: Strategies, Opportunities, and Planning for Recovery. National Academies Press (US); 2015 [cited 2021 Feb 16]. Available from: https://www.ncbi.nlm.nih.gov/books/NBK316524/

79. Puckrein GA, Egan BM, Howard G. Social and Medical Determinants of Cardiometabolic Health: The Big Picture. Ethn Dis. 25(4):521–4.

80. Schultz WM, Kelli HM, Lisko JC, Varghese T, Shen J, Sandesara P, et al. Socioeconomic Status and Cardiovascular Outcomes: Challenges and Interventions. Circulation. 2018 May 15;137(20):2166–78.

81. Mohammed SH, Habtewold TD, Birhanu MM, Sissay TA, Tegegne BS, Abuzerr S, et al. Neighbourhood socioeconomic status and overweight/obesity: a systematic review and meta-analysis of epidemiological studies. BMJ Open [Internet]. 2019 Nov 14 [cited 2020 Nov 17];9(11). Available from: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6886990/

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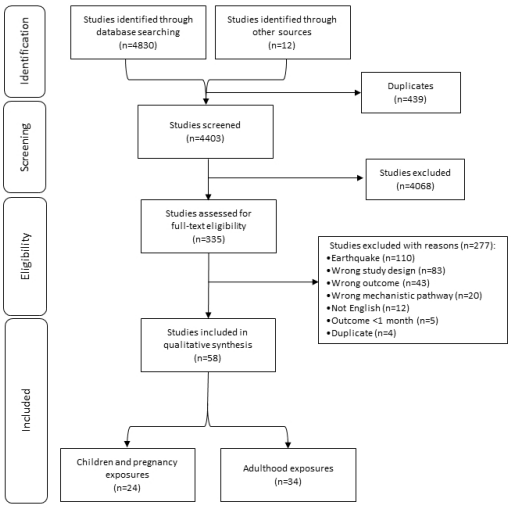


Figure 1. PRISMA flow chart

338x190mm (96 x 96 DPI)

APPENDIX

Table A1. Search strategy for Medline

1 "social isolation.mp." or Social isolation/	1479
2 quaratine.mp. or exp Quarantine/	5589
3 Pandemics/	7247
4 Epidemics/	10146
5 Coronavirus or coronavirus.mp.	19310
6 humanitarian crises.mp.	232
7 exp mass casualty incidents/ or exp natural disasters/	18883
8 public health emergency.mp.	1636
9 cardiovasucalr.mp. or exp Cardiovascular Diseases/	2605399
10 hypertension.mp. or exp Hypertension/	484321
11 exp Obesity/ or obesity.mp.	321847
12 body mass index.mp. or exp Body Mass Index/	236149
13 stroke.mp, or exp Stroke/	310109
14 myocardial infarction.mp. or exp Myocardial Infarction/	247417
15 angina.mp. or exp Angina Pectoris/	698186
16 diabetes.mp. or Diabetes Mellitus/	646582
17 9 or 10 or 11 or 12 or 13 or 14 or 15 or 16	3552066
18 Disasters/ or disaster.mp.	39131
19 1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 or 18	10917
20 17 and 19	3688
21 limit 20 to humans	2998

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Table A2. Critical appraisal using Joanna Briggs Institute (JBI) checklist for cohort studies by exposure period (n=41)

Author, Year	Were the two groups similar and recruited from the same population?	Were the exposures measured similarly to assign people to both exposed and unexposed groups?	Was the exposure measured in a valid and reliable way?	Were confounding factors identified?	Were strategies to deal with confounding factors stated?	Were the groups/participants free of the outcome at the start of the study (or at the moment of exposure)?	Were the outcomes measured in a valid and reliable way?	Was the follow up time reported and sufficient to be long enough for outcomes to occur?	Was follow up complete, and if not, were the reasons to loss to follow up described and explored?	Were strategies to address incomplete follow up utilized?	Was appropriate statistical analysis used?
Perinatal											
Dancause, 2012	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Dancause, 2013	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No	Yes
Dancause, 2015	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Unclear	Yes
de Rooij, 2007	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Dorn, 2007	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Unclear	Yes
Ekamper, 2015	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Goudet, 2011	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Unclear	Unclear	Yes
Huang, 2010	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Unclear	Unclear	Yes
Hult, 2010	Yes	Yes	Yes	Yes	Yes	Unclear	Yes	Yes	Unclear	Unclear	Yes
Kroska, 2018	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Unclear	Yes
Lei, 2016	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Unclear	Yes
Liu, 2016	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Unclear	Yes
Lumey, 2012	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mazumder, 2010	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Unclear	Unclear	Yes
Mryskyla, 2013	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Unclear	Unclear	Yes

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Painter, 2006	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Ravelli, 1999	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Roseboom, 2001	Yes	Yes	Yes	Yes	Yes	Yes	Unclear	Yes	Yes	Yes	Yes
Schrier, 2010	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Unclear	Unclear	Yes
Sotomayer, 2013	Yes	Yes	Yes	Yes	Yes	Unclear	Yes	Yes	Unclear	Unclear	Yes
Stein, 2006	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adult											
Baum, 2019	Yes	Yes	Yes	Yes	Yes		Yes	Yes	Yes	Yes	Yes
Brackbill, 2006	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Unclear	Yes	Yes
Dirkzwager, 2007	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Unclear	Unclear	Yes
Fonseca, 2009	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Not applicable	Yes
Gautam, 2009	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Unclear	Unclear	Yes
Huizink, 2006	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes
Jiao, 2012	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Unclear	Unclear	Yes
Jordan, 2011 (A)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No	Yes
Jordan, 2011 (B)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Unclear	Yes	Yes
Jordan, 2013	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Unclear	Yes	Yes
Joseph, 2014	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Unclear	Unclear	Yes
Lawrence, 2019	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Unclear	Not applicable	Yes
Moscona, 2019	Yes	Yes	Yes	Yes	No	Unclear	Yes	Yes	Unclear	Unclear	Yes
Ng, 2011	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Unclear	Unclear	Yes
Peters, 2013	Yes	Yes	Yes	No	No	No	Yes	Yes	Unclear	Unclear	Yes
Rey, 2007	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Unclear	Unclear	Yes

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Thethi, 2010	Yes	Yes	Yes	No	No	No	Yes	Yes	Yes	Unclear	Yes
Trasande, 2018	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Unclear	Not applicable	Yes
Vanasse, 2016	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Unclear	Not applicable	
Yu, 2018	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No	Yes

Table A3. Critical appraisal using Joanna Briggs Institute (JBI) checklist for cross-sectional studies by exposure period (n=7)

Author, Year	Were the criteria for inclusion in the sample clearly defined?	Were the study subjects and the setting described in detail?	Was the exposure measured in a valid and reliable way?	Were objective, standard criteria used for measurement of the condition?	Were confounding factors identified?	Were strategies to deal with confounding factors stated?	Were the outcomes measured in a valid and reliable way?	Was appropriate statistical analysis used?
Perinatal								
Bercovich, 2014	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes
Oni, 2015	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adult								
An, 2015	Yes	Yes	Yes	Yes	No	No	Yes	Yes
Bich, 2011	No	Yes	Yes	Yes	No	No	No	Yes
Husarewycz, 2014	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Katratzias, 2015	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Koroma, 2019	No	No	Yes	Yes	No	No	Yes	Yes

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Table A4. Critical appraisal using Joanna Briggs Institute (JBI) checklist for quasi-experimental studies by exposure period (n=10)

Author, Year	Is it clear in the study what is the 'cause' and what is the 'effect' (i.e. there is no confusion about which variable comes first)?	Were the participants included in any comparisons similar?	Were the participants included in any comparisons receiving similar treatment/care, other than the exposure or intervention of interest?	Was there a control group?	Were there multiple measurements of the outcome both pre and post the intervention/exposure?	Was follow up complete and if not, were differences between groups in terms of their follow up adequately described and analyzed?	Were the outcomes of participants included in any comparisons measured in the same way?	Were outcomes measured in a reliable way?	Was appropriate statistical analysis used?
Perinatal									
Xiao, 2019	Yes	Unclear	Not applicable	Yes	No	Yes	Yes	Yes	Yes
Adult									
Becquart, 2018	Yes	Unclear	Not applicable	No	No	Yes	Yes	Yes	Yes
Gerin, 2005	Yes	Yes	Not applicable	Yes	Yes	Yes	Yes	Yes	Yes
Hendrickson, 1996	Yes	Unclear	Not applicable	No	Yes	Yes	Yes	Yes	Yes
Kim, 2017	Yes	Unclear	Not applicable	No	Yes	Yes	Yes	Yes	Yes
Kong, 2019	Yes	Unclear	Not applicable	Yes	No	Yes	Yes	Yes	Yes
Lin, 2010	Yes	Unclear	Not applicable	Yes	No	Yes	Yes	Yes	Yes
McKinney, 2011	Yes	Unclear	Not applicable	No	Yes	Yes	Yes	Yes	Yes
Nagayoshi, 2015	Yes	Unclear	Not applicable	Yes	No	Yes	Yes	Yes	Yes
Silva-Palacios, 2015	Yes	Unclear	Not applicable	No	No	Yes	Yes	Yes	Yes

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	3
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	4
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	4
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	4
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	4
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	11 (Table 1)
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	5
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	5
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	5
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	5
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	5
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	5

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	N/A
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	N/A
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	6
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	6
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	10
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	6-10
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	N/A
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	N/A
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	N/A
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	10
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	12-13
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	14
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	1

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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